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Impact of seasonal changes in zooplankton population in the pond of Darbhanga district

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Abstract : The present study deals with the biological seasonal diversity and effect on ecosystem in zooplanktonic population in the Manth pond of Sunderpur area of Darbhanga, Bihar, India. The aim of investigation was to find out effect on ecosystem in zooplanktonic population due to seasonal diversity. The data collected in various months from March 2009- February 2010 revealed that the pond under study contains a much large number of zooplanktons: rotifers with 9 genera and 22 species, cladocera with 9 genera and 11 species and copepods with 6 genera and 6 species. With respect to seasonal variations and effect of ecosystem rotifers and cladocerans flourished well in warm weather months and copepods in cold weather months. The dominant zooplanktons were rotifers and shared 57.5% of the total zooplanktons whereas cladoceran shared 27.5% and copepods 15%.

Keywords : Zooplankton, pond, biological diversity

INTRODUCTION

Ponds are smaller bodies of still water located in natural hollows such as limestone sinks, or that result from the building of dams, either by humans as beavers. Ponds are found in most regions and may exist either seasonally or persist from year to year. Fresh water ecosystems usually contain a wide variety of life forms including bacteria, fungi and protozoa, bottom- dwelling organisms such as insect larvae, snails and warms and planktons, large animals like fish, amphibians, reptiles and birds. A pond or a lake ecosystem is a dynamic ecosystem since the boundaries are limited. The entry nutrients, sediments through the surface runoffs enter and remain in the system causing fluctuations in the physico-chemical characteristics of these ecosystems.

Animal species are considerably more diverse and numerous in inland waters than plants. Most of the major groups include terrestrial or marine species as well as fresh water forms. Apart from fishes, important groups with inland water species include crustacean (crabs, cray fishes and many more), mollusca, insects, sponges, flatworms, numerous microscopic forms. Bioindicators are evaluated through presence/ absence, condition, relative abundance, reproductive success, community structure (i.e. composition and diversity), community function (i.e. trophic structure). The presence or absence of the indicator species or indicator community reflects the environmental conditions of the water body under study.

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MATERIAL AND METHODS

The methods of collection, preservation and identification of population were as below

i) Collection- water was collected in a polythene container of 5 litre capacity from different place of Pond. For zooplanktonic counts, 50 litres of water were allowed to pass through No 30 HD filter cloth (100 meshes, cm-1) the filtrate containing planktons was transferred to a specimen tube, zooplanktons were killed by adding a few drops of formalin. The filtrate containing planktons were

concentrated by centrifugation at 3000-4000 r.p.m for 5 minutes. The concentrated sample was agitated to disperse organisms evenly. Zooplanktons were counted through Sedgewick Rafter cell under the microscope.

ii) Identification- with the help of monographs and relevant available literatures the zooplanktons were identified up to the species.

iii) Long term preservation- specimens were fixed and preserved in Formalin-Acetic acid (FAA) solution then specimens were transferred to a slide, one or two drops of Glycerine jelly was put over specimens. They were then covered with coverslip. The margin of cover slip was sealed by black paint and labeled.

RESULTS

The seasonal diversity and effect of ecosystem in community structure and population density of zooplankton collected from the pond of the present study during March 2009 to February 2010 were shown in Table 1 and 2. It was observed that the pond contains a much larger number of zooplanktons: i) Rotifers with 9 genera and 22 species ii) Cladocera with 9 genera and 11 species and iii) Copepods with 6 genera and 6 species.

With respect to seasonal variations and effect of ecosystem Rotifers and Cladocerans flourished well in warm weather months and copepods in cold weather months. The dominant Zooplanktons were rotifers and shared 57.5% of the total zooplanktons whereas Cladoceran shared 27.5% and Copepods 15%. These observations revealed that the seasonal variations and effect of ecosystem needs time to time investigations so far as fish culture in the pond is concerned.

Table 1- Biological seasonal diversity and effect on ecosystem in zooplanktonic population of the pond under study (March 2009- February 2010)

Zooplankton	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb
(a) Cladocera												
Alonopsis clongata	N	N	N	F	C	C	R	-	-	-	-	R
Alona intermedia	R	R	R	C	C	F	F	-	-	-	-	R
(b) Diaphenosomebrachyurum	R	R	R	R	F	C	-	-	-	R	C	R
Moina dubia	C	R	R	R	F	F	-	-	C	C	C	C
Moina macrocarpa	A	A	A	C	F	R	-	-	C	C	C	C
Moina daphnissp	-	-	-	F	F	F	C	C	C	-	-	-
Pleuroxus uncinatus	C	C	C	-	-	-	--	-	C	C	C	C
Pleuroxus bastatus	F	C	C	R	R	R	F	F	-	-	-	-
(c) Copepoda												
Cyclops viridis	R	R	R	F	F	F	F	C	A	A	A	L
Diaptomus blancli	R	F	C	C	C	-	-	-	A	A	A	S
Ectocyclops sp	A	A	C	C	C	-	-	-	-	-	F	F
Masocyclops sp	-	-	-	-	C	C	C	R	R	R	R	S
Masocyclops sp	C	C	C	C	C	C	A	A	A	A	A	C
Paracyclopina sp	C	C	C	C	C	C	A	A	A	A	A	C
(d) Rotifera												
Anuraeopsis fissa	C	A	A	A	R	R	R	F	F	F	R	C
Asplanchna sp	-	-	-	-	R	R	R	F	F	F	R	-
Brachionus plicatilis	C	C	C	C	F	F	S	S	R	R	F	F
B. variabilis	-	-	-	-	-	-	R	R	F	F	F	F
B.mirabilis	C	R	F	F	R	R	R	R	C	C	C	A
R. rubens	-	-	-	-	C	C	C	L	A	-	-	-
B. angularia	C	-	-	-	-	-	-	-	C	C	C	C
B. patulus	F	F	C	C	C	C	A	A	A	F	F	F
B. falcatus	R	-	-	-	C	C	C	A	-	-	-	R
B. calyciflorus	C	F	R	-	-	-	-	-	-	-	-	C
Filinia logiseta	-	R	R	-	-	F	L	S	F	R	-	-
F.teminalis	-	R	R	C	C	C	L	L	L	S	R	-
Gastropus sp	R	F	R	F	F	C	R	-	-	-	-	-
Karatella tropica	F	F	R	R	-	-	-	-	-	-	-	-
Keratella sp	F	R	F	C	C	-	-	-	-	-	-	-
Locane leontine	-	-	-	-	-	R	F	F	C	C	R	-
Locane (Monosttla) quadridentala	-	-	-	R	R	F	F	R	-	-	-	-
Lacane (M.) crenata	-	-	-	-	-	-	-	R	R	R	F	F
Loane (M.) angulata	C	-	-	-	-	-	-	F	R	C	C	C
Lecana elsa	R	F	F	F	R	-	-	-	-	-	-	R
Mytilina sp.	N	N	R	R	F	N	N	-	-	-	-	N
Tricocerca flageldata	N	N	F	F	F	F	-	-	-	-	-	-
Rotareia vulgaris	C	F	F	N	N	-	-	-	R	R	R	C
(e) Cariodaphnialacustris												
Daphnia pulex	N	N	N	R	C	A	A	F	F	R	R	-
Daphnia sp.	L	L	L	R	R	R	N	N	-	-	-	S

Table – 2 Biological Seasonal diversity and effect on Ecosystem in percentage composition of different Groups of Zooplanktonic Population of the Pond Umder study (March 2009- February 2010)

Months	Rotiers	Cladocera	Copepoda
March	78.00	40.0	15.00
April	80.2	50.00	13.5
May	102.5	65.2	11.5
June	90.00	65.00	9.5
July	94.5	55.40	10.4
August	85.00	54.50	4.00
September	85.00	46.00	4.00
October	80.00	45.00	40.00
November	75.5	40.2	45.50
December	46.6	35.4	45.60
January	45.0	35.6	44.5
February	56.2	36.3	30.00

DISCUSSION

In the present studied pond, 22 species of rotifers, 11 species of cladocera and 6 species of copepods have been recorded. There is a conflicting report of other authors in this regard to the recorded 10 species of zooplanktons observed by them. Shibu (1991)[1] recorded 10 species of rotifers, 15 cladocerans, and 10 Copepods from a fresh water tank of Bengal. Mahananda *et al.* (2010)[2] reported only few copepods and cladocerans from some ponds of South India. Jindal and Singh (2005)[3] reported 54 species Zooplanktons from some lakes. In temperate regions, Mendhe (1996)[4] recorded 9 rotifers. 7 Cladocerans and 10 copepods from studied lake. Grumbine (1994)[5] 25 species of rotifers, 5 cladocerans and 2 copepods in reservoir lake. Thus, it is obvious that tropical fresh water bodies do not always have fewer species of zooplankton fauna than temperate ones as held by Vaishali *et.al* (2007)[6]. Whereas Walter (2002)[7] observed the occurrence of high population of zooplankton during warm weather months and low during cold weather months. The present findings were in conformity with Nandan (1994)[8] who also recorded that rotifers constitute the most abundant group of zooplankton which shows almost rhythmic appearance and disappearance during the year. Pradeep *et.al*, (2010)[9] have pointed out that dens and poor population of zooplankton fauna depends upon a number of factors namely wave action, wind velocity and scarcity of vegetation presumably act as determinants for zooplanktons growth and reproduction.

CONCLUSION

The present works detail the study carried out at one water-body, designed for fish culture particularly, namely Manth pond. The analysis was done based on the standards methods prescribed by APHA. The biological parameter included Zooplanktons analysis. The detailed investigation of the parameters, biological seasonal diversity and effect on ecosystem in Zooplanktonic population as well as in percentage composition of different groups of Zooplanktonic population of the pond under study. In the present studied manth pond, 22 species of rofifers, 11 species of cladocera and 6 species of copepods have been recorded.

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CONFLICT OF INTEREST

Authors declare no conflict of interest regarding publication or any other activity related to this article.

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Isolation of a new flavone from *Negella sativum*

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Abstract : The plants produce a lot of organic compounds but some compounds like alkaloids, jerpenoids, phenolics, flavonoids, chromines and other minor chemicals have some medicinal use. A new flavonoids from *Nigella sativum* has been isolated and on the basis of U.V., MNR spectra and microanalysis the new flavone tephroriamin is characterized as {12" 3". dihidro - 5-methoxy-3" 2"- acetoxy 2"- methyl propylidene)- 2"-Oxo furyl (4".5". 7.8. flavones)

Keywords : Plants, isoloation, flavones, *Nigella sativum*

INTRODUCTION

The plants produce a lot of organic compounds but their essential components which metabolite are called primary compounds.

The primary compounds like carbohydrate lipids, nucleotides and peptides are shared by all living organism and are central to life processes. The secondary compound is also derived from essential components, but is not central to metabolism.

Secondary compounds include alkaloids terpenoids, phenolies, flavonoids, chromines and other minor chemicals. The so called secondary compounds, which have no known function in photo synthesis, growth or other aspects of plant physiology but help in protection against diseases, herbivores and environment stress. A lot of study has been done on the role of natural products from plants for the control of insect pests.[1,2]

Accordingly, chemical investigation of seeds *Nigella sativum* was taken up that resulted in the isolation of new flavone tephrorianin characterized as {2"3" di hydro - 5 Methoxy-3" (2" acetoxy 2" -methylpropylidene) - 2"-Oxofuryl 4",5" 7,8- flavone)

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EXPERIMENTAL

Mps (uncorrected) were recorded on electro thermal apparatus. U.V. spectra were recorded on sysfronics U.V. visible spectrophotometer 119. 1H NMR spectra were recorded in $CDCl_3$ on a Bruker WM400 MHz spectrometer. Using Me4si as an internal reference. Mass spectra were measured on Jeol JMSD-300 spectrometer at 70 eV. Micro analysis were performed by RSIC at CDIR Lucknow. Silica gel used for column chromatography was 60-120 mesh.

EXTRACTION

Oven dried *Nigella sativum* seeds (0.75Kg.) were extracted with hot $CHCI_3$; and concentrated extract (4.5 gm) on column chromatography yield among other unidentified compounds tephrorianin (10mg) in $CHCl_3$ - EtOAc (80:20).

Elemental Analysis

Required for the Molecular formula $C_{24}H_{20}O_7$	found experimentally
C,68.57%.	C,68.51%

H,4.76%. H,4.82%
O,26.66%. O,26.62%
Colure crystals from methanol (10mg)
M.P.-228.300
U.V.$CHCl_3$/nm:266,341
I.R(KBr)/cm-1:1796,1727,1651,1599 and 1496
1HNMR(400 MHzCD Cl_3) :1.90(1H,S,Gen-Me_2), 2.12(3H,S,$OCOCH_3$);4.05(3H,OCH3)6.65 (1H,S,H6) 6.68 (1H, S, H,3); 7.50-7.60 (3H,M. H-3.H-4' and H-5'); 7.80 - 7.84 (2H,m,H-2' and H-6');7.85 (1H.S.H-1")
Mass (70ev) m/z 419 ([M-1]+, 4.8(7.1); 376 (5.2); 360 (5-7); 359 (11.2); 358 (34.8);301 (5-1) 213 (100);156 (11.4) 102 (5.0)

DISCUSSION

Tephrorionin exhibited UV absorption maxima at 266 and 341 nm. The I.R. spectrum showed the presence of chromone carbonyl absorption at 1615 cm -1: actoxyl carbonyl absorption at 1727 cm-l and lactone carbonyl absorption at 1796 cm-1

The 1HNMR spectrum of tephorianin revealed the presence of an unsubstituted phenyl ring (7.50, 7.84 &, m, 5II), a gem-dimethyl group (1.90) and one methoxyl group (4.05). The C-3 Proton appeared as a singlet at 6.68. The spectrum further revealed two more singlets at 6.65 and 7.85 each integrating to one proton. The presence of lactones carbonyl group as indicated by I.R. together with an acetoxyl group, gem-dimethyl group and one proton singlet at 7.85 suggested the presence of an additional five member lactone ring system with an acetoxyl-2 methyl propylidene substituent. Thus additional ring system was placed at the C-7 and C-8 position on biogenetic grounds and the methoxyl group at C-5 position the singlet at 7.85 was assigned to C-1" proton and 2-acetoxyl- 2 methyl propylidene substituent[3] and the singlet at 6.65 was assigned to C-6 proton. Thus the structure of tepherorianin was established as [2", 3". di hydro-5 methoxy 3"(2" 'actoxy 2" methyl propylidene) 2"-Oxofury1-(4" 5" 7.8) flavones). The mass spectrum of tephororianin exhibited [M-1] + peak corresponding to 419. Flavones having either C-5 or C-8 methoxyl group normally exhibit [M -1] +Peak.[4,5]

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Studies on the effects of copper sulphate on liver protein content of an air-breathing fish *Clarias batrachus* (Linn.)

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Abstract : Among the various toxic pollutants, heavy metals are particularly severe in their action due to tendency of bio-magnification in the food chain. The global heavy metal pollution of water is a major environmental problem. With the advent of agricultural and industrial revolution, most of the water sources are becoming contaminated. Natural waters are often contaminated by untreated wastes of industrial, technological and agricultural origin containing various metallic compounds. Heavy metals due to their bio-accumulative and non biodegradable properties constitute a core group of aquatic pollutants. The direct or indirect bioaccumulation of metals in organisms in contaminated water is of great significance because it affects the food chain. In present investigation, the effect of copper sulphate on freshwater fish *Clarias batrachus* (Linn) has been shown. It was found that the biochemical parameters shown gradual decrease level in tissues exposed by copper sulphate.

Keywords : copper sulphate, liver, protein, fish, *Clarias batrachus.*

INTRODUCTION

The modern civilization, industrialization, urbanization and increased population have led to fast degradation of our aquatic environment. Water is polluted by domestic sewage, animal and industrial wastes. In addition, various chemical compounds like heavy metals pesticides, acaricides, fungicides, herbicides, algaecides and other chemical substances are used for agricultural production. These compounds load off the waterways with various chemicals that alter the nature of the water.[1]

Heavy metals are being utilized in a variety of ways in industries, agriculture and food processing and household in many forms. Metals are unique environmental and industrial pollutants in the sense that they are neither created nor destroyed by human beings but are only transported and transformed into various products.[2] Heavy metals play a vital role in the growth and development of plants. They may act as co-factors of some enzymes and help in the formation of intermediate metabolites. The primary anthropogenic sources of heavy metals are point sources such as mines, foundries, smelters, and coal-burning power plants, as well as diffuse sources such as combustion by-products and vehicle emissions. When excess amount of metals are absorbed by plants and animals, toxic effects are produced resulting in impairment of growth, inhibition of respiration and abnormalities in cell division and the extent of injury depends on the concentration of the metals. The heavy metals move from fields to ponds, lakes, rivers, estuaries and finally to sea, where they adversely affect the growth and survival of the non target species like fish and other aquatic animals.[3]

Among the various toxic pollutants, heavy metals

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are particularly severe in their action due to tendency of bio-magnification in the food chain. The global heavy metal pollution of water is a major environmental problem. With the advent of agricultural and industrial revolution, most of the water sources are becoming contaminated. Natural waters are often contaminated by untreated wastes of industrial, technological and agricultural origin containing various metallic compounds. Heavy metals due to their bio-accumulative and nonbiodegradable properties constitute a core group of aquatic pollutants. The direct or indirect bioaccumulation of metals in organisms in contaminated water is of great significance because it affects the food chain.[4,5]

For the past few decades, considerable attention had been focused on the fate of metals and their derivatives in the aquatic environment. (Environment Agency warning over *Clarias* Catfish found in River Tonge, Bolton. The Bolton News. 19 July 2017). The impact of heavy metals on aquatic animals can be evaluated in terms of toxicity and physiological alterations on aquatic animals. Various investigators studied intrusion of heavy metals and their salts in terrestrial and aquatic environments, and their toxicity to aquatic system. The heavy metal contaminations resulted in epidemic diseases such as "Minamata"[6] and "Itai Itai"[7]. The heavy metals were reported to cause massive fish kills and other aquatic animals.[8-13]

Heavy metals such as copper, lead, etc. are naturally occurring substances in the Earth's crust. Some are needed by humans in trace amounts to maintain optimum body function, whilst others such as mercury, arsenic, and cadmium can be harmful. All metals can be toxic if levels are too high.

Copper is one of 26 essential trace elements occurring naturally in plant and animal tissues and its availability is influenced by physico-chemical, hydrodynamic and biological factors. It makes its way into the receiving waters by extensive use in agriculture apart from usage in various industries like textile, tanneries, paints, battery, laundry, photography, copper ware and piping for water distribution systems. Copper ions are quite toxic to fish at various functional levels when environmental concentrations are increased.[14-16]

MATERIALS & METHODS

The fish were procured live from the local fish market with the help of fishermen. They were brought to the laboratory in wide mouthed large earthen pots half filled with natural water and covered with mosquito net. Every effort was taken to give least stress to the fish during transportation. They were first thoroughly washed and then rinsed in 0.1% $CuSo_4$ solution to remove any sort of dermal infections. Thereafter, healthy fish of an. average length (12-16 cm) and weight (80-100g) were transferred one by one with the help of small hand net to 40 litre rectangular glass aquaria for acclimation to various laboratory conditions for a fortnight.

The fish were not supplied any food for the first three days of acclimation. The fish were adjusted to natural photoperiod and ambient temperature. Running tap water was used in all the experiments and no aeration was done. They were fed with chopped goat liver every day Ad libitum at 11. 00 A.M. sharp. The feeding was stopped 24 hr before the start of experiment (Static acute bioassay) and no food was supplied to them during the period of experimentation. However, the feeding schedule was strictly followed during chronic experiments to avoid starvational effects.

TISSUE BIOCHEMISTRY

At the end of exposure period (day 30) the fish were anaesthetized with 1:4000 MS 222 (tricane methane-sulfonate, sandoz) for two minutes. The liver, muscle and gonads were quickly dissected out. Weighed and processed fur the quantitative estimation of total protein.

ESTIMATION OF TOTAL PROTEIN

The Folin-Lowry Method was followed for the estimation of total protein of liver, muscle and gonads. Tissues were homogenised in TCA buffer and then centrifuged at 3000 rpm for 10 minutes. The TCA precipitate the protein. After centrifugation the white precipitate so formed was analysed for the estimation of total proteins.

To 1 ml of the test solution was added 5 ml of alkaline solution (2% Na_2CO_3 in 0.1N NaOH + 0.5% $CuSO_4$ in 1% Na-K tartarate in equal volume). It was mixed thoroughly and allowed to stand at room temperature for 10 minutes. Thereafter 0.5 ml of diluted Folin-Ciocalteau reagent, which gives a coloured complex on reaction with protein, was added rapidly with immediate mixing. After 30 minutes rite optical densities were noted against the appropriate blank at 750 nm with the help of spectrophotometer (Spectronic-20).

Bovine serum albumin was used as standard in this test. After preparing a standard curve protein concentration of the test solution was estimated.

RESULTS AND DISCUSSION

In present investigation, the effect of copper sulphate on freshwater fish *Clarias batrachus* (Linn) has been shown. It was found that the biochemical parameters shown gradual decrease level in tissues exposed by copper sulphate.

Total protein

The changes observed in protein content exposed by copper sulphate on *Clarias batrachus*. It can be expressed as mg of protein/gram wet weight of tissue from control to treated set at 24 hr, 48hr, 72hr and 96 hours.

Freshwater catfishes treated with copper sulfate showed decline protein range from control set to 96 hours as in muscle 50.90+ to 30.16+2.30, in gills 60.05+2.16 to 31.30+1.52, in liver 65.05+1.71 to 42.40+1.19, in Gonads 55.49+1.71 to 42.36+1.19 (Table 1 and Fig 1).

Table 1- Levels of protein content in different tissues *of Clarias batrachus* exposed to copper sulphate

Sl. No.	Tissue	Control	Experimental			
			24 hrs.	48 hrs.	72 hrs.	96 hrs.
1.	Muscle	97.23±2.06	90.50±2.10	84.20±2.07	78.56±0.30	75.14±2.29
2.	Gills	80.40±2.30	74.52±1.19	71.49±1.20	67.55±1.90	62.30±1.52
3.	Liver	65.05±1.02	58.25±1.55	55.50±1.18	51.03±1.40	47.20±1.47
4.	Gonads	90.50±1.80	87.80±1.11	77.28±1.56	69.06±1.06	65.40±1.18

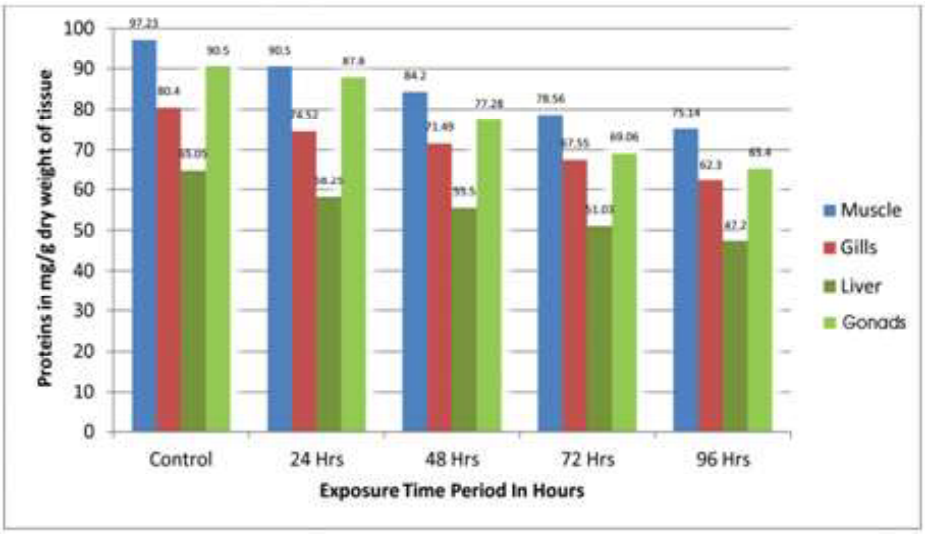


Fig- 1 – Levels of protein content of *Clarias batrachus* exposed to copper sulphate

Protein is most characteristic organic compound found in the living cell while the protoplasm of the cell is composed of protein. They play vital role in the process of interaction of cellular medium. During presents investigation increase in muscle and liver protein content of *Clarias batrachus* had been observed during 24 hours and 48 hours respectively after exposure to sub-lethal concentration of copper sulphate gonads also showed increased protein content during 48 hours. Increase in protein content occurs due to anaerobic metabolism which contribute major role in this situation. Proteins are main component for the building blocks of animal body.

In present investigation the total proteins were found progressively depleted in all the tissues throughout the exposed time. The percent responses appeared tissue-specific and the magnitude of depletion was more marked in the tissues of fish exposed to mercury chloride. The bio-accumulative heavy metal copper with biologically active constituents of the body such as lipids and proteins.

Another reason for increase in protein content is due

to increase in protein synthesis by increase the enzyme activity involved in protein synthesis. Toxic stress like copper sulphate treatment to the fish causes increase in the rate of protein synthesis. In experimentation, after 96 hours the decline in protein content was observed. This fall attributed to the constantly increasing contact of the copper sulphate with the bio-system which ultimately resulted in protein breakdown.

From the above results, it is clear that copper sulphate is very toxic and it lowers the rate of energy metabolic levels forcing the proposed fish to adapt to a sustained metabolic level, but shifting the gears of energy trapping from one segment to the other. The copper sulphate intoxicated proposed fish cannot rise to the high energy demands if necessary as there is limited energy production. Increasing fluoride excretion vit. C intake reduces its burden. Concluded that vit. C are useful in regulating fluoride induced improve the antioxidant status.

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Effect of growth regulators on two varieties of rice (Peizataifeng & Huayou 86)

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Abstract : In this study Gibberellic acid (GA3), Paclobutrozol (PBZ), 6- Benzylaaminopurine (6-BA) treatment and distilled water (control) were sprayed to two hybrid rice cultivars (Peizataifeng and Huayou 86) at the heading stage in the field experiments in both early and late season in 2007. In early season, grain yield of peizataifeng under PBZ and 6-BA treatment was remarkably higher than that of ok respectively. The higher yield was found for GA3, PBZ and 6-BA treatment, while the lowest one was found under control treatment.

Keywords : Gibberellic acid (GA-3), Paclobutrazol (PBZ), 6-Benzylaminpurine(6-BA)

INTRODUCTION

It is important to realize that the use of regulators in agriculture is much more extensive than is commonly realized the field is relatively old when one considers that ethylene has been used to degreen oranges since 1920's and auxins have been used to promote rooting of cuttings since the 1930's During this rather long history plant growth regulator research Indirectly gave a major boost to the herbicide industry. It was studies on rooting and auxins that lead ultimately to the discovery of 2,4-D. This compound at the first highly active organic herbicide with species selectivity was the keystone of the modern chemical herbicide industry at the present time there are some applications of growth regulators to major field crops. Plant growth regulators to play important roles in plant growth and development, but little are known about roles of plant growth regulators in yield, grain qualities and antioxidant enzyme activities in super hybrid rice.

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MATERIALAND METHOD

In this study gibberellic acid (GA3), paclobutrazol (PBZ), 6-Benzylamino purine (6-BA) treatment and distilled water (control) were sprayed to two hybrid rice cultivars (Peizataifeng and Huayou 86) at the heading stage in the field experiments in both early and rate season in 2007. Treatments were arranged in a split-plot design with four replications. Cultivars treatment with two newly developed super hybrid rice Peizataifeng and Huayou 86 were the main plots and plant growth regulators treatment were the subplots. Subplot treatment included (1) plots sprayed with distilled water (CK), (2) plots sprayed with 20 mg L-1 GA3 prepared using 95% ethanol as surfactant (GA3), (3) plots sprayed with 50 mg L-1 PBZ (PBZ), (4) plots sprayed with 30 mg L-1 6-ba (6-BA).

RESULTS

In early season grain yield of Peizataifeng under PBZ and 6-BA treatments was remarkably higher than that of CK, respectively. The higher yield was found for GA3, PBZ and 6-BA treatments, while the lowest one was found

under control treatment. Significant differences were found in the number of spikelets per panicle and grain filling percentage between GA3, PBZ, 6-BA treatment and CK in two cultivars of Peizataifeng and Huayou 86 in both early and late season in 2007. There was significant increase in the number of spikelets per panicle and grain filling percentage under plant growth regulator treatment in Peizataifeng and Huayou 86 compared with the CK, respectively. Higher grain yield under GA3, PBZ and 6-BA treatments was mainly ascribed to significant higher spikelets per panicle and grain filling percentage. On average, grain yield of Huayou 86 was higher than that of Peizataifeng. Average grain yield was higher in late season than that in early season.

Table 1 Effects of plant growth regulators on yield and its components of cultivars of *Peizataifeng* and *Huayou 86* in both early and late seasons in 2007

Treatments		Panicle number (m^{-2})	No. of Spikelets per panicle	Grain filling percentage (%)	1000-grain-weight (g)	Yield (kg ha^{-1})
2007 early rice						
Peizataifeng	CK	250 b	179.5 b	78.1 b	22.1 a	6807.7 b
	GA3	258 b	205.3 a	86.7 a	22.2 a	6565.0 b
	PBZ	258 b	198.0 a	85.8 a	23.1 a	7388.3 a
	6-BA	275 a	196.8 a	84.6 a	22.8 a	7626.7 a
	mean	260 A	194.9 A	83.8 A	22.5 A	7096.9 B
Huayou86	CK	228 b	140.6 a	80.4 b	19.8 b	7583.3 a
	GA3	271 a	122.9 b	86.2 a	20.0 b	7020.0 a
	PBZ	275 a	145.4 a	86.6 a	21.6 a	7930.0 a
	6-BA	258 ab	138.3 b	84.5 a	21.7 a	7940.7 a
	mean	258 A	136.8 B	84.4 A	20.7 B	7618.5 A
2007 late rice						
Peizataifeng	CK	273 a	135.0 b	80.8 b	21.7 b	7080.0 b
	GA3	276 a	141.4 ab	86.9 a	21.9 ab	7920.1 a
	PBZ	266 a	135.6 b	84.3 a	23.0 a	7740.0 ab
	6-BA	273 a	149.2 a	83.9 a	22.6 a	7680.7 ab
	mean	272 A	140.3 B	84.0 B	22.3 A	7605.0 B
Huayou86	CK	265 b	165.3 a	86.9 a	20.4 a	8280.0 a
	GA3	280 a	171.8 a	89.7 a	20.7 a	9040.0 a
	PBZ	269 b	168.3 a	88.9 a	20.4 a	8406.0 a
	6-BA	283 a	168.4 a	88.4 a	20.2 a	8620.0 a
	mean	274 A	168.5 A	88.4 A	20.5 B	8586.5 A

Within a column for two groups of genotypes, common letters are not significantly different at the 5% level.

Table 2 Effects of plant growth regulators on brown rice rate, milled rice rate, head rice rate, chalkiness and amylose content of cultivars *Peizataifeng* and *Huayou 86* in both early and late seasons in 2007

Treatments		Brown rice (%)	Milled rice (%)	Head rice (%)	Chalkiness (%)	Amylose content (%)
2007 early rice						
Peizataifeng	CK	81.5 a	73.8 a	63.4 b	9.8 b	11.8 b
	GA3	81.6 a	74.3 a	66.2 ab	13.2 a	12.5 b
	PBZ	81.9 a	75.9 a	68.9 a	10.1 b	17.1 a
	6-BA	82.0 a	75.4 a	66.5 ab	13.6 a	17.6 a
	mean	81.7 A	74.9 A	66.3 B	11.7A	14.7 A
Huayou86	CK	81.4 a	75.4 a	66.0 b	12.9 a	15.5 a
	GA3	82.5 a	75.6 a	67.5 b	12.4 a	15.1 a
	PBZ	82.7 a	75.3 a	74.7 a	10.7 a	16.5 a
	6-BA	81.3 a	74.0 a	67.2 b	11.4 a	14.3 a
	mean	82.0 A	75.1 A	68.9 A	11.9 A	15.4 A
2007 late rice						
Peizataifeng	CK	82.0 a	76.3 a	67.7 a	8.3 ab	14.5 b
	GA3	82.4 a	77.2 a	70.0 a	10.1 a	17.9 a
	PBZ	82.1 a	76.8 a	69.7 a	9.2 a	17.2 a
	6-BA	82.0 a	77.4 a	69.7 a	6.7 b	17.2 a
	mean	82.1 A	76.9 B	69.3 A	8.6 A	16.7 A
Huayou86	CK	81.9 a	78.6 a	68.9 a	7.0 b	14.0 b
	GA3	81.7 a	78.6 a	70.1 a	8.7 b	18.5 a
	PBZ	82.2 a	79.0 a	71.3 a	12.1 a	18.7 a
	6-BA	81.2 a	79.0 a	70.1 a	10.1 ab	18.0 a
	mean	81.7 A	79.0 A	70.1 A	9.5 A	17.3 A

CONCLUSION

Overall, foliar application of plant growth regulator at heading stage can increase grain yields in Peizataifeng and Huayou 86 in both early and late seasons in 2007. In early season, grain yield of Peizataifeng under PBZ and 6-BA treatments was remarkably higher than that of CK, respectively. The higher yield was found for GA3, PBZ and 6-BA treatment, while the lowest one was found under control treatment. Significant differences were found in the number of spikelets per panicle and grain filling percentage between GA3, PBZ, 6-BA treatment and CK in two cultivars of Peizataifeng and Huayou in both early and late season in 2007. There was significant increase in the number of spikelets per panicle and grain filling percentage under plant growth regulator treatments in Peizataifeng and Huayou 86 compared with the CK, respectively.

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Hematological profile of male and female *Anabas testudineus*

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Abstract: The haematological profile of air breathing fish *Anabas testudineus* male and female was studied. The parameters studied were Haemoglobin (Hb), haematocrit (HCT), mean corpuscular volume (MCV), mean corpuscular haemoglobin (MCH), mean corpuscular haemoglobin concentration (MCHC), erythrocyte red blood cells (RBC) or erythrocytes, lymphocytes, monocytes, neutrophils and eosinophils. Most haematological parameters indicated sex variations. Higher Hb concentration, RBC counts and HCT percentages were found in male compared to their counterparts but higher MCH values, WBC counts and ESR levels were detected in female than in the male. In contrast, higher level of ESR was observed in female than male. Total counts of RBC observed in both male were significantly higher than in the female. MCV was found higher in male and female. However, no significant differences were observed in neutrophils, lymphocytes and eosinophils. Findings of this study showed sex variations in the haematological parameters between *Anabas testudineus*.

Keywords : Blood, air breathing, *Anabas testudineus*, haematology

INTRODUCTION

The blood volume of fishes varies between 2-8% of their body volume.[1] One third to one half of the total blood volume in fish consists of blood cells, the rest is fluid plasma.[2] In spite of systematic diversity all fishes possess two main types of blood cells erythrocytes (red cells) and leucocytes (white cells), a property shared by the land living vertebrates, which are derived from early fish like ancestors. A bibliography on fish hematology was assembled by Hawkins and Mawdesley-Thomas (1972).[3] The major proteins of fish blood plasma are globulin to a large extent antibodies- immunoglobulin. A peripheral blood serum from a healthy fish shows a preponderance of erythrocytes. Other cells are lymphocytes, neutrophils, monocytes, eosinophils, basophils and thrombocytes or platelets. Blood parameters of fishes may have species specific variations in cellular counts and biochemical values.

Air breathing climbing perch in India especially in Bihar is a commercially important high value fish. However, although a few literatures are available on the haematological parameters in various fishes of Bihar in which haematological parameters of climbing perch have never been reported. Banerjee (1966)[4] has reported on the haematological profiles of *Anabas testudineus*. This study, therefore, was designed to see the variation between male and female in hematological parameters.

MATERIALS AND METHODS

Sexually matured *Anabas testudineus* were used as experimental animals purchased from the local market of Madhepura of Bihar.

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The experimental variables were the various blood parameters such as haemoglobin (Hb), erythrocyte sedimentation rate (ESR), red blood cells (RBC), haematocrit (HCT), mean corpuscular volume (MCV), mean corpuscular haemoglobin (MCH), mean corpuscular haemoglobin concentration (MCHC), white blood cells (WBC), differential neutrophils, lymphocytes, monocytes, eosinophils and basophils counts.

Samples of blood were drawn by severing the caudal peduncle. A sample of 2 ml blood was collected by using 3 ml plastic syringes treated with an anticoagulant: EDTA from (n=5) and held in sample test tube containing K3 EDTA solution. Blood parameters were assayed by using Mythic 18 Automated Haematology Analyzer (China). This instrument analyzes the blood parameters using three detector blocks and two types of reagents. To analyze haemoglobin by automated methods, the cyanmet hemoglobin method or oxyhaemoglobin method was applied.

Table 1: Blood parameters of *Anabas testudineus*

Sl. No	Parameters	Male	Female
1	Haemoglobin (g d/l)	13.27± 0.69	9.8 ± 0.12
2	ESR (mm,1st h)	1.67± 0.33 m	42.67 ±0.33
3	RBC (m/ml)	3.03± 0.23	1.9 ± 0.10
4	HCT (%)	50.67 ± 3.76	30.33 ± 0.88
5	MCH	44.00 ± 2.65 pg	52.00 ±2.08 pg
6	Total WBC ($\times 10^4$ cu/mm)	5.04 ± 0.32	9.05 ± 0.36
7	Neutrophils (%)	21.67 ± 1.33	18.33 ± 2.03
8	Lymphocytes (%)	75.33 ±2.60%	71.33 ± 1.86%
9	Monocyte (%)	2.33 ± 0.33%	75.33 ±2.60%
10	Eosinophils (%)	4.33 ± 0.67%	3.67 ± 0.67%

RESULTS AND DISCUSSION

Higher Haemoglobin concentrations were found in male (13.27 ± 0.69 g d/l) than female (9.8 ± 0.12 g d/l)

The observed higher Hb concentration found in male compared to females indicates sex differences. Males are more active and aggressive than the female, which is in agreement with Smith *et al.*(1952).[5] Hb concentration in different fishes living in similar water bodies may vary with each other considerably, but the variation in the same species is very little. The observed similar Hb concentration in the same sex between two fish is expected.[6] Levels of Hb found in both sexes koi was above 9 g d/l, which denotes the occurrence of no anaemic condition[7]. The concentration of Hb found in *C. batrachus* ranged 8.0 - 19.2 g d/l with an average of 14.24 ± 0.66 g d/l.[8]

The female (42.67±0.33 mm, 1st h.) had significantly higher ESR levels than in the than male (1.67± 0.33 mm, 1st h.).

The observed level of ESR in male was fewer than 10 mm, 1st h which indicates the normal physiological condition. Higher levels of ESR detected in female than male could be because of the production of large number of eggs compared to the sperms in male. Similar observation was found by (Mishra *et al.* 1977)[9] in *Amphipnous cuchia*. The observed higher ESR in females indicates the chances of occurrence of infections because at high ESR chronic inflammatory conditions, connective tissue diseases, anaemia etc. may attack fishes. ESR is used as the indicator of the general health of the fish.[10] It has also been used for the early detection of infections in *Salvelinus fontinalis.*[11]

While comparing, total counts of RBC observed in male (3.03± 0.23 m/ml) were significantly higher than in the female (1.9 ± 0.10 m/ml).

RBC containing hemoglobin plays a vital role in respiration described by Wells *et al.*1980.[6] The present data of RBC count was similar to the range found by Banerjee, 1966[4] in the same fish. Present study showed higher RBC counts in male than the female which denote sex variations. The observed higher RBC counts in male perhaps because of their hyper activeness compared to the female. RBC counts have proven to be a highly variable blood parameter among fishes and showed a remarkable difference in their sex. Svobodova *et al.* 2008[12] also found higher RBC in males of seven fish species.

HCT (%) observed in male (50.67 ± 3.76%) were significantly higher than in the female (30.33 ± 0.88%).

Males had significantly higher value of HCT than in the female that perhaps because of relatively higher RBC counts in male. The likely another reason could be of sex difference where male may remain stressed from hyperactivity in nature.[7] A lower haematocrit value in females indicates iron deficiency anaemia and relatively low RBC count etc.

The MCV value was found more in male (167.33±1.20 fL) than female (160.00 ± 4.16 fl).

MCV value reflects the size of red blood cells by expressing the volume occupied by a single red blood cell. Normal value varies according to the species, sex, age etc.

MCH levels observed in female (52.00±2.08 pg) were significantly higher than the levels measured in the male (44.00 ± 2.65 pg).

The higher MCH value in female than in the male indicates higher likelihood of occurrence of macrocytic anaemia in females than in male.[13]

The MCHC found in female (32.33± 0.67 g/dl) was significantly higher than in the male (26.33 ± 0.67 g/dl).

The observed higher level of MCHC in indigenous female than their male counterparts could be explained by sex differences. High levels of MCHC indicate more Hb in a unit of RBCs.[12] In case of fish MCHC there is no prescribed normal range. In teleosts this has been found species specific. The levels of MCHC have been detected 38.57, 29.93 and 27.16 g/dl in *Labeo rohita, Clarias batrachus* and *Channa punctatus*, respectively.[14]

Total WBC counts detected in both female (9.05 ± 0.36×104 cu/mm) were significantly higher than the levels measured in the male (5.04 ± 0.32 × 104 cu/mm).

Observed WBC in female were significantly higher than the levels measured in the male, which indicate egg carriage, infection, adverse condition.[15] Levels of WBC in climbing perch are considerably higher than the level of man (Normal range: 4000-11000 cu/mm) as expected.[16] In fishes the significance of WBC and their biological function are not clearly understood. The number in some species of fish varies greatly with age, season and maturation.[17]

Percentage composition (%) of neutrophils found in both male and female male (21.67 ± 1.33%) were similar (18.33± 2.03%).

Neutrophils are the most important of the leucocytes as they show the greatest sensitivity to change in the environment. Their characterization and identification is, therefore, of significance for assessing the changes in the physiological state of fishes.

Sexually matured male and female (71.33 ± 1.86%) had similar levels of lymphocytes count (75.33 ± 2.60%). Similarly, no difference was detected in the lymphocyte (%) composition within male and female groups.

Lymphocytes were regarded as multi potential haemopoietic stem cells.[18] But now they are regarded as highly differentiated cells with immunological stimuli in various ways and known as immune-competent cells. This similar immune level could be because of healthy fish farmed in farms and natural environment.

Similar level monocyte (%) was measured both in male (2.33 ± 0.33%) and in the female (1.33± 0.33%) and within groups.

Differential leucocyte counts in koi confirmed the presence of monocytes in teleost blood which is in agreement with Ellis (1977)[19]. Similar leucocytes count in both sexes of Thai and indigenous koi indicates that leucocytes count do not vary between sexes. The percentage of monocyte found in some fishes were 37% with an average of 4.89 ± 1.94%) in *Clarias batrachus*.[8]

Both (4.33 ± 0.67%) male and female (3.67 ± 0.67%) had similar composition of eosinophils.

According to Ellis (1977)[19] presence and absence of fish eosinophil are contradictory. While the previous workers opted for their absence, recent workers opted for their presence. During present investigation eosinophil was detected which deny the absence of eosinophil opted by earlier workers. The observed no difference in eosinophil counts between sexes shows similar physiological condition in terms of WBC differential counts. Eosinophil percentages detected in *C. batrachus* is 0-3% with an average of 0.72 ± 0.18 %.[8]

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Effect of different concentration of chromium salt on growth of some blue green algae

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Abstract : Effect of chromium salt was tested on 4 local strains of blue green algae *Microcysts aeruginosa, Spirulina princep*, *Oscillatoria amphibian* and *Anabaena macrospora* on their growth. Upto the concentration of 2% growth increased but after that growth gradually decreased.

Keywords : Chromium salt, *Microcysts aeruginosa, Spirulina princep*, *Oscillatoria amphibian* and *Anabaena macrospora*

INTRODUCTION

Hexavalent chromium compounds are genotoxic carcinogens. Chronic inhalation of hexavalent chromium compounds increases the risk of lung cancer. (The lungs are the most vulnerable, followed by the fine capillaries in kidneys and intestines).[1] Soluble compounds, like chromic acid, are much weaker carcinogens. Chromate-dyed textiles or chromate-tanned leather shoes can cause or exacerbate contact dermatitis. Ingestion of chromium VI can also cause irritation or ulcers in the stomach and intestines.[2] Of 2,345 unsafe products in 2015 listed by the EU Commission for Justice, Consumers and Gender Equality some 64% came from China, and 23% were clothing articles, including leather goods (and shoes) contaminated with hexavalent chromium.[3-5]

Hexavalent chromium is transported into cells via the sulphate transport mechanisms, taking advantage of the similarity of sulphate and chromate with respect to their structure and charge. Trivalent chromium, which is the more common variety of chromium compounds, is not transported into cells.[6] Inside the cell, Cr (VI) is reduced first to metastable pentavalent chromium (Cr (V)), then to trivalent chromium (Cr (III)). Vitamin C and other reducing agents combine with chromate to give Cr(III) products inside the cell.[7]

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MATERIAL & METHODS

Four local strains of Blue Green Algae *Microcysts aeruginosa, Spirulina princep, Oscillatoria amphibian* and *Anabaena macrospora* collected from local area were selected to evaluate the effect of Chromium on their growth. Axenic culture of each strain was prepared and cultured in BG11 medium.

To evaluate effect of chromium salt on growth of selected Blue-green algae, different concentrations of potassium di chromate was prepared (0.5% to 5%) and mixed in culture medium. The effect was observed on 3rd, 6th, 9th, 12th and 15th day and also effect of different concentrations after 15 days.

RESULT

Growth response of different selected Blue Green Algae was observed in different concentration of Chromium salt on 3rd, 6th, 9th, 12th and 15th day and also after 15th day. Up to the concentration of 2% Chromium salt the growth linearly increased. In higher concentration, growth decreased. At the concentration of 5% the growth of all test algae decreased linearly from 3rd day to 15th day. The result is shown in Fig no. 1 to 8.

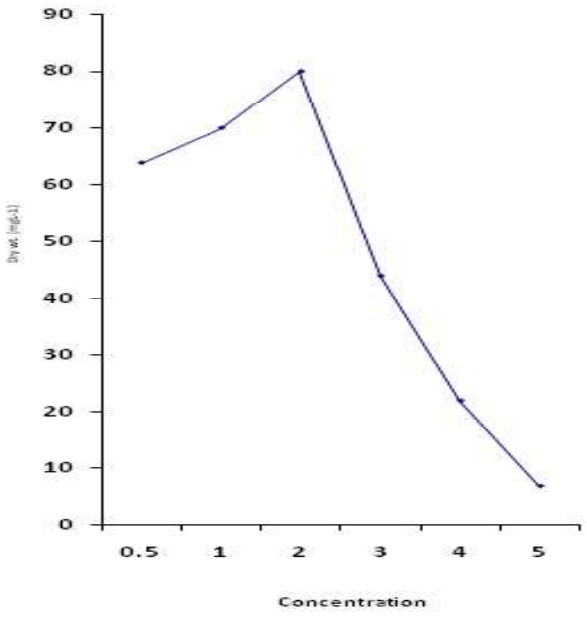


Fig. No. 1 Growth response of *Microcystis aeruginosa* in different concentration of Chromium salt on 15th day

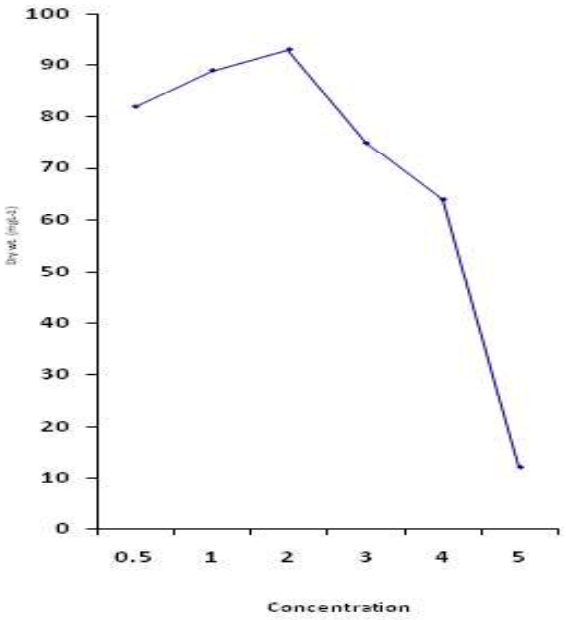


Fig. No. 3 Growth response of *Spirulina princep* in different concentration of Chromium salt on 15th day

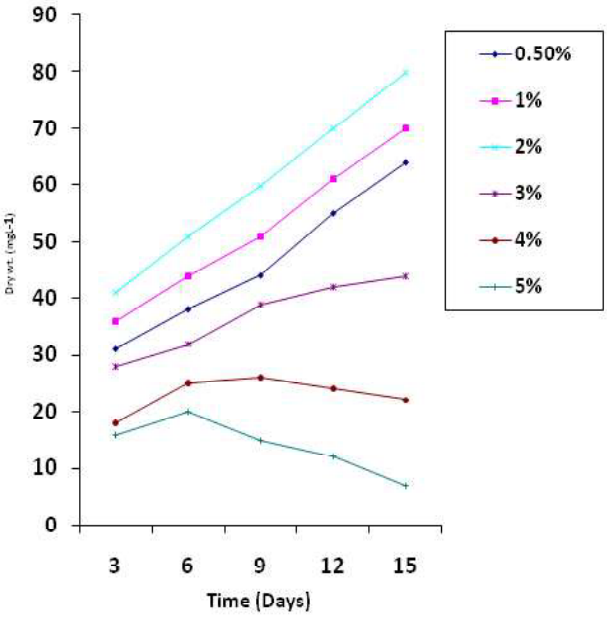


Fig. No. 2 Growth response of *Microcystis aeruginosa* in different concentration of Chromium salt

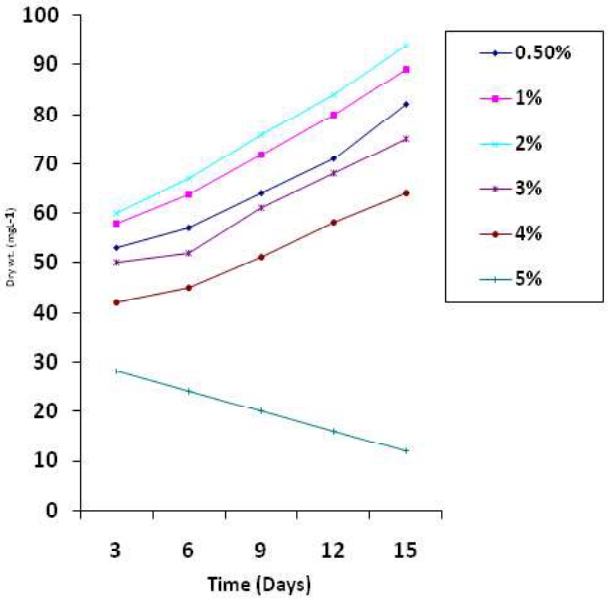


Fig. No. 4 Growth response of *Spirulina princep* in different concentration of Chromium salt

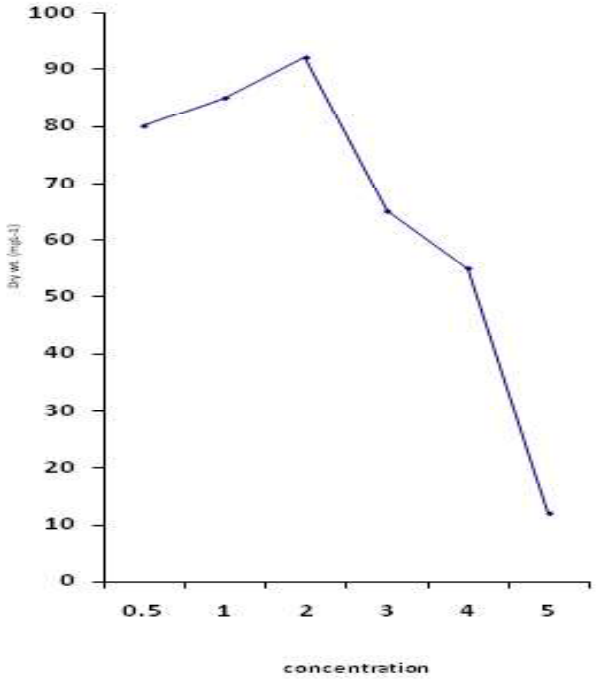


Fig. No. 5 Growth response of *Oscillatoria amphibia* in different concentration of Chromium salt on 15th day

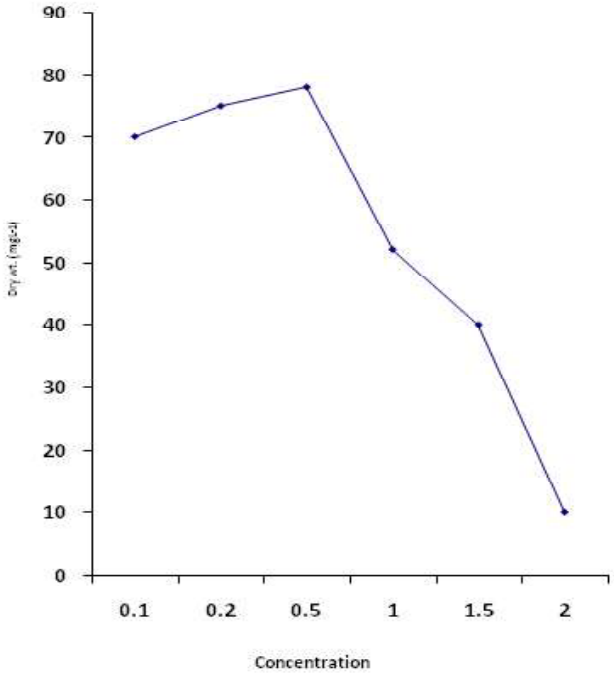


Fig. No. 7 Growth response of *Anabeana macrospora* in different concentration of Chromium salt on 15th day

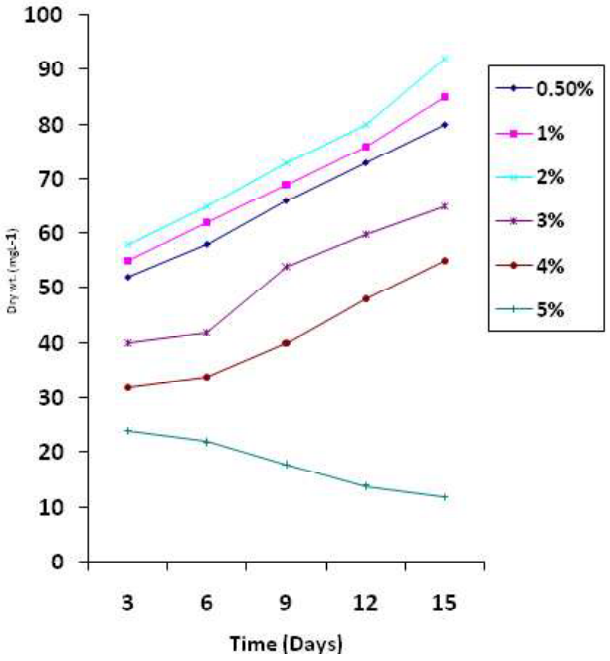


Fig. No. 6 Growth response of *Oscillatoria amphibia* in different concentration of Chromium salt

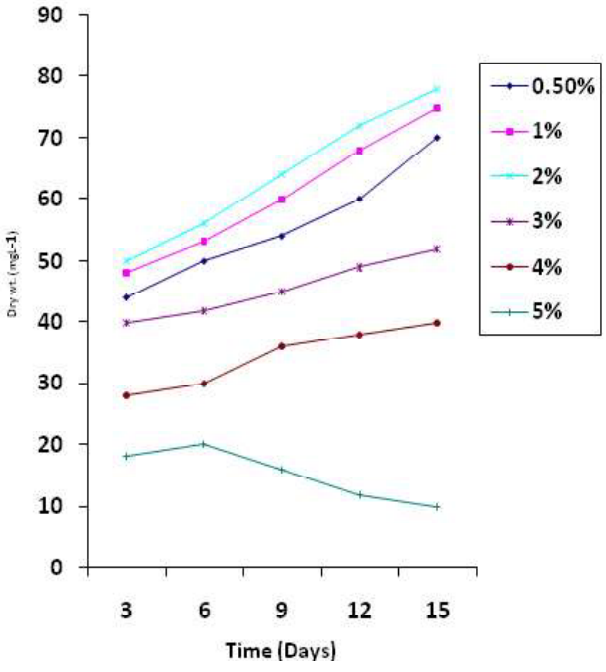


Fig. No. 8 Growth response of *Anabeana macrospora* in different concentration of Chromium salt

CONCLUSION

Most of the blue green algae are capable of absorbing heavy metal salt from the water up to a limited concentration. Chromium salts are carcinogenic and damage lungs. These salts are also absorbed by some blue green algae. In the present study, it was observed that four local strains of blue green algae *Microcysts aeruginosa, Spirulina princep, Oscillatoria amphibian* and *Anabaena macrospora* are capable of absorbing Chromium salt up to 2% but higher concentration of the salt have adverse effect on the growth of these blue green algae.

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Lamina propria macrophages of intestine of Guinea Pig

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Abstract : The macrophages of lamina propria were commonly found both in the small and the large intestine of guinea pigs, mice, rats, and in human duodenum. The frequency of macrophage occurrence was highest in the lamina propria of villous apices of the small intestine and beneath the lining epithelium of the large intestine. In the cytoplasm of lamina propria macrophages localized in those regions. Feulgen-positive (DNA-containing) granules could be observed particularly in the guinea pig intestine. Lamina propria containing the highest accumulation of macrophages i.e. at the apices of villi of the small intestine and beneath the lining epithelium of the large intestine. Labeled nuclei of macrophages were observed scarcely and much less frequently than were labeled cytoplasmic granules. It is suggested that the lamina propria macrophages may play a role in the phagocytosis of some migrating cells of the intestinal mucosa, most probably of the sheath-fibroblasts and/or intraepi-thelial lymphocytes.

Keywords : Lamina propria, Guinea Pig, macrophages

INTRODUCTION

Lamina propria macrophages (LPM) have been described both in the small and large intestine of human beings and rodents.[1,2] It was found that LPM are granular cells occurring most frequently in the lamina propria close to the intestinal lumen. The latter finding together with the tendency to occurrence of LPM in the close neighborhood of plasma cells, led to the conclusion that LPM are the first, or perhaps the second line of defense against intestinal bacteria and their toxins.[3,4]

While studying cell proliferation in the guinea pig it was found that in the cytoplasm of LPM localized beneath the lining epithelium numerous Feulgen-positive DNA containing granules. In the LPM situated in the deeper layers of mucosa, e.g., at the level of crypt bottoms, the Feulgen-positive granules could scarcely by observe. Therefore, it was hypothesized that LPM may play a role in phagocytosis of nuclear material originating from migrating cells of the intestinal mucosa.[5,6]

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MATERIALS AND METHODS

The studies were conducted in 16 guinea pigs, weighing 230 to 300 g. Three days before and throughout the experiment the animals were kept on standard laboratory diet and received water ad libitum. The guinea pigs were injected intraperitoneally at 10:00 to 11:00 AM. With 0.5 pCi of [6-°H] thymidine ["HlTdR, 1, 2, 3, 4, 5, 6, and 7 days after the injection. Specimens, each 4 cm long, of duodenum, middle, jejunum, middle ileum, and ascending colon were excised. Those used for autoradiography. Sections 1 pm thick as well as ultrathin sections, were stained, the former in loo toluidine blue, the latter in lead citrate and uranyl acetate."Additionally the appearance and the reaction for acid phosphatase of macrophages of both the small and large intestine of the mice rats. The Granular cells of lamina propria of either the small or large intestine were identified.

RESULTS AND DISCUSSIONS

LPM was found both in the small and the large intestine of guinea pigs duodenum. They were observed as granular cells occurring most frequently in the apical regions of the villi of the small intestine, and beneath the lining epithelium of the colon. Both in small and large intestine of the guinea pig the frequency of LPM occurrence was higher than in those of mice, rats, and human duodenum.[7,8] The guinea pig LPM was also larger and was heavily loaded with granules. The LPM could be particularly well observed in 1-pm thick sections, although in 5-pm thick paraffin sections of intestine they could also be seen. The LPM had the cytoplasm loaded with lysosomes, residual bodies, and phagosomes. In some phagosomes dark deposits could be seen. LPM were also found to stain intensively for acid phosphatase by the Gomori method

Autoradiographs of the intestine of guinea pigs killed 1/2 hr after injection revealed that the label could be found in the crypts, in nuclei of both epithelial cells and intraepithelial lymphocytes, in the sheath- fibroblasts surrounding the crypt and in some unidentified connective tissue cells throughout the lamina propria. Careful examination of epithelial cells and sheath- fibroblasts in the villi as well as of LPM situated in various parts of lamina propria showed no labeling at all. One and 2 days after injection the labeled epithelial cells and sheath-fibroblasts could be seen at the level of the crypts and up to one-half of the villus length, whereas LPM remained unlabeled. Three and 4 days after injection most labeled epithelial cells were observed in the apical region of the villi of the small intestine and in the lining epithelium of the colon. At that time after injection, in LPM situated in the apices of the villi of the small intestine and beneath the lining epithelium of the colon, whereas in these very cells the nuclei remained unlabeled. Labeled nuclei of LPM were rarely observed, although their occurrence seemed to be more frequent at 7 days than at 3 to 4 days after injection.

It is noteworthy that LPM localized in deeper parts of the mucosa, e.g., at the level of the crypt or at the base of the villi showed no labeling of their cytoplasmic granules at any time after injection. Five days after injection labeling of LPM cytoplasm occurred less frequently and 7 days after it ceased, although at that time, Feulgen-positive granules of LPM cytoplasm as well as some slightly labelled epithelial cells could still be found.

The results of the present investigation indicate that in lamina propria of both the small and the large intestine of guinea pigs the granular cells identified as macrophages. The macrophages are most frequently localized in the upper parts of the villi in the small intestine and beneath the lining epithelium in the large one. These findings are in agreement with the earlier data published by Trier *et al.*[9], Deane[3], and Donnellan[10]. The present study may reveal, therefore, the ultimate fate of at least some of these cells in small and large intestine of the guinea pig.

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Intestinal protozoan diversity of the human population of Patna district

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Abstract : Intestinal parasitic infections continue to be an important cause of morbidity and mortality in the developing countries including India. A study was carried out on intestinal protozoan parasites in Patna district during the year 2015 to 2016 with the purpose of achieving a better understanding of the incidence of the disease. A total of 588 stool samples were collected randomly from the inhabitants of selected areas and examined by simple saline preparation technique. In addition stained preparation by Lugol's iodine solution was also used for the identification of nuclear character of the cysts. Chi-Square test (χ^2 - test) was used to compare relative frequencies between sex specific incidence. 34% of the population (198/588) was found to be harbored with parasites. The incidence of infections for various protozoan parasites was as follows *Entamoeba histolytica* (19.5%) *Entamoeba coli* (7.6%) and *Giardia lamblia* (3.4%). About 3% of the samples contained multiple infections. The study revealed that the prevalence of intestinal protozoan parasites in Patna district is still so high and required special control measures.

Keywords : Intestinal parasites, morbidity, mortality, protozoan, prevalence, Patna

INTRODUCTION

Parasitic infections are prevalent throughout the world including India. It is estimated that more than 3 billon people are infected with intestinal parasites[1,2] and remain significant cause of morbidity and mortality in the developing countries.[3] The prevalence of parasitic infections depends on different variables such as poverty, illiteracy, lack of adequate sanitation, drinking water, health care, prevailing climatic and environmental conditions.[4] Besides these, socioeconomic conditions and cultural habits are also responsible for transmission of parasites.[5] The most common parasitic infections reported worldwide are *Ascaris lumbricoides, Ancylostoma duodenale, Trichuris trichiura, Enterobius vermicularis, Entamoeba histolytica, Entamoeba coli* and *Giardia lamblia*.[6,7] Most of the populations chronically affected with these parasites are responsible to increased risk of iron deficiency and nutritional anaemia, malnutrition, growth deficit[8] and sometimes mentally related disorder.[9] Protozoans are microscopic single celled organism that can be either free living or parasitic in nature. The most common protozoan parasites are *Entamoeba histolytica*, *Entamoeba coli* and *Giardia lamblia.* The disease caused by these parasites are known as amoebiasis and giardiasis and they are associated with diarrohea.[10] *Entamoeba histolytica* recorded worldwide and is more prevalent in the tropics and subtropics in colder climate.[11] *Entamoeba histolytica* can cause amoebiasis with an incidence rate of 5%-81% and estimated to affect approximately 480 million people.[12] Amoebiasis is the third leading cause of death from parasitic disease worldwide. The WHO estimates that approximately 50 million people worldwide suffer from invasive amoebic infections each year, resulting in 40-100 thousand death

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annually.[13,14] *Giardia lamblia* also has worldwide distribution with an incidence rate ranging from 11% to 30%. In USA, it is now considered to be the most common parasite and the leading cause of diarrohea.[15]

In India, the overall prevalence rate ranges from 12.5% to 66% with the prevalence rate for individual parasites varying from region to region.[16] However, studies on the overall parasitic load in the state of Bihar are still so scanty whereas studies related to soil transmitted helminths among school children in Bihar have been published in recent years.[17] Another study reported that incidence of helminth infection is greater than protozoan ones in eastern part of Bihar.[18] Therefore the present study was undertaken to determine the incidence related diversity of different protozoan parasite infections in the socioeconomically backward population of Patna.

MATERIALS AND METHODS

The survey was carried out from March 2015 to Feb 2016. A total of 588 fresh fecal samples were collected in wide mouthed, sterilized container with and without 10% formalin bearing necessary code. Other details regarding age, sex, religion, food habits, living status, source of drinking water and laboratory facilities were also recorded.

A macroscopic examination of all the collected samples was carried out for determination of color, consistency and presence of mucus and blood. For microscopic examination, Saline wet mount and Lugol's iodine wet mount were prepared as per standard protocol[19] for the identification of eggs, cysts and trophozoites of intestinal parasites. The data was statistically analyzed by applying Chi-Square test ($\chi 2$ - test).

RESULTS

The result of the overall survey has been summarized in Fig 1. Out of a total 588 fecal samples examined 198 (34%) of individual were found to be infected with intestinal protozoan parasites. Although females show slightly higher infection rate (52.1%; 103/225) than male (47.9%; 95/363) and the difference was found to be significant ($p < 0.01$).The incidence of intestinal protozoan was highest (41.6%; 75/180) in the age group of 6 to 12 years followed by (38.8%; 35/90) in age group 1 to 5 years, (29.4%; 25/85) in 20 to 30 years, (28.6%; 43/150) in 13 to 19 years and the least incidence was seen in the age group >30 years (24%; 20/83) (Table 1). Among different species of protozoan parasite the highest rate of infection was of *Entamoeba histolytica* 19.5% (115) followed by *Entamoeba coli* 7.6% (45) and *Giardia lamblia* 3.4% (20). About 3%(18) multiple infections were present (Table 2).

DISCUSSIONS

This study shows a much higher prevalence of intestinal protozoan parasites. In this study 34% of healthy subject revealed positive results for intestinal parasites and three species of protozoans are detected. The high rate of infection may broadly attribute to poverty, illiteracy, lack of safe drinking water, poor hygienic condition of the surroundings and malnutrition. The incidence and diversity of intestinal protozoan parasitic infections are known to show spatial (place specific), demographic, community, ethnic and temporal (season) variations.[20-21] In sex related incidence, females show higher infection rate than males. Higher prevalence among females may be attributed to their prolonged contact with prevailing unhygienic surroundings, low literacy rate, lack of knowledge of proper hygiene and carelessness about health. With respect to different species of protozoan *Entamoeba histolytica* (19.5%) and *Entamoeba coli* (7.6%) showed relatively higher incidence than *Giardia lamblia* (3.7%). These results are similar to those reported in other parts of India including Bihar 40.4%[22-24] but they are not in line with those obtained by other authors who found that *Giardia lamblia*[25] was the most prevalent protozoan species. The high rate of *Entamoeba histolytica* and *Entamoeba coli* in our study could be due to the existence of resistant cysts of both parasites in the study area.

On the basis of our results it is evident that the incidence of intestinal protozoan parasitic infection is relatively high in the study area. Poor sanitation and inadequate environment constitute the main determining factors that predisposed this population to parasitic infection. Therefore, this findings suggests the need of the hour that preventive measures should be taken to educate social and medicinal personnel as well as NGO's which can further educate community for hygiene practice. Our study also suggests that steps should be taken rigorously to prevent health damage through intestinal parasitic infection which causes the financial burden of the nation.

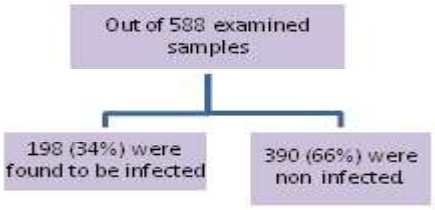
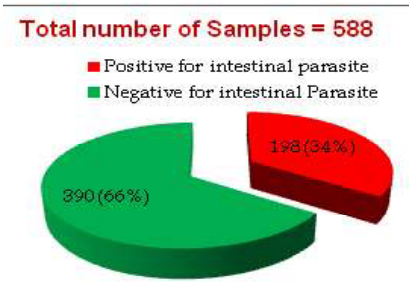


Fig.1.



Pie Chart with reference to Fig.1.

Table 1. The incidence of intestinal parasitic infection out of 588 samples studied.

Variables	Total no. of samples Studied (588)	No. of infected samples (198)	Prevalence % (34)	*P value*
Sex				
Male	363	95	47.9	
Female	225	103	52.1	<0.01
Age(Years)				
1 – 5	90	35	38.8	
6 – 12	180	75	41.6	
13 – 19	150	43	28.6	
20 – 30	85	25	29.4	
>30	83	20	24.0	

Table 2. Species wise prevalence of protozoan infection out of 588 samples studied.

Parasites	No. of infected case (198)	Prevalence (%)
E. histolytica	115	19.5
E. coli	45	7.6
G. lamblia	20	3.7
Multiple infections	18	3.0

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Evaluation of some metals toxicity in *Channa punctatus* from river Kosi, Bihar

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Abstract: This study was aimed to evaluate the bioaccumulation and concentration level of two heavy metals such as Cadmium (cd) and Chromium (cr) in a edible fish *Channa punctatus* obtained from River Kosi, Bihar. These heavy metals are released into the environment through industrial activities, municipal urban runoff and coal burning fertilizers etc. In order to estimate the effect of five concentrations of Cadmium and Chromium (10 m M, 1 mM, 0.1mM, 0.01mM and 0.001mmM) was studied at 24°C. All concentrations of Cadmium decreased the rate of glucose transport. Maximum decrease was recorded with 10 mM of Cadmium. The rate of transport decreased with an increase in the concentration level of Cadmium used. Chromium increased glucose absorption rate at all concentrations evaluated. The highest rate of absorption was recorded at 0.001 mM of Chromium.

Keywords: Heavy metals, bioaccumulation, industrial activities, Kosi River

INTRODUCTION

Cadmium and Chromium are important heavy metals that have been used in the manufacturing of household materials, plastic and electronic products[1], Cadmium (Cd) Chromium (Cr) and Mercury (Hg) are usually added to polymers and pigments, filters, Uv stabilizers and flame retardants.[2] Also Chromium, lead, chromate and cadmium salts are used in various household items like kitchen utensils, house painting produce colour that vary from red to orange and yellow. Due to various industrial and consumer wastes some heavy metals can infiltrate into the water, moreover, contaminated the entire aquatic system.[3] As per the estimate Central Pollution Control Board (CPCB) of India. The plastic consumption in India is 8 million tones per annum and annually about 5.7 million tones of plastic is converted into non-biodegradable wastes[4] which is matter of serious concern due to excessive release of these heavy metals into the water reservoirs in the form of coloured plastic and E-wastage. Further, biomagnificated into aquatic food chain and produce many times more toxic materials into the system.[2] The chromium has carcinogenic properties and is highly toxic in nature which generated many diseases such as sinus, cancer and skin ulcers.[5] However, Itai-itai disease and renal failure is observed due to toxicity of Cadmium.

The toxicity of Cadmium to teleost fish is well documented. Mc Carty *et al.* (1978)[6] studied the toxicity of cadmium to gold fish, *Carassias auratus* in hard and soft waters. Cadmium has been reported to produce hepatic storage alterations of Vitamin B12 in some fresh water fishes,[7] hypocalcemi in rainbow trout[8]. Pickering (1980)[9] reported toxic effects of Chromium, to fat head Minnow (*Pimiphalus promelas*), there is great concern about the potential risk to human health which is directly

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related to the consumption of such fishes which have accumulated these toxic metals through food chain. Thus this study has been undertaken to examine if, cadmium and chromium produce any alteration in intestinal absorption of glucose along with heavy metals.

MATERIALS & METHOD

The fish *Channa punctatus* measuring (20 + 4 cm. long, 70 + 10 gram) were collected from Koshi River and maintained in laboratory in glass aquaria. Fishes were allowed to acclimatize to laboratory conditions for one week, five concentrations (10 mM, 1 mM, 0.1 mM, 0.01 mM and 0.001 mM) of Cadmium and Chromium inkrebs ringer- bicarbonate solution containing 40 mM of glucose were prepared, intestinal sacs were filled with these solution according to the method of Musacchia and Bramante. In control fish glucose solution alone was used, the luminal fluids of both experimental and control fish were collected after one hours at 24ºC. The concentration of glucose in the recovered solutions was determined by adopting the method of Park and Johnson (1963)[10]. The dry weight of the intestine was recorded by keeping the tissues at 100ºC in an oven until the weight became constant. The t test described by Fisher (1950)[11] was used to estimate the significant level of difference between control and experimental values.

RESULTS AND DISCUSSION

Heavy metals like mercury, lead, Cadmium and Chromium are being used and their refuse is disposed in the environment which reaches through many ways in fresh water sources and are harmful to aquatic fauna. Heavy metals get enter in the body of fishes through the food chain and may affect digestion and absorption of food material. In the present study transport of glucose by the intestine of *Channa punctatus* was reduced by inclusion of cadmium in the instilled solution. All five concentrations cadmium produced significant decreases in the rate of absorption of glucose but the decreases were not proportional to the increases in the cadmium concentrations (Table 1).

Further, the decreases in the rate of transport of glucose were more marked in the cadmium concentration range of 1-10mM. Sugar and amino acid transport through the intestine is carrier mediated. According to Newcomer (1973)[12], a Protein carrier in the cell membrame, releasing them into the cytoplasm. Csaky (1965)[13] and Alvardo (1966)[14] have suggested a common carrier mechanism for amino acids and sugars in the intestine. Transport of sugar by carrier protein is sodium dependant. Heavy metals are heaving strong affinity for ligands like phosphate and the cysteinyal and histidyl side chains of proteins, can bind with carrier protein molecules resulting inhibition of sugar and amino transport. Miller *et al.* (1980)[15] have shown that transport of glucose from 2 day and 21 day old is markedly reduced by mercuric chloride. Further, evidence for inhibition of glucose transport by heavy metals comes from the work of Wapnir *et al.* (1979)[16]. Who demonstrated inhibition of intestinal glucose transport in rats that received 20mg of lead acetate per kg body weight. In contrast to cadmium, all five concentrations of chromium increased the rate of glucose absorption by the intestine of *Channa punctatus* (Table 2).

The rate of glucose transport was highest at 0.001mM of chromium increases in concentrations of chromium in the medium gradually decreased the rate of absorption, this indicated that lower concentrations of chromium are more effective in producing elevated glucose absorption rates. Contrary to present findings Stokes and Fromon (1965) reported inhibition of glucose uptake by intestinal epithelial cells of rainbow trout exposed to chromium. The mechanism of increased intestinal glucose transport in *Channa punctatus* exposed to chromium is not clear from the present study further study may give more clear result and conclusion regarding this study.

Table 1- Effect of cadmium (Cd) on the rate of transport of glucose by the intestine of *Channa punctatus*

Concentration of Cadmium(mm)mM	(n)	Rate of Transport (μ mol glucose absorbed/g/dry wt/h) Control	Experimental result
10	18	14.10± 1.33	10.80 ±10.45[b]
1	18	12.19±0.20	10.05±0.26[b]
01	14	11.65 ±0.35	9.70 ±0.17[a]
0.01	14	12.91 ±0.17	11.33±0.40[a]
0.001	18	10.28± 0.10	9.60 ±0.10[b]

Values are mean + S. E., a $p<0.01$, b$p< 0.001$

Table 2- Effect of chromium (Cr) on the rate of transport of glucose by the intestine of *Channa punctatus*

Concentration of Cadmium(mm)mM	(n)	Rate of Transport (μ mol glucose absorbed/g dry wt/h) Control	Experimental result
10	22	10.07 ± 0.24	11.09 ± 0.26[a]
1	22	9.24 ± 0.30	10.36 ± 0.28[a]
01	19	8.07 ± 0.24	9.62 ± 0.22[b]
0.01	18	8.75 ± 0.20	10.85 ± 0.45[a]
0.001	20	7.66 ± 0.14	11.22 ±0.10[b]

Values are mean + S. E., ap<0.01, bp< 0.001

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Inventory & population diversity of fresh water crabs of Kanwar lake, Begusarai, Bihar, India

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Abstract: A first hand preliminary inventory of fresh water crabs found in the Kanwar lake of Begusarai has been prepared on the basis of random sampling through aquatic net (having big mesh size). The population count of each crab species in each & every net sweep has also been done in order to prepare the species level population diversity. The article deals with statistical approach of crab species abundance & richness computed through the fundamental sampling data of the lake. The available food resources useful for the crabs and present in the lake have also been investigated. This exercise provides information about the ecological wellbeing of the lake with respect to the crab's population & their available food resources.

Keywords : Inventory & population diversity, fresh water crabs, species abundance & richness, food resource, ecology of Kanwar lake, Begusarai

INTRODUCTION

Freshwater crabs are considered as one of the diverse group belonging to the phylum Arthropoda & infraorder Brachyura, with very short projecting "tail"(abdomen) which is entirely hidden under the thorax. Crabs come under the order decapoda and class crustacean & shows great variation in size among all arthropods. Globally, 1394 species of freshwater crabs in five different families have been seen.[1] These crabs are referred as primary freshwater crabs, which lack a larval stage and complete life cycle independent of marine environment.[2] Freshwater crabs are one of the most diverse groups among brachyurans with a distribution limited to most of the tropical and subtropical waters. They occupy almost all freshwater bodies, including streams, rivers, ponds, swamps, and rice fields besides semi-terrestrial or terrestrial habitats like tree holes, leaf axils, and leaf litters. They are also considered as a good source of food to different communities of the people as well as play an important role in the food chain of aquatic ecosystem.

The present study was conducted to prepare an inventory of different crab species along with their population diversity present in the Kanwar lake of Begusarai, Bihar. The habit & habitat along with the food choise of the crabs present in the lake have also been observed in order to make a holistic ecological study which is considered as one of the Asia's largest freshwater lake also known as oxbow lake. It is approximately six times the size of the Bharatpur Sanctuary. Kanwar jheel, as it is locally called, is located 22 km north-west of Begusarai town and is extended over an area of 7,400-hectare. It is a residual oxbow lake, formed due to meandering of Gandak River, a tributary of Ganga, in the geological past.[3] Ecologically, it is one of the most important wetlands in

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the state, hosting 106 species of resident birds and providing a nesting ground for 59 species of migratory birds. In addition, the wetland supports a large number of flora and fauna round the year. Economically too, the lake is significant because it yields about two tones of fish everyday and is the single biggest source of irrigation in the area. The wetland is used simultaneously for rice cultivation, fishing, and many other uses. Agriculture is the most important use of the wetland and the basic source of income in the area.

MATERIAL & METHODS

Collection of the sample & its statistical analysis

Samples were collected from the different points of the Kanwar Lake in Begusarai during the morning hour from 10 am to 12 pm. The crabs were sampled statistically from sixty randomly selected sites of above mentioned water body. Sampling was made twice in a week. About 388 specimens were chosen for the statistical study. Crabs were collected by using aquatic net. Each of the collected specimen were kept in a plastic container & brought in the laboratory where these crabs were preserved in 5-10% formalin & identified using secondary data.[4-6] Thus the statistical analysis of crabs population during various seasons has been given in table no 1-4 & in fig 1 along with its data representing its relativity in graph 2-5.

Table 1: Seasonality in the crab population of Kanwar lake during rains 2016.

Si. No.	Name	n	Relative abundance %(n/N* 100)	pi(n/N)	log pi	pi*log pi	H
1	*Sartoriana spinigera*	82	21.2	0.212	-0.673	-0.1426	0.8927
2	*Barytelphusa guerini*	16	4.1	0.041	-1.387	-0.0568	
3	*Barytelphusa lugubris*	22	5.6	0.056	-1.251	-0.0700	
4	*Inglethelphusa fronto*	48	12.4	0.124	-0.906	-0.1123	
5	*Paratelphusa masoniona*	65	16.8	0.168	-0.774	-0.1300	
6	*Paratelphusa spinigera*	43	11.1	0.111	-0.954	-0.1058	
7	*Potamon lugubre*	62	16	0.160	-0.795	-0.1272	
8	*Barytelphusa cunicularis*	28	7.2	0.072	-1.142	-0.0822	
9	*Vanni travancorica*	20	5.1	0.051	-1.292	-0.0658	

Table 2: Seasonality in the crab population of Kanwar lake during winter 2016

Si. No.	Name	n	Relative abundance% (n/N* 100)	pi(n/N)	log pi	pi*log pi	H
1	*Sartoriana spinigera*	65	23.7	0.237	-0.625	-0.1481	0.8301
2	*Barytelphusa guerini*	10	3.6	0.036	-1.443	-0.0519	
3	*Barytelphusa lugubris*	16	5.8	0.058	-1.236	-0.0716	
4	*Inglethelphusa fronto*	30	10.9	0.109	-0.962	-0.1048	
5	*Paratelphusa masoniona*	40	14.5	0.145	-0.838	-0.1215	
6	*Paratelphusa spinigera*	32	11.6	0.116	-0.935	-0.1084	
7	*Potamon lugubre*	55	20	0.200	-0.698	-0.1396	
8	*Barytelphusa cunicularis*	15	5	0.054	-1.267	-0.0684	
9	*Vanni travancorica*	11	4	0.040	-1.397	-0.0158	

Table 3: Seasonality in the crab population of Kanwar lake during autumn 2016

Si. No.	Name	n	Relative abundance %(n/N* 100)	pi(n/N)	log pi	pi*log pi	H
1	*Sartoriana spinigera*	60	26.9	0.269	-0.570	-0.1533	0.8428
2	*Barytelphusa guerini*	8	3.5	0.035	-1.455	-0.0509	
3	*Barytelphusa lugubris*	11	4.9	0.049	-1.309	-0.0641	
4	*Inglethelphusa fronto*	23	10.3	0.103	-0.987	-0.1016	
5	*Paratelphusa masoniona*	32	14.3	0.143	-0.844	-0.1206	
6	*Paratelphusa spinigera*	25	11.2	0.112	-0.950	-0.1064	
7	*Potamon lugubre*	48	21.5	0.215	-0.667	-0.1434	
8	*Barytelphusa cunicularis*	9	4	0.040	-1.397	-0.0558	
9	*Vanni travancorica*	7	3.1	0.031	-1.508	-0.0467	

Table 4: Seasonality in the crab population of Kanwar lake during summer 2016

Si. No.	Name	n	Relative abundance %(n/N* 100)	pi(n/N)	log pi	pi*log pi	H
1	*Sartoriana spinigera*	30	34.8	0.348	-0.458	-0.1593	0.7561
2	*Barytelphusa guerini*	2	2.3	0.023	-1.638	-0.0376	
3	*Barytelphusa lugubris*	4	4.6	0.046	-1.337	-0.0615	
4	*Inglethelphusa fronto*	1	1.1	0.011	-1.958	-0.0215	
5	*Paratelphusa masoniona*	14	16.2	0.162	-0.790	-0.1279	
6	*Paratelphusa spinigera*	12	13.9	0.139	-0.856	-0.1189	
7	*Potamon lugubre*	18	20.9	0.209	-0.679	-0.1419	
8	*Barytelphusa cunicularis*	3	3.4	0.034	-1.468	-0.0499	
9	*Vanni travancorica*	2	2.3	0.023	-1.638	-0.0376	

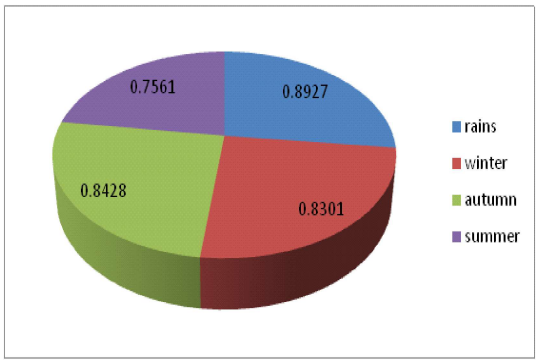
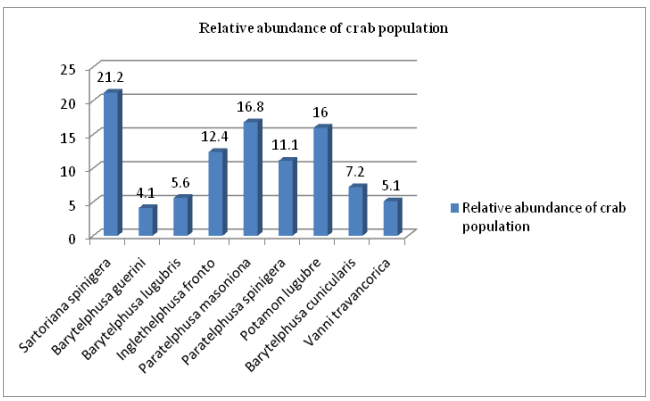
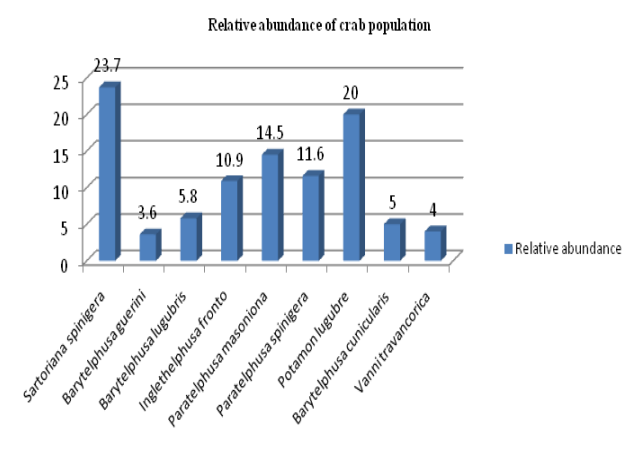


Fig 1: Pie chart showing seasonal variation in cumulative diversity (&) of fresh water crabs of Kanwar lake, Begusarai(2016)

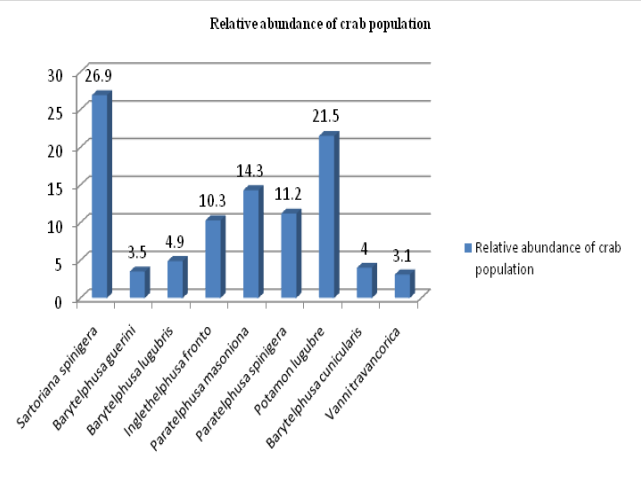
Graph 1: Relative abundance of freshwater crabs in Kanwar lake of Begusarai during rains



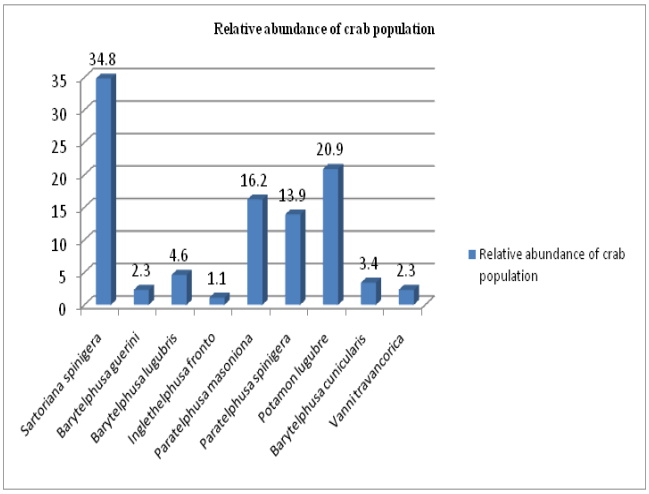
Graph 2: Relative abundance of freshwater crabs in Kanwar lake of Begusarai during winter



Graph 3: Relative abundance of freshwater crabs in Kanwar lake of Begusarai during autumn



Graph 4: Relative abundance of freshwater crabs in Kanwar lake of Begusarai during summer



RESULT & DISCUSSION

On the basis of present investigation nine species of fresh water crab have been recorded in the aquatic habitats of Kanwar Lake, Begusarai. These species includes *Sartoriana spinigera, Barytelphusa guerini, Barytelphusa lugubris, Inglethelphusa fronto, Paratelphusa masoniona, Paratelphusa spinigera, Potamon lugubre, Barytelphusa cunicularis & Vanni travancorica.* These species were at its highest peak during the rainy season, however during summer they were rare in number (Table no. 1-4). While the findings on relative abundance of these species shows that *Sartoriana spinigera* were relatively among the most abundant species found in all the four seasons as mentioned above (Graph no. 1-4) in those habitat which were usually meant for human and aquaculture feed. Rural people used them as fried food and as the medicine of Asthma diseases. Nowadays, also used in poultry feed and as a fertilizer. While *Barytelphusa guerini* were among least species present which may be due to unbalanced aquatic habitat for the survival of these species of crabs. However *Paratelphusa masoniona, Potamon lugubre* were among those species which were moderately distributed among this habitat. Thus the above findings as given above reflects that freshwater habitat of Kanwar Lake in Begusarai is ecologically fit for the aquatic dwelling organisms and also acts as rich food resources for the survival of these species.

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Seasonal variations in the bimodal oxygen uptake in *Clarias batrachus*

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Abstract: In the present work an attempt has been made to study bimodal oxygen uptake in relation to body weight and sex in freshwater siluroid fish, *Clarias batrachus* (Linn.). Among most aquatic air breathers O_2 uptake is divided between air and water fraction given to each mode varies depending on physical conditions such as aquatic O_2 tension and temperature. The percentage of aquatic: aerial O_2 uptake were 55.45 in June while these values were 74.78: 25.22 in December. Similarly aquatic/aerial quotient values were 1.22 in June and 2.96 in December. E.E.U. values were 0.025 K. cal/h in June and 0.005 kcal/h in December.

Keywords: Seasonal variations, bimodal oxygen, uptake, *Clarias batrachus*

INTRODUCTION

Air-breathing fishes, inhabiting tropical freshwater ponds, lakes and rivers employ two modes of respiration namely aquatic respiration using gills and aerial respiration using highly vascularised air-breathing organs. The fishes depend on aquatic and aerial respirations to different degrees in relation to their environmental conditions and the efficiency of their respiratory organs. The relative importance of the two modes of respiration differs in obligate and facultative air breather which is reflected structurally as well.

Though the patterns of bimodal oxygen consumption has been investigated in a number of air-breathing fishes, the number of species so far explored appears too small compared to the total member of species (about 140) of air-breathing fishes available. A wide survey of respiratory patterns in these unexplored species is considered profitable contributing to our better understanding of the pattern of bimodal respiration and the relative importance of aquatic and aerial gas exchange in these fishes.

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The metabolic rate, as expressed in terms of oxygen consumption in fishes, is influenced by a number of factors. The knowledge about the structure and respiratory surface areas of the gills, the blood water barriers, diffusion capacity and the structural modifications in these parameters with respect to the habitat of fish will certainly throw some light on the adaptation of respiratory mechanism of fishes to a variety of environment. Such information's are of prime Importance for better understanding of the respiratory physiology of any particular species of fish. The fresh water air breathing fishes of tropical countries inhabit waters of low O_2 and high CO_2 contents. Moreover, the pools, ponds, swamps, creeks and also the torrential streams inhabited by them often after dry out during summer or their water becomes muddy, highly hypoxic and hyperbaric which is unsuitable for gaseous exchange across the gills. Under such unfavourable environmental conditions, these fishes depend on aerial respiration alone until the commencement of the next rainy season. However, while some fishes resort to air breathing only under adverse conditions, others such as the climbing perch adopt bimodel respiration even in normoxic water. The adaptive features of the air-breathing

fishes inhabiting hypoxic and hypercarbic waters have been described by Carter (1957)[1] and Johansen (1970)[2] anal several others Until later 1950, much attention was paid to the structural modifications and adaptations of air breathing fishes.[3,4]

In the present work an attempt has been made to study bimodal oxygen uptake in relation to body weight and sex in freshwater siluroid fish, *Clarias batrachus* (Linn.).

Among most aquatic air breathers O, uptake is divided between air and water fraction given to each mode varies depending on physical conditions such as aquatic O_2 tension and temperature.

MATERIALS AND METHODS

The live specimens of *Clarias batrachus* (Linn) were procured from local fish dealers at Purnea (Bihar) and were transported and acclimatized in the laboratory conditions for ten days in big glass aquaria. The fishes were fed on pieces of goat liver; small prawns and earthworms, daily at 11.00 a. m. during a minimum acclimatization period of ten days in the laboratory. They were kept fasting for 24 hrs before experiments. No feeding was done during the experiment. Five fishes from each w group were selected for measuring the oxygen consumption Bimodal oxygen consumption from air and still water was measured in a closed glass respirometer containing 8L. water (0, content 6.22-6.5 mg/1; pH 7.22-7.3) and 01.51 of air.

Live specimens of *C. batrachus* (weight range - 55-659; Av 60g) were procured from local fish dealers in the first week of every month for a period of twelve months. They were maintained in large glass aquaria for a week for acclimatization before use for experimental work. The fishes were fed daily on pieces of goat liver, earthworms and fish meal. The fishes were kept fasting for 24 hours before experiment. No feeding was done during experiment.

The datails of the methods employed in the study of biomodal oxygen uptake from air and still water has already been discussed in earlier para graphs. The concentration of dissolved oxygen in the water was estimated by Winkler's volumetric method.[5]

Linear relationship between ambient water temperature and day length vs. bimodal oxygen uptake from air and still water was established on the basis of available data using the general equation Y = a + bx. Here Y stands for oxygen uptake from air and still water X is the independent variables (temperature and day length). b = slope of the regression line and is the intercept i.e. value of oxygen uptake at 1 degree centrigrade or 1 hr. day length O_2 (change per 10 degree centigrade rise in temperature) value for oxygen uptake from air and still water were calculated by the following formula.

$$\log Q_{10} = \frac{(\log M_2 - \log M_1)}{T_2 - T_1}$$

Where M_1 and M_2 are VO_2 at temperature T_1 and T_2 respectivley.

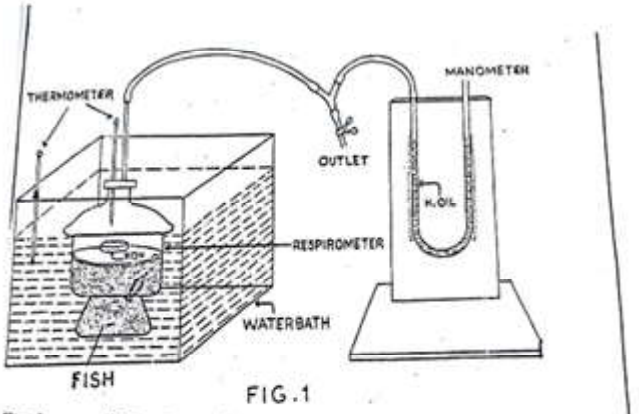


FIG.1

Fig I : Experimental set up for the measurement of bimodal oxygen uptake in *Clarias batrachus* (Linn.)

RESULTS AND DISCUSSION

The data showing av. water temperature (°C), av. day length (h), aquatic, aerial, total O_2 uptake (mlo,/h & cc/kg/h). percentage of aquatic and aerial O, uptake, aqautic/aerial quotient and E E.U. in different months of the year in *C. batrachus* are summarised in Table 1.

The aquatic, aerial and total NO, uptake showed wide range of variations in different months of the year. These values were minimum being respectively 0.838, 0.283 and 1.121 amlo/h (or 18.62, 6.28 & 24.90 cc/kg/h) in December and maximum being respectively 2.888. 2.363 & 5.251 mlo,/h (or 64.18. 52.51 and 116.69 cc/kg/h) in June. The O_2 uptake values were higher during summer & breeding season while these values were lower during winter and nonbreeding periods.

The percentage of aquatic : aerial O_2 uptake were 55.45 in June while these values were 74.78: 25.22 in December. Similarly aquatic/aerial quotient values were 1.22 in June and 2.96 in December. E.E.U. values were 0.025 K. cal/h in June and 0.005 kcal/h in December.

Table 1- Seasonal variation in bimodal oxygen uptake, percentage of aerial and aquatic in *Clarias batrachus*

Sl. N0.	Months	Water Temp. 0C	Day length (n)	Oxygen Consumption						Percentage		Aquatic	EEU Kcal/h
				Aquatic		Aerial		Total		Aquatic	Aerial		
				ml/h	Cc/kg/h	ml/h	Cc/kg/h	ml/h	Cc/kg/h				
1	January	22.0	10.29	0.922	20.49	0.321	7.13	1.243	27.62	74.18	25.82	2.87	0.006
2	February	23.5	10.59	1.710	38.00	0.632	14.04	2.342	52.04	73.02	26.98	2.71	0.011
3	March	25.5	11.29	2.574	55.96	1.210	26.30	3.784	82.26	68.02	31.98	2.13	0.018
4	April	29.5	12.29	2.605	57.89	1.737	38.60	4.342	96.49	60.00	40.00	1.50	0.021
5	May	31.0	13.14	2.861	62.19	2.248	48.89	5.109	111.08	66.00	44.00	1.50	0.024
6	June	32.1	13.29	2.888	64.18	2.363	52.51	5.251	116.69	55.00	45.00	1.22	0.025
7	July	30.0	13.14	2.421	51.56	1.422	30.25	3.843	81.81	63.00	37.00	1.70	0.018
8	August	30.0	12.44	1.790	39.78	0.964	21.42	2.754	61.20	65.00	35.00	1.86	0.013
9	Sept.	29.0	12.14	1.637	36.38	0.889	19.75	2.526	56.13	64.81	35.19	1.84	0.012
10	Oct.	29.5	11.14	1.716	36.51	1.051	23.51	2.767	60.02	62.02	37.98	1.63	0.013
11	Nov.	25.0	10.29	1.603	35.62	0.673	14.95	2.276	70.43	70.43	29.57	2.38	0.011
12	Dec.	19.0	10.14	0.838	18.62	0.283	6.28	1.121	74.78	74.78	25.22	2.96	0.005

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Inventory of blue green algae in island of Kosi division

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Abstract: On field sampling of different wetland pockets(six selected sampling sites) of Saharsa district falling under Kosi division, as many as 15 different species of chlorophyceae as members of blue green algal flora have been recorded that reflect the species richness of the group in this habitat. As regards the species abundance of these species, the individuals of the species have also been numerically counted from the algal samples in the net. The data has been statistically computed to investigate its relative abundance (RA %=Σn/N).

Keywords : Blue green algae, cyanobacteria, Kosi division, relative abundance

INTRODUCTION

Blue green algae which is also known as cyanobacteria are a group of photosynthetic bacteria, some of which are nitrogen-fixing, that live in a wide variety of moist soils and water either freely or in a symbiotic relationship with plants or lichen-forming fungi.[1] They may be present in the form of unicellular to filamentous and also in colonial form. Colonies may also be present in the form of filaments, sheets, or even hollow spheres. Some filamentous species can differentiate into several different cell types. Blue-green algae (BGA) constitute the largest, most diverse & the most widely distributed group of prokaryotes that per- forms oxygenic photosynthesis. They commonly obtain their energy through oxygenic photosynthesis, which produces the oxygen gas in the atmosphere of Earth.[2,3] The name cyanobacteria comes from their color, giving them their other name, "blue-green algae", though some modern botanists restrict the term *algae* to eukaryotes.[4,5] They appear to have originated in freshwater or a terrestrial environment. Unlike heterotrophic prokaryotes, cyanobacteria have internal membranes. These are flattened sacs called thylakoids where photosynthesis is performed.[6,7] Excessive growth of blue green algae in ponds, lakes and reservoirs has become a serious water quality problem and threatens human and animal health. Nutrient rich eutrophic water bodies promote growth of blue-green algae that produce toxic chemicals.[8] The algae hamper the treatment of water for drinking, prevent recreational use and clog pipes. Therefore, monitoring of water bodies for phytoplankton and planning preventive measures accordingly are needed for controlling the algal bloom in water bodies which are in a good and moderate status. If the water body found to be already colonized or invaded by algae, then an assessment followed by urgent actions are needed to bring them into improved status.

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MATERIAL & METHODS

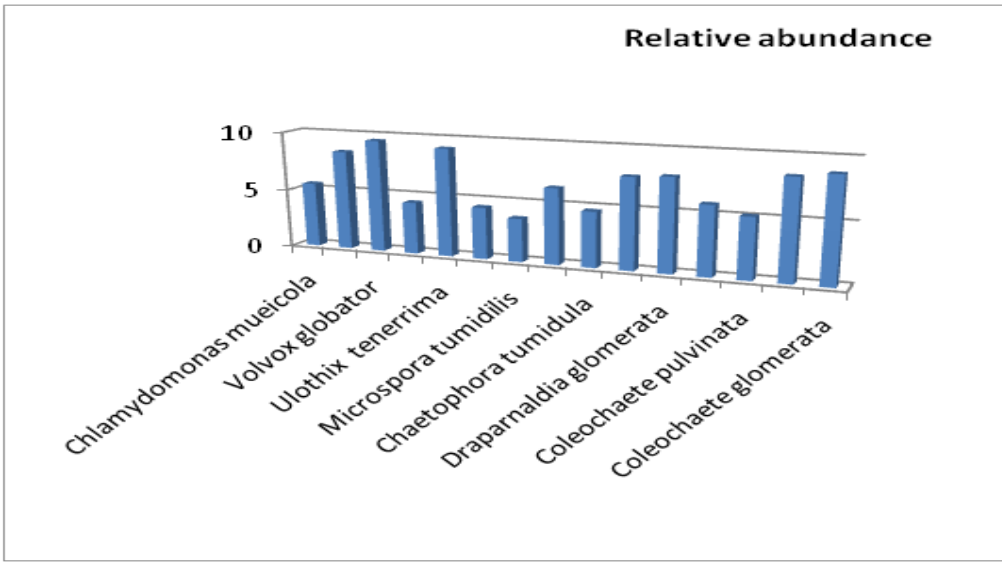
Collection of the blue green algae which is also known as cyanophyceae was carried out from randomly sampled wet land water of five different sites (Mehshi, Mohanpur, Salkhua, Bangaon & Pahapur) falling in Saharsa district of Kosi division for a period of one year from 2013-2014.

The collection was done with the help of planktonic net which were kept in the plastic &brought to the botany department in Madhepura for its further analysis & for computing the data statistically as well as for its identification. The statistical analysis performed for its study on relative has been given in table 1 and also presented in graph 1.

Table 1- Relative abundance of blue green algal species sampled from five selected wetland sites of Saharsa district in Kosi division

Sl. No	Name of the individual	No. of individual	Sampling sites	Relative abundance %(n/N* 100)
1	*Chlamydomonas mueicola*	150	Mehshi	5.498
2	*Pandonina morum*	230	Mohanpur	8.431
3	*Volvox globator*	260	Salkhua	9.530
4	*Ulothrix aequator*	120	Bangaon	4.398
5	*Ulothix tenerrima*	250	Pahapur	9.164
6	*Microspora amoena*	120	Bangaon	4.398
7	*Microspora tumidilis*	100	Mohanpur	3.665
8	*Stigeoclonium lubrium*	175	Mehshi	6.414
9	*Chaetophora tumidula*	128	Salhua	4.692
10	*Chaetophora incrassate*	210	Mehshi	7.697
11	*Draparnaldia glomerata*	215	Pahapur	7.881
12	*Coleochaete divergens*	160	Bangaon	5.865
13	*Coleochaete pulvinata*	140	Mohanpur	5.131
14	*Coleochaete orbicularis*	230	Bangaon	8.431
15	*Coleochaete glomerata*	240	Mehshi	8.797
	Total	2728		

Graph 1- Histogram showing the RA % of blue green algae sampled from five selected wetland sites of Saharsa district in Kosi division



RESULT AND DISCUSSION

Fifteen species of blue green algae belonging to family chlorophyceae were identified which has been given in table no 1that also depicts the relative abundance of major groups of cyanophyceae recorded during the study period in the reservoir of the selected wetland water body. Among these taxonomic groups, the *Volvox globator* sampled from Salkhua site were among the most abundant groups contributing about 9.530% to the total population which is followed by *Ulothix tenerrima* (9.164%), sampled from Pahapur, *Coleochaete orbicularis* (8.431%) from Bangaon, *Coleochaete glomerata* (8.797%) from Mehshi. However, *Microspora tumidilis* (3.665%) sampled from Mohanpur & *Ulothrix aequator* (4.398%) sampled from Bangaon were rarely found in these reservoir. The above finding reflects that the five selected wetland sites namely Mehshi, Mohanpur, Salkhua, Bangaon & Pahapur are having diverse distribution of blue green algae with their specific roles in the aquatic ecosystem chiefly related to nitrogen fixation & eutrophication .

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A brief study on gaseous pollutants (NOx and SOx) in ambient air in Ranchi City

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Received : 18th June, 2017; Revised : 15th July, 2017

Abstract: Air pollution is a serious problem in cities. Rapid industrialization and urbanization causes increased concentration of sulfur oxides and nitrogen oxides. In the rainy season the concentration of these pollutants is at the minimum level.

Keywords : air pollution, oxides of nitrogen and sulfur, concentration, mobile sources

INTRODUCTION

The quality of ambient air is deteriorating because of the rapid industrialization and urbanization. Manmade and natural sources of emissions have created toxic substances, which pollute the air. Countries all over the world have set stringent ambient air quality objectives, guidelines and regulations in order to protect both the general population and those thought to be most at risk e.g., children, the elderly, and those with pre-existing cardio-respiratory disease. Ambient air pollution levels have declined in many developed and developing countries because of these regulatory efforts.[1,2]

Criteria air pollutants including CO, NOx, SO_2 & PM are the quality parameters of ambient air. The main sources of nitrogen oxides NOx (NO + NO_2) is the combustion process. Mobile sources are responsible for an important part of total NOx emissions. Sulphur oxides are produced by burning of coal, vehicle emissions and emissions from oil/gas fields and refineries.[3,4] They are mainly emitted from stationary combustion sources.

The dispersion and transport of pollutants in the atmosphere is of prime concern for the environmentalists. The dispersion is affected by various meteorological parameters, such as wind speed, direction, temperature, humidity, etc.[5]

In the present study an attempt has been made to assess the prevailing concentration of SOx and NOx in the rapidly growing of Ranchi city. The motivation for doing this work emanates from the fact that as per the National Ambient Air Quality Monitoring Data, the SOx and NOx levels in most of the cities in India have been found to be on the rise and in many cities their concentration is above the permissible limit.

STUDY AREA

Ranchi city, the capital of Jharkhand State lies at 23°22'N latitude 85°20'E longitude. Its municipal area is 175.12 square kilometers, and its average elevation is 651 m above sea level. Three specific zone of Ranchi city which were chosen for the present study represent three regions of the city: commercial zone (Kantatoli), residential zone (Ashok Nagar), and industrial zone (Tupudana).

MATERIALS AND METHOD

The sampling of ambient air quality was done simultaneously in each month between December 2009 and November 2011 at the three select sampling sites. The respirable dust sampler (RDS) fitted with gaseous

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sampling attachment (APM 411) was used for monitoring of SOx and NOx values. SOx and NOx concentration was observed continuously for 8 hours in a day. The sampling for SOx and NOx was done using absorbing reagent. On site meteorological data was gathered by using Wind Monitor (Envirotech WM251).

The meteorological data showed that during the year 2009-10, the average wind in the winter months (December- March) was 3.45 m/s, in summer or pre-monsoon (April-June) it was 6.53 m/s , in rainy season (July –September) 5.83 m/s, and in the post-monsoon or autumn period (October- November) it was 3.9 m/s. The corresponding figures for the year 2010-11 are 4.0 m/s, 5.37 m/s, 5.13 m/s, and 4.1 m/s, respectively. Around 80 percent of precipitation was received during the months of June, July, August, and September. During the post monsoon period the precipitation was around 9 percent, while it was 11 percent between the months of January and May. The average temperature in the first experimental season ranged between 15.6 and 32.0^{o}C; and between 15.5 and 29.5^{o}C in the second sampling year.

The average relative humidity ranged between 24.1 and 84.0% during the first sampling year (December 2009 to November 2010) and between 36.5 and 88.3% during the second sampling year (December 2010 to November 2011).

RESULT

The maximum concentration of sulphur dioxide occurs just during the winter months which can be attributed to low ambient air temperature. The SOx values during the pre-monsoon period (April to June) in the commercial zone (CZ), residential zone (RZ), and industrial zone (IZ) was found to be between 27.53 – 28.84 μg/m^3, 10.60- 11.81μg/m^3, and 20.41-27.77 μg/m^3, respectively in the year 2009-2010 (December to November), and 25.67-28.25μg/m^3, 7.27-25.26 μg/m^3, and 25.67-28.25μg/m3, respectively in the year 2010-2011 (December to November).The post monsoon period also shows increase in SOx values which has also been studied in other localities. In the post-monsoon period (October to November), SOx values recorded in the CZ, RZ, and IZ are between 27.84-28.04μg/m^3, 13.92-15.30μg/m^3, and 30.44-30.85 μg/m^3, respectively in the year 2009-2010, and 27.84-28.04μg/m^3, 12.83-14.36 μg/m^3, and 27.57-29.9 μg/m^3, respectively in the year 2010-2011. During the monsoon period the amount of SOx in the ambient air decreases which is in accordance with the theoretical behaviour of SOx. During the winter season (December to March) the value of SOx in the CZ, RZ, and IZ was between 25.70-27.62 μg/m^3, 6.64-16.09μg/m^3, and 20.92-33.85 μg/m^3, respectively in the year 2009-2010, and 25.24-30.69μg/m^3, 11.87-20.18 μg/m^3, and 17.14-33.87μg/m3, respectively in the year 2010-2011. The data shows that SOx pollution in the city is more because of domestic activities than the industrial activities. The SOx shows an increasing trend in concentration from the month of October. The monsoon rains lashes Ranchi city from the early part of June and continues till the month of September. The SOx concentration during the monsoon period (July to September) of 2009-2010 in the CZ, RZ, and IZ was found to be between 26.31 – 28.35μg/m^3, 11.87– 12.33 μg/m^3, and 22.97-27.77μg/m^3, respectively. During the year 2010-2011, the corresponding values were between 14.34-21.48μg/m3, 10.39-21.2μg/m^3, and 12.97-25.04μg/m^3, respectively. The more or less consistent values of SOx values throughout the year indicate that either the dispersal of SOx through the meteorological mechanisms is not very efficient or that the generation of sulphur dioxide in the city balances the dispersion.

The NOx values for the three select sampling sites, namely, CZ, RZ, and IZ during the winter season was between 45.90-55.22μg/m^3, 10.39-25.71 μg/m^3, and 41.68-59.39μg/m^3, respectively in the year 2009-2010. The corresponding values in the year 2010-11was 43.78-64.8μg/m^3, 21.02-37.16μg/m3, and 34.15-59.39μg/m^3.

In summer of sampling period 2009-2010, the values in the CZ, RZ, and IZ ranged between 53.45-56.38μg/m^3, 9.57-16.21 μg/m^3, and 41.00-46.36μg/m^3, respectively.

The corresponding values in the year 2010-11 was 43.78-46.81 μg/m^3, 22.68-34.08μg/m^3, and 29.32-37.16 μg/m3. The values of NOx show a remarkable decline in the monsoon period. In the sampling year 2009-2010, the values obtained in the CZ, RZ, and IZ was between 49.60-55.76 μg/m^3, 18.74-24.12μg/m^3, and 33.82-48.87 μg/m^3, respectively. In 2010-2011, the corresponding value was found to be between 22.85-38.2μg/m^3, 16.4-26.48μg/m^3, and 33.82-43.63μg/m3. The value of NOx in the CZ, RZ, and IZ during post-monsoon period was found to be between 55.58-55.87 μg/m^3, 22.69-25.40 μg/m3, 51.37-57.64 μg/m^3, respectively. In 2010-2011, the corresponding value was found to be between 55.58-55.87 μg/m^3, 21.07-22.68 μg/m^3, and 41.03-46.81μg/m^3. The trend of NOx values is similar to that of NOx.

Table 1: SOx & NOx profile at different locations of Ranchi City, 2009-10

Month	Commercial		Residential		Industrial	
	SOx	NOx	SOx	NOx	SOx	NOx
December	27.62	45.90	11.09	19.20	33.85	59.39
Januray	25.70	55.00	16.09	25.71	20.92	47.37
February	26.25	52.91	9.57	16.40	24.06	41.68
March	27.12	55.22	6.64	10.39	23.01	42.74
April	27.53	54.33	10.60	9.57	20.41	41.00
May	28.84	53.45	11.71	15.68	24.12	46.36
June	27.77	56.38	11.81	16.21	27.17	48.87
July	26.31	49.60	12.03	24.12	26.31	47.61
August	26.31	52.64	11.87	20.16	22.97	33.82
September	28.35	55.76	12.33	18.74	27.78	48.28
October	28.04	55.58	15.30	25.40	30.44	57.64
November	27.84	55.87	13.92	22.69	30.85	51.37

Table 2: SOx & NOx profile at different locations of Ranchi City, 2010 -11

Month	Commercial		Residential		Industrial	
	SOx	NOx	SOx	NOx	SOx	NOx
December	30.69	64.8	15.16	21.02	33.87	59.39
Januray	27.57	51.03	11.87	26.93	29.07	42.69
February	27.78	55.05	14.36	37.16	21.03	34.15
March	25.24	73.78	20.18	23.02	17.14	47.12
April	26.76	46.81	16.09	25.71	25.67	34.34
May	25.67	43.91	25.26	34.08	28.22	37.16
June	28.25	43.78	7.27	22.68	28.25	29.32
July	21.48	22.85	12.75	24.52	16.84	39.13
August	14.34	36.13	10.39	16.4	12.97	33.82
September	16.87	38.2	21.2	26.48	25.04	43.63
October	28.04	55.58	12.83	21.07	27.57	41.02
November	27.84	55.87	14.36	22.68	29.90	49.81

CONCLUSION

The data on NOx/SOx ratio shows that the atmospheric pollutants originated mostly from traffic emissions. The lower NOx/SOx ratio typically characterizes the point sources using high content sulphur coals SOx and NOx are water soluble which dissolved in the rain water and washes away. Hence in the rainy season the concentration of these pollutants is at the minimum level.

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Acute organophosphate (carbamate) toxicity causing anemia in female *Cirrhinus mrigala* (Ham.)

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Abstract : Indirect and continuous entry of carbamate (organophosphate) pesticide in the system of *Cirrhinus mrigala* amounts to acute toxicity that hamper the normal profile of haematological counts of the fish. The present article deals with the impact of acute carbamate toxicity especially on the RBC counts & haemoglobin quantitative composition in the experimentally treated female fishes with two concentrations variants (0.47%& 0.94% v/v) for five different durations of exposure. Statistically significant findings have been found on the gradually depleting standard values of the blood parameters considered. The result has been discussed in the light of clinical symptoms of anemia in the female fishes which are responsible for egg production & carrying out future generation of its race. Additionally these fishes also have edible value, so the anemic individuals may not be good for food from nutritional aspect.

Keywords: Organophosphate toxicity, RBC count, haemoglobin%, anemia, *Cirrhinus mrigala*

INTRODUCTION

The existence of different pollutants in the environment and ecosystem due to their extensive use in agricultural and industrial applications are one of the major problems acting as toxicity for all the living organisms, terrestrial as well as aquatic. Majority of these chemicals are present in the form of heavy metals that are non-biodegradable and thus leading to endogenous binding of the molecules as they enter the biological system causing toxic effect both in the terrestrial & aquatic habitat.[1] Pesticides are widely used in modern agriculture to aid in the production of high quality food. However, some pesticides have the potential to cause serious health and environmental damage. Carbamates are a class of insecticides structurally and mechanistically similar to organophosphate (OP) insecticides. Carbamates are N-methyl Carbamates derived from a carbamic acid and cause carbamylation of acetylcholinesterase at neuronal synapses and neuromuscular junctions. Though the pesticides are applied to enhance agricultural production while the indiscriminate and contaminate the biota. Subsequent to the translocation of pesticides to aquatic environment the non-targets such as fish are exposed to low concentration over a long period and affect the efficiency of various life parameters and seem to produce many physiological and Biochemical changes in fish. Fish have been valued for many years as excellent indicators of water quality. High usage of pesticide in the field, it affects both biotic and abiotic environment. The oxygen consumption (biotic) is a very sensitive physiological process and the change in respiratory activity has been used as an indicator of stress in animals exposed to

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toxicants. A number of investigations on the effect of pesticides on the oxygen consumption of fish have been reported. [2-4] Stresses and pollutants generally cause relatively rapid changes in Blood characteristics of fish.[5,6] A reduction in haemoglobin content and erythrocyte population resulting in anaemia have also been suggested as reason for drop in oxygen uptake in fish *Channa punctatus* exposed to lethal concentration of Deltamethrin.[7] Though, the biochemical, Physiological and enzymatic parameters are the common biomarkers of exposed fish to toxicity of pollutant. It can even cause countless abnormalities and reduces the life span of organisms.[8,9] Discontinuous use of these pesticides may also lead to serious environment as well as aquatic & land dwelling animals. Haematological studies on fishes have assumed greater significance due to the increasing emphasis on pisciculture and greater awareness of the pollution of natural freshwater resources in the tropics.

MATERIAL & METHODS

Sample collection

About 30 species of freshwater fish *Cirrhinus mrigala* were bought from the local market having the length 12-14cm and 20-25 g respectively. These species were then kept in laboratory for few days and fed with commercial fish food twice a daily. After acclimation, 30 aquaria each containing 30L water were stocked with ten fish per aquarium. All the fishes were then fed with different carbamate concentration depending on the given exposure of time-1day, 7 days, 14 days and 28 days. After completion of the respective duration these fishes were then examined for their statistical analysis for the presence of RBC and haemoglobin count.

RESULT AND DISCUSSION

Carbamate insecticide entering indirectly into the system of *C.mrigala* in the paddy field may cause continuous depletion of haemoglobin content after breaking the RBC. Day specific treatment of different concentrations of the insecticide with the fish in the laboratory has produced significant decrease in the haemoglobin content causing anemia. The plausible reason of decrease of haemoglobin may be the detachment of peptide chain from the central metal iron by the molecular leaching of carbamate which enters into RBCs by breaking their cell boundary. Therefore, it is recommended that the fish culture personnel concerned with quality fish production must take into consideration the standard & treated profiles of haemoglobin in fishes in general & that in *C.mrigala* in particular in order to protect the fishes from getting pushed into the anemic zone below (Table no. 1)

Table 1- Profile of RBC count and haemoglobin % in male *C. mrigala*

Treatment no.	Exposure time	Variations in RBC counts%			Variations in Haemoglobin %		
		Control	0.47%(v/v) crbm. treated	0.94%(v/v) crbm. treated	Control	0.47%(v/v) crbm. treated	0.94%(v/v) crbm. treated
1	1	5.19± 0.54	5.08±0.33	5.02±21	12.57±0.23	12.23±0.26	12.09±0.57
2	7	5.21±0.29	4.67±0.17	3.88±0.06	12.63±0.25	10.73±0.25	10.27±0.64
3	14	5.20±0.44	3.85±0.11	3.10±0.11	12.50±0.30	10.40±0.75	9.47±0.60
4	21	5.27±0.38	2.40±0.28	2.20±0.08	12.67±0.45	9.13±0.84	8.90±1.37
5	28	5.17±0.28	2.73±0.06	2.14±0.08	12.40±0.56	9.50±0.61	9.23±0.55

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Life cycle of bean beetles *Callosobruchus maculatus* (F.)

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Abstract: Life cycle of bean beetle, *Callosobruchus maculatus* (F.) is theoretically described to observe holometabolous pattern of metamorphosis. However, host specific life cycle of this stored grain pest propagating on stored dry gram needs to be carefully observed in order to record the duration of life cycle in different seasons. The number of larval instars also undergoes variation which has been reported in this article. The pattern of life cycle of the pest also determines the number of generations observed in one year which is also called voltine cycle. The voltinism is responsible for regulating the degree of damage caused by the pest to the host. In this study, the pest has been found to be tetravoltine with pentainstar holometabolous metamorsphosis.

Keywords: Holometabolous metamorphosis, tetravoltine, pentainstar, life cycle, bean beetle (*C.maculatus*)

INTRODUCTION

Bean beetles, *Callosobruchus maculatus* (F.) are insect pests of Africa and Asia that probably range throughout the tropical and subtropical world. This species is also known as the 'Southern Cowpea Weevil'. The larvae of this species feed and develop exclusively on the seed of legumes (Fabaceae) hence the name 'been beetle'. The adults do not require food or water and spend their limited life span (one-two weeks) mating and laying eggs on beans.[1,2]

The systematic placement of bean beetles is as follows: *Callosobruchus* is one of the genera in the family Bruchidae (seed weevils) that is in the super family Chrysomeloidea. This group is part of the order of beetles, Coleoptera.[3]

Bean beetles (*C. maculatus*) are small in size, about 1/8 inch long. They are reddish – brown, slightly elongate beetles compared to the typical rounded appearance of other members of this family (bruchids). Although weevil like, they are not true weevils (Curculionidae) and do not have heads prolonged into a long "snout". The head is small with a blunt rostrum, long conspicuous serrate antenna. The prothorax is well developed. Wing covers (elytra) are marked with black and grey and these are two black spots near the middle. The elytra are short and do not cover the abdomen completely. The last abdominal segment also has black spots visible.[4,5]

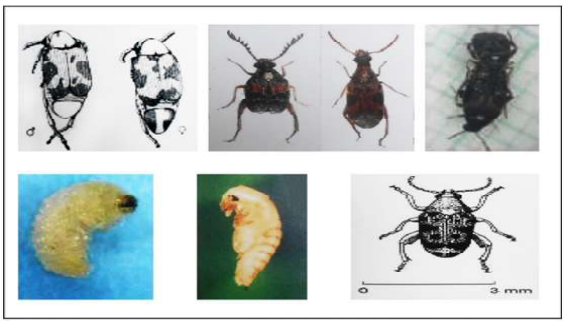


Figure showing the life cycle of *Callosobruchus maculatus* (F.)

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Bean beetles exhibit two adult forms (morphs), a sedentary (flightless) form and a dispersal (flying) form. The dispersal morph is induced by high larval density in stored grams or laboratory cultures and is caused by density dependent micro-habitat temperature increases. These two morphs have very different life history characteristics such as longer adult lifespan in the dispersal morph and significantly reduced fecundity compared to the sedentary morph. In the sedentary form, the sexes are highly dimorphic and readily distinguished but sex differences are very subtle in the dispersal form.

Male and female gram beetles (of the sedentary morph) are easily distinguished from one another by general appearance. The most distinguishing characteristic is the colouration on the plate covering the end of the abdomen. In the female, the plate is enlarged and is darkly coloured on both sides. In the male, the plate is smaller and lacks stripes. In some strains, females are larger in size than males. Also, females are black in colouration and males are brown.[6,7]

They attack practically all whole pulses, gram, beans and grain. Once inseminated, adult females lay single fertilized egg on the external surface of gram. Individual eggs are about 0.75 mm long, oval or spindle shaped and grayish white in colour. The eggs are firmly glued to the gram surface. The hatching period is about 5 days. Larva hatches out from egg. Penetrated into the pod or grain. The egg larva is fleshy, curved, white creamy in colour with black mouth parts. The larva feeds on the gram endosperm and embryo, undergoes a series of molts and burrows to a position just underneath the seed coat prior to pupation. Larval stage lasts about 30 days. Although the seed coat of the gram is still intact, a round 1-2 mm window is apparent at the location where the beetle is pupating. Pupal period lasts about 4 days in summer and about 2 weeks in winter. The adult that results from pupation chews through the seed coat and emerges from the gram. The elapsed time from newly laid eggs to the emergence of adult beetles varies from 3-4 weeks in a 30°C incubator and average 30% RH to 5-6 weeks in 25°C incubator and average 50% RH. The adults are fully mature 24 to 36 hours after emergence. Neither male nor female adults require food or water during their short adult lifetime (10-14 days).

MATERIAL & METHODS

Freshly emerged adult beetles from the stored gram were randomly collected from the store houses as the test insects. Both infected & fresh grams were also procured from the local market as host on which the pest was allowed to propagate in the laboratory. For this purpose at least 5 pairs of both male & female beetles were taken in different plastic jars containing at least 50 gms of host gram seeds. Constant observation on the egg laying, larval development & emergence of the new adults were made to carefully record the life cycle.

RESULT & DISCUSSION

The beetle has been found to complete its life cycle in 28 days at 30°C & 30% RH at laboratory temperature & humidity repectively whereas the life cycle duration was prolonged upto 40-45 days at lower temperature of 25°C & higher humisity of 50%.

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Environmental strategy of survival & perpetuation in garden lizard *Calotes versicolor* Madhepura, Bihar, India

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Abstract: Garden lizard is currently falling in the red list of endangered reptiles. However the species, *Calotes versicolor* is trying to protect its racial survival by adopting several environmental strategies which is purely behavioral & scientific observation on the ethology of the garden lizard with respect to its protection needs to be genetically verified & established. Nevertheless, there is evidence that the *Calotes versicolor,* the lizard, is now established in Madhepura (Bihar), and it is reported to be dispersing away from its original point of introduction. Collected data on its habitats, feeding habits, predators and parasites show that this species is adaptable, prolific and omnivorous, and it is considered to be an invasive alien species that competes with or feeds on native biota in some parts of the Bihar state, such as Purnea and Saharsa. These population need to be studied to prevent their potential ecological threat.

Keywords: Environmental strategies, survival, perpetuation, environmental threat, *Calotes versicolor,* Madhepura

INTRODUCTION

Environmental strategies for survival & perpetuation are one of the most remarkable etho-ecological phenomenons observed by flora & fauna in any eco-system. Large number of species crowdiong in the particular habitats at community level in one hand & some getting spread over wide range of geographical stretch undergo continuous species change & turn over in the population. The lizard, *Calotes versicolor* is a strong candidate for the status of most widespread non-gekkonid lizard in the world. Gunther (1864)[1] noted: "This is one of the most common lizards, extending from Afghanistan over the whole continent of India to China; it is very common in Sri Lanka" Its present distribution stretches from right across southern and south-east Asia to Indo-China to the east, the Maldives, Reunion, Mauritius, including Rodrigues, Seychelles and Florida in the United States.[2-8]

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With growing evidence that *C. versicolor* is an invasive species with potentially harmful ecological impacts in the areas where it is introduced largely because of its omnivorous feeding habits, it is necessary to gather together the available, albeit limited, information on the status of its presence in Madhepura, Bihar, India.[9,10] At the same time the opportunity is taken to review the existing literature on the ecology of the species with a view to summarizing the available data as a basis for further research on its status as a potential threat. It has various common English names used for this species garden lizard, bloodsucker, crested tree lizard etc.

Due to urbanization and increase in human population modification of the natural ecosystem is taking place at alarming rate. As a result of which the flora & fauna are facing great challenges for adaptation bringing alteration in natural habitats, predator communities, predation intensities, composition, distribution as well as availability of food resources.[11,12]

As we know that reptiles are a group of tetrapod, cold-blooded animals that are found almost in entire world, Squamata (lizards) are one of these. These species play a very important role in ecosystem regarding food chain, bio-monitoring pest and also acts as ecological indicator due to their sensitiveness. Even a little change in any of the climatic condition such as temperature, humidity etc may lead to extinction of these species. Therefore serious concern should be taken for the adequate survival of these species.

MATERIALS & METHODS

Collection of the sample

Collection of *C.versicolor* was done from different urban sites of Madhepura district which is situated in the Plains of River Koshi and located in the Northeastern part of Bihar at latitude between 25°.34 to 26°.07' and longitude between 86°.19' to 87°.07' during 2017-2018.[10] Female lizards in early gestation were identified from the different areas and were taken to the Dept of Zoology, B.N.Mandal University Madhepura for its further study. When the eggs reached the stage of 27-28 embryos, they were taken out from the lizard because oviposition occurs usually at 27-28 embryonic stage. The average duration of incubation at 24°C, 27°C, 30°C, and 33°C was 82.1 days, 60.5 days, 51.4 days, and 50.3 days, respectively. Incubation temperature affected hatching success, energy expenditure for embryonic development, and several hatchling traits examined, but it did not affect the sex ratio of hatchlings. Each of the lizards seemed to yield 20±3 eggs. Over all 130 eggs were collected which were then incubated in the laboratory till its hatchling. Altogether 12 individuals were obtained, 7 females and 5 males. As the hatchlings emerged out both the sexes were identified & were separated from each other which were then kept in individual terraria. Growth rate, size of the snout, length of the head was measured periodically (1, 6 & 12 month after the emergence of hatchlings) to study their difference.

Graph 1:Relationship between clutch & egg mss

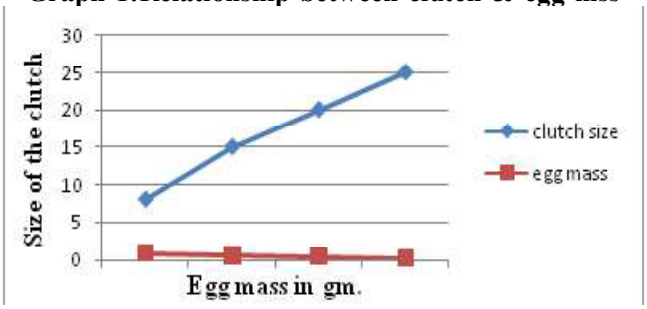


Table no 1:

Months	SVL size		Length of head	
	Male	Female	Male	Female
Hatching	23.52±0.6	24.80±0.7	9.26±0.2	9.50±0.3
First	36.34±0.5	32.02±1.3	11.30±0.3	10.90±0.3
Sixth	71.12±0.8	58.45±1.3	26.5±0.6	20.00±0.6
Twelfth	118.33±2.1	105.20±2.1	39.2±0	30.01±0.8

RESULT AND DISCUSSION

The study related to the survival and perpetuation of garden lizard shows that the mass of egg when compared to clutch size, relationship between both showed the negative correlation i.e. increase in the clutch size lead to decrease in the mass of the egg which has also been reported in the graph 2. While when the SVL size and length of the head was measured in both the male and the female after the hatching, the hatchling of both the sex showed variable differences. The length of the SVL of male & female showed very minute amount of differences, further while calculating the head the female had larger than the males. In first month hatchlings both the sex had rather similar amount of variation in SVL size and head length. In sixth month and twelve month hatchlings although much variation is not found in length of the head however great differences has been found out in the SVL length in both the sexes of the lizards.

Thus the present study reports that temperature, other environmental factors are the major cause for change in the developmental strategy of the garden lizards not only the lizards but also the other reptiles. Since the reptiles are ectothermic, cool temperature usually lowers down the development of the reptiles while warm temperature speeds up the development.

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Histopathological changes in the suprabranchial chamber of fresh water fish *Channa gachua* exposed to a fungicide sedaxane

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Abstract: The present investigation is conducted to assess histopathological alternations caused in suprabranchial chamber of an air breathing teleost *Channa gachua* (Bloch) after exposure of sedaxane (sublethal concentration of sedaxane for 96 hours is 0.045 ppm). Light microscopic studies showed many alternations in the suprabranchial chamber of fish. These alternations are observed as pathological biomarkers indicating the load of pollution of toxic stress. The effect of sedaxane can be observed in the form of hyperplasia of mucous cells found in the epithelial lining of SBC. The higher degree of wear and tear caused sloughing off of epithelial cells and thus the non-tissue space is formed in the respiratory epithelium. Breakdown of vascular papalliae caused disintegration of vascular system. The blood capillaries of SBC also bulged out extensively. These observations are thus indicative of the toxic affect caused by sedaxane at histopathological level in suprabranchial chamber of *Channa gachua* (Bloch).

Keywords: Histopathology, suprabranchial chamber, *Channa gachua*, sedaxane, toxicity

INTRODUCTION

Pesticides encompass a wide range of chemicals used to kill insects, fungi or weeds for massive production in agriculture. Pesticides are extensively used in world for pest control. Pesticides are used as important component in agricultural development and protection of public health in India, as the climate of India is very suitable for plant breeding.[1] Use of Pesticide in agricultural field to control pests is very toxic to non-target organisms like fish and affects fish health through impairment of metabolism, sometimes leading to mortality.[2] Among all pesticides, sedaxane is widely used to control pests. It inhibits respiration by binding to the succinate dehydrogenise complex of fungi. These insecticides enter into the aquatic ecosystem and then food chain and are responsible for adverse effect in aquatic ecosystem and finally human.

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The accessory respiratory organ in some fishes has developed to fulfill the high oxygen demand of the body even in the adverse conditions. The main respiratory organ in fishes is no doubt the gill but some fishes have evolved air- breathing organs as an adaptive feature to supplement optimum oxygen from the atmospheric air. The air breathing organs helps the fish in surviving even in very low oxygen containing water bodies as its enables the fish to absorb oxygen from the atmospheric air and helps in sustaining their life for certain period of time out of water. It is an adaptive feature associated with life in hypoxic and hypercarbanic ambient water have be well documented by Carter (1957)[3]. Thus the air breathing fishes differ from others in having accessory respiratory organs, blood physiology feeding habits and in their breaking water behavior.

The accessory respiratory organs in fishes are usually newly evolved which are accommodated within the limits

of buccophyrax or even beyond it in some cases. The major problem related with neomorpic air-breathing organ is to develop a respiratory epithelium suitable for gas exchange with air. Once it is achieved the immediate task before a fish is to increase the respiratory surface area which can be accommodated in a minimum of space .Some common Indian air- breathing fresh water teleost that have neomorphic air-breathing organs are *Anabas testudineus*, *Colisa fasciatus* and *Osphronemus nobilis* (Anabantidae); *Heteropneustes fossilis* (silurooidae); *Clarias batrachus* (Claridae); *Amblyceps mangois* and *Amblyceps murray stuarti* (Amblycepatidae); *Channa marulius* and *Channa gachua* (Channidae) and *Monopterus cuchia* and *Monopterus albus* (symbranchidae). Besides, the fresh water forms a number of estuarine genera and species belonging to the family gobiidae are air breathers like *Psudapocryptes lanceolatus*, *Perioptha*, *Imodanseholosseri* and *Boleophthamus boddarti*.

The histopathological study on different fish part are quite common but the work on SBC is scanty and mainly restricted to a limited number of air-breathing fishes. The origin structure and function of accessory respiratory organs differs in various species of air breathing fishes hence the degree of impact on them also varies the main target of various toxicants in the SBC is their epithelial lining and thus it undergoes histopathological alterations after getting their exposure. The severity of damage depends upon the variety and concentration of hazardous material along with some ambient physico chemical factors. Hence, the efforts have been made to know the degree of hazardous effect of Sedaxane on the air breathing organs of *Channa gachua*.

MATERIALS AND METHODS

Adult healthy specimen of *Channa gachua* (Bloch) of length 7-10 cm and weight 10-12.5 gm were taken without any sex differentiation from paddy field of my home town Kanti, Muzaffarpur, Bihar. After that these fishes were transported to the laboratory in aerated containers and then these fishes were treated with 0.5% $KMnO_4$ solution for 10 minutes to remove dermal infection. After that these fishes were acclimatized in the tank for 15 days and provided with chopped earthworm or fish tone. To maintain dissolve O_2 concentration water was changed on alternate day. During experiment no food was given to them. Important parameters to maintain the fish were strictly followed by as par condition recommended by APHA (2005)[4]. After that fishes were exposed to sedaxane. After exposure kidney was isolated from the control and experimental fish and then they were cleaned and fixed by using saline solution for 24 hours. This vital organ was kept in formaldehyde and then dehydrated, embedded in paraffin wax and then sliced by using microtome of thickness 5 micrometre. They were stained by using double staining technique (Haematoxylin and Eosin) and viewed in compound microscope.

RESULTS

SBC of control fish:-

It was found that the lining of suprabranchial chamber (SBC) is formed by the active participation of ectoderm, mesoderm and endoderm during the embryonic development. The SBC develop dorsal to the gill arches in the pharyngeal roof and lateral to the prootic bone of the auditory capsule. On the dorsal side it is protected by the skull bone while on the ventral side it is lined by bronchial epithelial and the opposite wall with supra bronchial line. Each suprabranchial chamber is divided into an interior small and a posterior large compartment by a self like transverse outgrowth. The pharynx open into the interior compartment and there is no shutter present to guard this opening. The interior compartment is freely connected with posterior compartment. The pharyngeal epithelium covers interoventrally with taste buds and supra bronchial lining posteriorly the suprabranchial chamber from below the original pharyngeal epithelium thus bucco pharynx become able to draw oxygen from engulf air. The bucco pharyngeal lining or epithelial lining consist numerous taste buds. The excellent opening serves as an exit for expired air to pass from SBC to opercula chamber which is located at the dorsal part of first gill silt. An inhalant opening is found at the anterior ventral position which allow the posterior compartment of SBC to communicate with pharyngeal chamber and it is guarded by a shutter. The supra bronchial chamber is formed by many layer of ectodermal cells and basophilic cells derive from gill mass and becomes folded posteriorly and remain buldges out into opercula chamber. The constrictor (suprabranchialis) muscle fiber is present in the wall of SBC consisting vascular (islets) consisting of a number of rosettes and non-vascular (lanes) areas. Each rosett is form by a group of blood capillaries projecting on the surface epithelium. The two blood capillaries in vascular papillae gets separated like structures and in these vascular papillae mainly the

gaseous exchange take place. The lanes are covered by stratified epithelium consist polygonal mucous secreting cells (Figure 1, 2).

Figure 1.T.S of SBC of *Channa gachua* control showing blood vessels and respiratory epithelium.H/E X 150

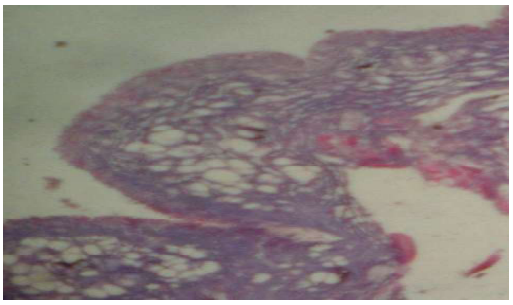
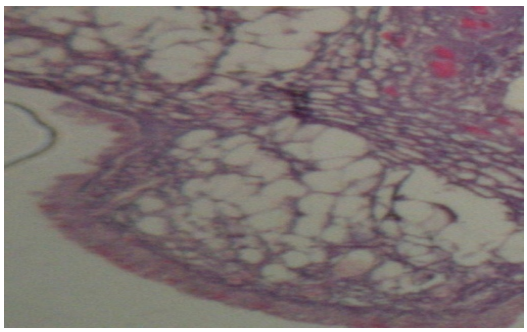


Figure 2. Same is enlarged to show blood vessels. H/E X 400



Histopathlogical changes in SBC due to Effect of Sedaxane:-

Though the structure and physiology of SBCs varies in different species of air breathing fishes and thus the toxic impact of stressors are also different. The SBC was lined by a stratified epithelium situated above a layer of connective tissues which were highly vascularised. Numerous fine sub-epithelial blood capillaries reaches to the surface of SBC. Minute blood channels (BCs) forming the vascular papillae (VP). The projected aerial surface of papillae was covered with a thin respiratory epithelium. Numerous mucous cells (MCs) were present towards the outer surface of the epithelium probably to protect them by forming a slimy layer over it. After getting the exposure of Sedaxane for about 2-6 hours the blood capillaries of the vascular papallie get swollen and bulged out due to accumulation of RBCs. The blood capillaries get congested and forms finger like structure at some places. Enlargement of mucous cells had been seen when the exposure time gets enlongated. At some places blood spots were also seen on the surface of SBC probably due to the rupture of highly stressed blood capillaries. At some places detachment or lifting of the respiratory epithelia from the underline vascular region. A large number of wear and tear of the epithelial lining and severe necrosis of epithelial cells can also be observed (Figure 3). However the SBC of the fish do not lie in the direct contact of toxicant mixed in their surrounding water but it harms the SBC by reaching there through the blood. The effect of sedaxane can be observed in the form of hyperplasia of mucous cells found in the epithelial lining of SBC. The higher degree of wear and tear caused sloughing off of epithelial cells and thus the non-tissue space is formed in the respiratory epithelium. Breakdown of vascular papalliae caused disintegration of vascular system. The blood capillaries of SBC also bulged out extensively, probably to compensate the damage caused by the toxicants to the gills by taking part in the process of gaseous exchange. It has reduced the blood atmospheric air barrier distance (Figure 4).

Figure 3. SBC Treated with sedaxane showing breakdown of blood vessels in *Channa gachua* after 96 hours. H/E X 600

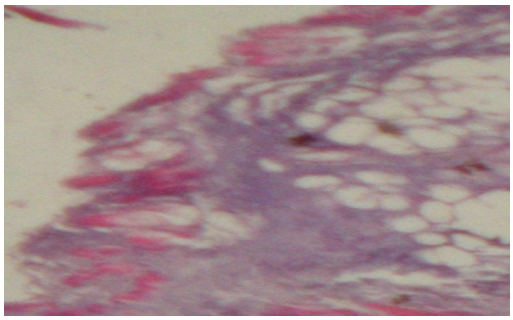
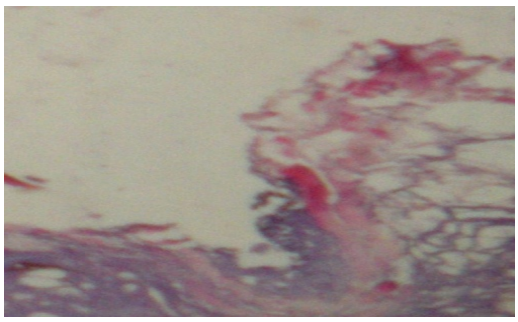


Figure 4. Total breakdown of epithelial layer of sbc and blood capillary in *Channa gachua* after the sedaxane exposure. H/EX800



DISCUSSION

Channa gachua regularly visits the water surface to engulf air for aerial respiration but the exposure to toxic water hampered the bronchial respiration and thus the fish

were compelled to take oxygen from the air. At the initial stage the surface visiting behaviour increases but as the toxic material reaches to blood or come in direct contact to the body part induced the mucous secretion. The inner epithelial lining of the air sac often gets stick together causing a decrease in respiratory area and their volume. The similar effect in *H.fossilis* was also observed by Johansen (1970)[5]. The accumulation of mucous and blood materials in the air sac also lead to the disturbance in the gaseous exchange. The mucous cells show the quantitative as well as qualitative alterations. After getting the exposure as the saline secreted contains a mixture of neutral acidic and sulphated glycol proteins, the finding matches with the work of others.[6-10] During the earlier stage of exposure fish showed restlessness and respiratory distress the opercula activity gets enhanced. Haemorrhage due to the rupture of lamellar system leads to seepage of blood materials would not only chock it but my also result in anaemia leading to reduced oxygen carrying capacity of the blood. A wide range of functions has been attributed to the fish mucous including protection against environmental contaminants and UV radiation while some are having the opinion that the mucous consists a metal binding property while the many more have observed the hyperplasia of mucous cells in the suprabranchial chamber, the dendritic plates is also closely associated with gills the operation of the shutter seems to be effected by the pesticides as the respiratory failure is the main cause of fish dead however the failure of nervous system also assists in it.[11] The available data on the toxic impact of different stessors on the air breathing organs indicates that the effect was not vary serve however the alterations noticed include immense congestion of the blood cells which project out from the respiratory surface.[12]

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Effect of atrazine (Herbicide) on blood parameters of common carp (*Cyprinus carpio)*

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Abstract: In the present study an attempt was made to investigate the acute toxicity of atrazine (ATR) a herbicide on an economically important fish, *Cyprinus carpio*. In which, 24 h median lethal concentration (24 h LC_{50}) of herbicide, atrazine to *C. carpio* was 18.5 ppm. Due to fish exposure to 24 h LC_{50} of atrazine for 24 h, red blood cells (RBCs) count (-63.17%), hemoglobin (-27.35%), plasma glucose (-6.78%) and plasma protein (-18.73%) levels were decreased; whereas white blood cells (WBCs) count (+3.73%) was enhanced. The differences in haematological and biochemical values were statistically significant ($p < 0.05$). However, WBC count was not significantly changed. The alterations of the above parameters could be used as an important tool for the assessment of pathological conditions of fish.

Keywords : Atrazine, acute toxicity, hematology, biochemical, *Cyprinus carpio*

INTRODUCTION

The rapid industrialization, application of synthetic fertilizers and use of various insecticides and pesticides, the natural water resources are fast degrading in the water quality. Aquatic ecosystems that run through agricultural or industrial areas have high probability of being contaminated by runoff and ground water leaching by a variety of chemicals.[1] Agricultural pesticides are released into the atmosphere by the spray drift, post application, volatilization and wind erosion of soil.[2] Pesticides presents in aquatic environments can affect aquatic organisms in different ways.[3] In India, more than 70% of the chemical formulations are employed in agricultural practices and to find their way to freshwater bodies, ultimately affect non target organisms.[4]

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The use of herbicides to control aquatic weeds has applied in fish management where they are used in aquatic habitats especially rice fields and some fish farms.[5] ATR has been one of the most widely used herbicides to control broadleaf weeds in corn or crops, including green vegetables.[6] After spraying on crops, it can enter watercourses, because of its high mobility through soil.[7] It is also pointed out that ATR reaches aquatic environments due to proximities of the agricultural country sides to the water places, or directly due to the careless application in such environments.[8] After reaching the environment the atrazine or triazine based herbicides are not degraded by microbial or hydrolytic process.[9] However, reports pointed out that atrazine can be degraded in surface water by photolysis and microorganisms and the half- lives of 20-50 days at 20-25°C have been found under laboratory conditions and increasing at lower temperatures.[10] It has

been detected in natural and surface waters at concentrations exceeding 0.1 gl^{-1} in some areas and also accumulated in a variety of tissues.[11-14] Many authors have reported the impact of atrazine on the physiology and metabolism of aquatic organisms particularly on fishes.[15-18]

Fish are one of the most widely distributed organisms in an aquatic environment and being susceptible to environmental contamination may reflect the extent of the biological effects of environmental pollution in waters. Monitoring of blood parameters, both cellular and non-cellular may have considerable diagnostic value in assessing early warning signs of pesticide poisoning.[19] In India, ATR still is one of the most widely used herbicides controlling broad leave weeds and grasses. The possible effects of ATR on aquatic eco-system have stimulated studies to understand the mechanisms and measurements of the toxic effects of it to aquatic organisms. Hence, the present study is designed to study the acute effect of ATR as a herbicide on blood parameters of a freshwater teleost fish *C. carpio*.

MATERIALS AND METHODS

C. carpio was selected for the present investigation and the healthy specimens were collected from local market of Madhepura. Fish were acclimatized to laboratory conditions for about 15 days before the commencement of the experiment. During this period, fish were fed *ad libitum* with rice bran and oil cake in the form of dough daily. Water was replaced every 24 h after feeding in order to maintain a healthy environment with enough oxygen. The aquarium water was analyzed for physico-chemical characteristics according to APHA (1998)[20].

Five hundred fish were stocked in a large cement tank (4m×6 m×3m) after cleaning and disinfected with potassium permanganate. Fish with an average weight of 6 g and length about 7-8 cm were selected for the experiment. The LC_{50} value of the herbicide atrazine (ATR), (2-chloro-4-ethylamino-6-isopropylamino- 1,2,3-triazine) (18.5 ppm) to fish was calculated following the method of Finney (1978) with a confidence limit of 95%. The acute toxicity experiment was carried out in two circular glass tanks, filled with 40 l of water. A normal pH (6.3) and ATR concentration of 18.5 ppm (LC_{50} 24h) were maintained throughout the experiment. Twenty fish, which were already with held from feeding for 48h, were introduced into each tub. The control was maintained in two circular glass tanks with 20 fish per tub. After 24 h, 40 randomly sampled fish, 20 each for the control and ATR treated groups were used for the haemato-biochemical assay. Care was taken to minimize disturbances to the animals. After the stipulated time period (24 h) fish from control and ATR treated tanks were sacrificed and blood was collected by cardiac puncture using heparinised syringes and kept at low temperature (4°C). All analyses were performed on pooled blood samples. Whole blood was used for the estimation of red blood cells (RBCs), white blood cells (WBCs) and haemoglobin (Hb) content. RBCs and WBCs were counted by the method of.[21] Haemoglobin was estimated by cyanmethemoglobin method.[22] Then the pooled blood samples were centrifuged for 15 min at 10 000 rpm, the plasma was withdrawn and transfer- red into clean vials for plasma glucose and protein estimation. Glucose was estimated by *O*-Toluidine method of Cooper.[23] Protein was estimated according to the method.[24] For the experiment and control group (toxicant free water), three replicates were maintained.

RESULTS AND DISCUSSION

The calculated LC_{50} for 24 h with a confidence limit of 95% of atrazine (ATR) to the carp, *C. carpio* was 18.5 ppm indicating the moderate toxicity of ATR to the fish.

During above exposure period the fish shows various behavioral responses like increased opercular movement, mucous secretion, jerky movement, floating on the sides, hypersensitivity showing violent erratic and fast swimming etc. The abnormal behaviour of the fish indicates the toxic effect of ATR on central nerves system (CNS) and cardiovascular system as suggested by Antychowicz (1979)[25]. Documented effects of ATR in fish include a slowdown in reflexes, swimming activity and feeding. Puigdoller (1978)[26] reported that fish Atlantic salmon exposed to 100 gl^{-1} ATR had reduced food consumption after 10 and 15 days of exposure. These behavioral changes were the result of decreased acetyl cholinesterase activity.

The effects of environmental stressors on the peripheral blood of fishes are well documented in the literature. ATR is toxic; often bioaccumulative and persistent.[27] The blood alterations of carp *C. carpio* exposed to acute concentration of ATR is shown in Figure 1. During acute treatment RBC count (-63.17%), hemoglobin (-27.35%), plasma glucose (-6.78%) and plasma protein (-18.73%) levels were lowered when

compared to that of their control group whereas WBC count (+3.73%) was increased. The differences in haematological and biochemical values were statistically significant ($p < 0.05$). However, WBC count was not significantly changed. Hussein *et al*. (1996) reported decreased RBCs number, hemoglobin concen- tration and haematocrit percentage of *Oreochromis nitoticus* and *Chrysichthyes auratus* when exposed to 3 and 6 mg/l ATR. There also occurs significant increase in hematocrit in Atlantic salmon when exposed to ATR. found that damage of the gill lamellae causes decreased respiratory capacity in *Tilapia mosambica* exposed to 1.1 mgl^{-1} ATR.[28] Found a decline in the number of erythrocytes after exposure of *C. carpio* to 0.1 mg/l atrazine.[29] A high frequency of micronuclei and nuclear abnormalities in *O. niloticus* exposed to different concentrations of ATR is also shown by other authors.

Erythrocyte level was found to be depressed in fishes subjected to stressful conditions. Changes in the erythrocyte profile suggest a compensation of oxygen deficit in the body due to gill damage and the nature of the changes shows a release of erythrocytes from the blood depots.[30] Inhibition of erythropoiesis and increase in the rate of erythrocyte destruction in hematopoietic organs is the cause of decrease in RBC count found significant decrease of RBCs, Hb and packed cell volume (PCV) in ATR exposed fish species and indicated the toxic effect of ATR on spleen, liver and anterior kidney.[31] In the present study, the significant decrease in RBCs and hemoglobin content might have resulted from the lowering of the oxygen content of the water due to the presence of atrazine in the test media. Further reduction in the total erythrocyte count (TEC) might have attributed to a decrease in the erythropoietic activity of the kidney or to the haemodilution resulting from impaired osmoregulation across the gill epithelium. Leucocytes are involved in the regulation of immunological function and their numbers increase as protective response in fish to stress. Such an increase in total leucocyte count (TLC) occurs by the increase in lymphoperisis and/or enhanced release of lymphocytes from lymphoid tissues.[32] Other scientists also reported that WBC increase could be due to an induced proliferation as a result of the chemical toxicity, of pluripotential hematopoietic cells that in turn may be a consequence of a depletion circulating differrentiated cells.[33] The increase in WBC count in the present study indicates the stress condition of the fish caused by ATR which might have produced hypoxia and gill damage. Changes in carbohydrate metabolism measured as plasma glucose can be used as general stress indicators in fish. Reduction in serum glucose levels after exposure to toxicants appears to be caused by hypoxic conditions leading to an excess utilization of stored carbohydrates. Significant decrease in serum glucose was also found. This decrease could be attributed to the toxic effect of ATR on the liver.[34] In the present study, the decreased level of plasma glucose during acute treatment might have resulted from hypoxic condition caused by the herbicide ATR. Further, it is seen that reduction in food intake of ATR treated fish could be also considered another possible reason for decrease in plasma glucose.

The concentration of protein in the serum of fish has been used as an indicator of their general state of health. Reported that the higher energy demand might have triggered an increase in protein catabolism, a process in which both blood and structural protein are converted to energy, thereby reducing serum protein.[35] They further reported that dilution of plasma volume after haemolysis and shrinkage of RBC could also cause a small reduction in protein percentage in serum. Failure of haemopoiesis is a characteristic indicator of kidney damage. Kidney damage causing increased renal excretion of blood protein may also have contributed to the depletion of serum protein in the fingerlings. However, various reports also indicates decrease of total protein in ATR treated fish *Oreochromis niloticus* and *Chrysichthyes auratus* was mainly due to globulin, explaining the toxic effects of ATR on the immune system of these fishes. In the present study, the reduction of plasma protein of fish from acute treatment indicates the toxic effect of ATR on spleen, liver and kidney. Various other researchers also found that carp exposed to 100 gl^{-1} ATR for 72 h had significantly lower plasma protein concentrations, which suggested a haemodilution effect operating in these fish.

From the present study, it is concluded that, atrazine (ATR) has a profound influence on the blood profiles of the treated fish. Hence, the use of the herbicide ATR should be minimized and these parameters could be effectively used as potential biomarkers of herbicide toxicity to *C. carpio*.

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Efficacy of black pepper (*Piper nigrum*) as pesticide against *Callosobruchus maculatus* (F) in stored gram

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Abstract : Gram beetle *Callosobruchus maculatus* is a serious pest of stored legumes with polyphagous & polyvoltine biological features. Control of the pest is a big challenge to curtail the average annual loss of post harvested legumes upto 250 crores in India (FAO). In the present investigation, effect of black pepper powder (*Piper nigrum*) in controlling the pest in the form of sprinkle & emulsifier pesticide has been studied on the gram host experimentally subjected to the *Callosobruchus maculatus* infestation. Very satisfactory result has been obtained which was found statistically significant at both 1% and 5% levels of significance.

Keywords: Stored gram pest, *Callosobruchus maculatus, Piper nigrum* powder, control efficacy

INTRODUCTION

Black pepper, commonly known as "Golmirch" or Kalimirch in Hindi is generally used as spices and flavouring agent by people. It consists of dried unripe fruits of *Piper nigrum* Linn. it belongs to family Piperceac. Piperine (2.5%) a pungent alkaloid is the major chemical constituent of Black Pepper.[1,2] The insecticidal role of black pepper has been reported by many workers.

Insect pest (*Callosobruchus maculatus*) – Bean beetles, *Callosobruchus maculatus* (F) are insect pest of Africa and Asia that Probable rang throughout the tropical and subtropical world. This species is also known as 'Southern Cowpea Weevil'. The larvae of this species feed and develop exclusively on the seed of legumes (Fabaceae). Hence the name "bean beetle" The adults do not require food or water and spend their limited life span (one - two weeks) mating and laying eggs on gram bean etc. Black pepper was obtained from the local market. It was washed thoroughly, dried and then put into incubator for 8 hours at 60°C to make it free from any organism. The dried black pepper was ground into powder by a high speed rotary mill. The powder was again dried and then collected into dry bottle for experimental work.[3]

In this experiment pesticidal role of Black Pepper Powder (BPP) against *C. maculatus* infesting stored gram was tested.

MATERIAL & METHODS

BPP was taken from the laboratory stock maintained for this experiment. 8 doses of BPP i.e. 0.5gm, 1gm, 1.5gm, 2gm, 3gm, 3.5gm and 4 gm were taken each container containing gram was treated with a particular doses of BPP. Containers with BPP treated gram were tightly closed and shaken vigorously for 10 minutes for optimum coverage of the grain surface. Five pairs of freshly emerged test insect, *C. maculatus* from the laboratory culture were then released in each container. The same numbers of insects were also released in the control also. Adult mortality oviposition and emergence of insects was recorded, tabulated.

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RESULT & DISCUSSION

Black Pepper Powder treatment showed clear cut effect on adult mortality, egg laying and population growth of insect pest. *Callosobruchus maculatus* infesting stored gram. In the highest dose i.e. 4gm. Treatment 100% adult mortality was observed on 10 DAI. Whereas it control it was on 16 DAI. In the lowest does i.e. 0.5gm treatment adult mortality was observed from 2 DAI but in achieving 100% mortality it was similar to control i.e. on 16 DAI. 1gm, and 1.5gm, treatment lowered down this period i.e. on 14th day. In 2gm, 2.5gm, 3gm and 3.5 gm treatment 100% mortality was observed on 12 DAI. 1 gm, 1.5gm and 2gm showed 73% to 80% mortality upto 8 DAI. It was noticed that BPP started death of insect pests from the very beginning i.e. 2 DAI in all the treatments. In 2.5gm and 3gm. BPP treatments 83% to 86% mortality was observed on 8 DAI and 93% on DAI.

Oviposition was observed only in 0.5gm and 1gm treatment. In 0.5gm treatment there was no reduction in egg laying in comparison to the control. In 1gm treatment there was 31.2% reduction in egg laying than the control. In 1.5gm, 2 gm, 2.5 gm, 3 gm, 3.5 gm and 4 gm, treatment there was no egg laying.

The observation for population growth on 45th day revealed that it was similar to the control (400) in 0.5gm, treatment, whereas it was reduced to 56.25% in 1gm treatment. In 1.5gm, 2gm, 2.5gm, 3gm, 3.5gm and 4gm treatment there was no growth at all.

Effect of Black Pepper Powder (BPP) Treatment

BPP/ 100 gm. of gram	Adult Mortality Percentage							
	2 DAI	4 DAI	6 DAI	8 DAI	10 DAI	12 DAI	14 DAI	16 DAI
Control					30	63	80	100
0.5 gm.	20	20	26	30	30	80	90	100
1 gm.	50	53	70	73	76	86	100	
1.5 gm.	50	60	70	80	80	90	100	
2 gm.	60	60	70	80	80	100		
2.5 gm.	60	66	70	83	93	100		
3 gm.	70	70	83	86	93	100		
3.5 gm.	70	70	80	88	95	100		
4 gm.	70	80	80	93	100			

DAI = Days After Infestation

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Relation in oxygen consumption and body weight in a hill-stream fish *Glyptothorax telchitta* (Ham.)

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Abstract : Oxygen uptake through the gills of a hill-stream fish *Glyptothorax telchitta* has been studied in relation to different body weights. The oxygen uptake was determined at 24.0±1°C using a cylindrical glass re spirometer with continuous water flow system. The aquatic oxygen uptake through the gills increased from 1.048 to 8.115 ml oxygen/kg/hr within the body weight range of 3.215 to 31.355 g. A fish of 17.052 g average body weight consumes 4.461 ml oxygen while the weight specific oxygen uptake was found to be 267.915 ml $O_{2/}$ kg/hr.

Keywords : Hill-stream fish , *Glyptothorax telchitt*a, oxygen uptake

INTRODUCTION

Oxygen uptake is considered as a measure of energy requirements for metabolism. Generally, with an increase in metabolic activity, the rate of oxygen uptake also increases. Oxygen uptake in a fish depends on various extrinsic and intrinsic factors viz. temperature, level of dissolved oxygen and carbon dioxide in ambient water, pH, salinity, season, nutrition, sex of fish, level of hormones, body weight.[1,2] Several workers have studied the relationship between oxygen uptake and body weight in fishes.[3-8]

In the present study an attempt has been made to study the routine oxygen uptake in a hill-stream fish, *Glyptothorax telchitta* of different body weight to establish the relationship between the weight and oxygen uptake.

MATERIALS AND METHODS

Live specimens of *G. telchitta* of different weight groups were collected from the local market and maintained in a plastic drum of 250 liter capacity. The plastic drum was connected with a pipe for continuous supply of cold water supplied from the stream. At the bottom of the drum, pebbles collected from the river bed were kept to provide a natural condition to the fish. Just above two inches from the bottom of the drum, an outlet with a tap to control the water level was fitted with. The fishes were fed with earthworm and pupae of white ant twice a day. Water of the drum was changed completely once in every 24 hrs. Feeding was stopped at least 24 hrs before experimentation.

The oxygen uptake rate from water was measured in a closed glass respirometer as designed by Munshi and Dube (1973)[9]. The flow of water from the reservoir (a plastic drum of 30 l capacity) to the respirometer was maintained according to the size of the fish to avoid suffocation and stress. The respirometer was covered with a piece of black cloth leaving a small space with a view to keep the fish stress free and count the opercular frequency. Two sets of water samples were taken i.e., inspired (collected before the respirometer) and the expired i.e.,

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the water coming out of the respirometer. Oxygen concentration of inspired and expired water was measured by means of Winkler's volumetric method.[10] The difference between the oxygen contents of the inspired and expired water together with the rate of water flow (ml/h) through the respirometer, were used to calculate the oxygen uptake rate. Three to four readings were taken at every half an hour interval for each fish. The experiments were performed at 24.0±1°C.

RESULTS AND DISCUSSIONS

Glytothorax telchitta is a benthic hill-stream fish is a purely water breathing Sisorid. Its gills are respiratory organs for the extraction of oxygen from water. Measurements of the rate of routine oxygen uptake from the flow-through water in eight (8) weight groups of *Glyptothorax telchitta* were made and the data is summarized in Table 1. The data were analyzed using logarithmic transformations for the allometric relationship.

Table 1. Means value of the rate of oxygen uptake (VO_2) per unit time ($mlO_2.h^{-1}$) in relation to body weight (W) in *Glyptothorax telchitta.*

Body weight	Opercular beating	Oxygen uptake (ml/hr)	
		Observed value	Estimated Value
3.215±0.723	98±1	1.048±0.353	0.950
5.650±0.000	88±0	1.48±0.000	1.600
10.093±3.192	90±1	2.608±0.113	2.750
10.093±3.192	90±1	2.608±0.113	2.750
19.405±1.188	87±2	4.948±0.184	5.050
24.310±0.000	87±3	6.517±0.000	6.230
27.200±1.626	87±2	7.136±1.116	6.920
31.355±0.000	85±0	8.11±0.000	7.900

As soon as the fish was introduced into the cylindrical glass respirometer containing little water, the fish showed erratic movements for a while. When the respirometer was completely filled up with the reservoir water, the fish moved to and fro in search of hiding place. The fish seemed to be under stress. The opercular movement was so high that it could not be counted accurately through eyes; however, the fish gradually became quiet after about 30 to 40 minutes. Then the flow rate of inspired water was so adjusted that the fish did not feel any stress and could move freely within the cylindrical chamber.

The atmospheric temperature recorded during the experimental time ranged from 25°C to 26°C whereas, the temperature of water ranged from 23°C to 25°C, the pH of water ranged from 6.8 to 7, similarly the oxygen content of ambient water ranged from 14.18 to 15.62 mg-

In *Glytothorax telchitta,* the oxygen uptake gradually increased from 1.048 to 8.115 $mlO_2.h^{-1}$ with increase in body weight from 3.315g to 31.355g. A fish of 19.026g average body weight consumed 4.344 $mlO_2.h^{-1}$.

The weight specific oxygen uptake ($mlO_2\ g^{-1}.h^{-1}$) decreased with increasing body weight (Table 1). The weight specific oxygen uptake in *G. telchitta* decreased from 0.326 to 0.258 $mlO_2g^{-1}.h^{-1}$ with increase in body weight from 3.22 g to 31.36 g. The weight specific oxygen uptake of a fish of 19.039 average body weight was found to be 0.26 $mlO_2\ g^{-1}.h^{-1}$

The rate of oxygen uptake in fish depends upon various abiotic and biotic factors.[11] Of the different biotic factors, body weight plays a very important role and it is well known that a normal increase in oxygen uptake is associated with an increase in body weight

In the present study, the oxygen uptake per unit time in hill-stream Sisorid catfish, *Glytothorax telchitta* was estimated to be 0.930. This value is higher than the exponent values reported for a number of purely aquatic breathing fishes.[12-15]

The intercept value 'a' in case of *G. telchitta* was found to be 0.320 which is higher than the values for many purely aquatic breathing fishes but lower than the intercept values for some aquatic breathing fishes.

For 1 g fish, the computed value of intercept 'a' in case of *G. telchitta* found to be 0.320 which is higher than the values reported for many purely aquatic breathing fishes. This is an indication to a higher rate of metabolic decrease with unit increase in body weight in *G. telchitta* but this intercept value is lower than the values reported for some other purely aquatic breathing fishes (Table 5).

Table 2. Computed data on aquatic oxygen uptake rate ($mlO_2.h^{-1}$) for different fish species of 1 g body weight for comparison with *G. telchitta*

Fish species	Oxygen(ml/h)	References
Ictalurus nebulosus	0.084	Jager & Dekkers (1975)
Clarias batrachus	0.134	Munshi *et al*. (1974)
Macrognathus aculeatum	0.140	Ojha & Munshi (1974)
Cyprius casrpio	0.186	Jager & Dekkers (1975)
Macrognathus auleatum	0.213	Ojha & Munshi (1974)
Heteropneustes fossilis	0.215	Munshi *et al*.(1978)
Catostomus commersonii	0.216	Jager & Dekkers (1975)
Catla catla	0.263	Kunwar *et al*. (1989)
Cirrhinus mrigala	0.251	Roy & Munshi (1984)
Anguilla annquilla	0.277	Iager & Dekkers (1991)
Rita rita	0.287	Singh *et al*. (1991)
Glyptothorax relchitta	0.320	Subba (1999)
Glossogobius giuris	0.321	Singh & Munshi (1985)
Channa gachua	0.394	Ojha *et al*. (1978)
Salmo trutta	0.368	Jager & Dekkers (1975)
Samlo gairdneri	0.369	Jager & Dekkers (1975)
Mystus cavasius	0.532	Ojha & Singh (1981)
Anabas testudineus	0.545	Munshil & Dube (1973)
Nemacheilus rupicola	0.575	Rooj (1984)
Acipencer stellatus	0.577	Jager & Dekkers (1975)
Garra lamta	0.816	Rooj (1984)

The intercept values 'a' reported for certain air breathing fishes *Clarias batrachus* (0.134), *Heteropneustes fossilis* (0.215) and *Anguilla anguilla* (0.277) were less than the intercept value of *G. telchitta* but it was found to be higher in *Channa gachua* (0.394) and *Anabas testudineus* (0.545) in the same condition.

The oxygen uptake for 1 g in hill-stream fishes, *N. rupicola* (0.628) and *G. lamta* (0.813) are two to three times more than the intercept 'a' value for *G. telchitta*. This suggests that the metabolic rate of 1 g *N. rupicola* and *G. lamta* are almost two to three times more than that of a *G. telchitta* of similar body weight.

From the studies of oxygen uptake rate of *G. telchitta* it can be concluded that the rate of oxygen uptake is neither high nor low.

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Effect of nimbecidine on caterpillar, *Diacrisia obliqua* (Walker)

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Abstract: The biological activity of eco-friendly insecticide, nimbecidine was evaluated on the mortality, growth and feeding responses of jute hairy caterpillar, *Diacrisia oblique* (Walker). The results showed that all the larval instars of jute hairy caterpillar were highly susceptible to nimbecidine and the susceptibility gradually increased with the increase of concentration, exposure time and decreased with the increase of larval age. Growth and feeding responses of jute hairy caterpillar to nimbecidine were assessed. Observation on growth response indicated that nimbecidine prolonged larval and pupal period, reduced larval and pupal weight and also reduced percent pupation. In addition, nimbecidine also induced morphological abnormalities on survived larvae, pupae and adults. Results on feeding response showed that nimbecidine significantly reduced the food consumption of treated larvae as compared to control. It is clear that nimbecidine possesses insecticidal, growth retardant and antifeedant properties and may be useful in the management of hairy caterpillar.

Keywords : Nimbecidine, *Diacrisia obliqua*, insecticide

INTRODUCTION

Diacrisia obliqua causes huge damage to agricultural crops, there is an urgent need for safe but effective and biodegradable pesticides with no toxic effects on non-targeted organisms. There are about 2000 plant species to possess pest control properties.[1] Among all of the effective botanical insecticides, compounds from neem plant (*Azadirachta indica*) play a vital role in insect pest management. It has extensively been used and has proved its pest controlling efficacy against several insect pests both in field and storage.[2] The present experiment was, therefore, undertaken to investigate the biological activity of the neem based insecticide 'nimbecidine' against hairy caterpillar *Diacrisia oblique* .

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MATERIALS AND METHODS

The research works were conducted to evaluate the biological activities of an eco-friendly insecticide nimbecidine on the mortality, feeding and growth responses of hairy caterpillar, *Diacrisia obliqua* in the laboratory of the Department of Zoology, B. N. Mandal University, Madhepura. The neem based botanical insecticide 'nimbecidine' used in this experiment was purchased from market. During the course of evaluating the activity of nimbecidine, collected fresh and healthy jute leaves were treated separately with different concentrations of nimbecidine by dipping method. These treated leaves were air dried before offering to the insects. In control treatment, leaves were treated with distilled water only. The activity of nimbecidine on the larval mortality of *Diacrisia oblique* was determined by exposing fourth, fifth and sixth instar larvae to food treated with nimbecidine at 2.0, 1.0, 0.5,

0.25 and 0.125% concentrations. For this purpose, newly moulted fourth, fifth and sixth instar larvae were separately placed in petridish (15 cm. dia.) containing treated food. Thirty larvae were maintained in each petridish. Larvae were transferred to new petridish containing fresh treated food daily. Control treatments were done side by side for each instar separately. The larval mortality was recorded each day.

The activity of nimbecidine on the growth and development of *Diacrisia obliqua* was observed by exposing fourth instar larvae to the treated and untreated jute leaves. These leaves were treated with nimbecidine at 0.1, 0.05, 0.025 and 0.0125% concentrations. From laboratory culture newly moulted fourth instar larvae were individually transferred into petridishes (9 cm.dia.) containing treated and untreated leaves. Ten petridishes were maintained for each concentration. The larvae were transferred daily into new petridishes containing fresh treated and untreated food. Rearing of the larvae were continued till death or pupation. The duration of each larval instar, larval weight, larval mortality, total larval period, percent pupation, pupal period and pupal weight were recorded. In addition, morphological abnormalities occurred due to action of nimbecidine were also recorded.

Similar to growth response, the activity of nimbecidine on the feeding response of *Diacrisia obliqua* was also observed by exposing fourth instar larvae to the treated and untreated jute leaves. Before offering to the larva, jute leaves were cut into a piece of 4800 mm^2 leaf disc and treated with different concentrations of nimbecidine. One to two leaf discs were offered to each larva in each petridish and ten petridishes were maintained for each concentration. Treated food materials were replaced by treated leaf discs daily and data on food consumption were recorded daily till death or pupation. The food consumption of larvae were measured by using square millimetre graph paper. The experimental data were statistically analyzed in accordance with one factor Completely Randomized Design (CRD). The mean values were separated by Duncan's Multiple Range test (DMRT) under Microsoft statistical programme in a computer.

The mortality data were analyzed by probit analysis originally designed by Finney (1971)[3] using Mstat statistical package programme. Before analysis, mortality data were corrected[4] as follows:

$$\text{Corrected mortality} = \frac{\text{Treated mortality} - \text{Control}}{100 - \text{Control mortality}} x\ 100$$

RESULTS AND DISCUSSION

Table 1: Toxicity of nimbecidine treated against fourth instar larvae of *Diacrisia obliqua* at different days after treatment.

Days after treatment	LD_{50}	95% Fiducial limit	
	(%)	Lower	Upper
D_4	1.89	0.98	4.58
D_5	1.78	0.65	3.87
D_6	0.91	0.44	1.90
D_7	0.38	0.22	0.63

Table 2: Toxicity of nimbecidine treated against fifth instar larvae of *Diacrisia obliqua*

Days after treatment	LD_{50}	95% Fiducial limit	
	(%)	Lower	Upper
D4	2.74	1.26	5.96
D5	2.15	0.98	3.91
D6	1.54	0.74	2.95
D7	0.89	0.35	1.25

Table 3: Toxicity of nimbecidine treated against sixth instar larvae of *Diacrisia obliqua*

Days after treatment	LD_{50}	95% Fiducial limit	
	(%)	Lower	Upper
D_4	3.69	2.07	6.53
D_5	2.98	1.58	3.59
D_6	1.89	0.81	3.49
D_7	1.05	0.96	1.79

Table 4: LT_{50} values of nimbecidine against *Diacrisia obliqua*

Concentration (%)		LT_{50} (days)	
	Fourth instar	Fifth instar	Sixth instar
0.125	5.25	6.59	7.89
0.25	5.09	6.15	7.58
0.5	4.87	5.75	6.65
1.0	3.98	4.12	5.59
2.0	3.82	4.25	4.53

Effect of nimbecidine on growth response of *Diacrisia obliqua*

Effect on growth and development

The activity of nimbecidine on the growth response of *Diacrisia obliqua* was evaluated by exposing fourth instar larvae to food treated with nimbecidine at different concentrations and feeding was continued throughout the remaining larval period. Observations made on the activity of nimbecidine on the larval and pupal development of *Diacrisia obliqua* are presented in Tables 5 and 6 respectively.

Nimbecidine had significant effect on larval development. The larval period was prolonged on treated food compared to control treatment. The longest larval period (17.00 days) was recorded in 0.1% concentration which was statistically identical with the remaining concentrations of nimbecidine. Significantly lowest (11.00 days) larval period was recorded in control treatment (Table 5). The weight of surviving sixth instar larvae were reduced in treated food over control. Significantly lowest larval weight (0.17g) was recorded in 0.1% concentration and significantly highest larval weight (0.45g) was recorded in control treatment. The second lowest larval weight (0.22g) was recorded in 0.05% concentration which was statistically identical with 0.025 and 0.0125% concentrations. Larval mortality was increased with the increase of concentration of nimbecidine. The larval mortality was recorded as 20, 40, 50, and 70% in 0.0125, 0.025, 0.05 and 0.1% concentrations respectively whereas no mortality was recorded in control treatment. Chowdhury *et al.* (2001)[4] reported that azadirachtin, the major constituent of neem exhibited significant growth inhibition against *Spilarctia obliqua* larvae. Mosaddeque (1995)[6] also observed that neem oil markedly affected the larval growth and development of *Diacrisia obliqua*.

Pupal period was also prolonged with the treatment of nimbecidine. The longest pupal period (14.25 days) was recorded in 0.1% concentration which was statistically different from the remaining concentrations of nimbecidine. Significantly shortest (8.80 days) pupal period was recorded in control treatment. Pupal periods were recorded as 13.00, 12.40 and 10.50 days in 0.05, 0.025 and 0.0125 % concentrations and which were statistically different (Table 6).The weight of pupae developed from nimbecidine in treated larvae were reduced over control. Significantly lowest pupal weight (0.10g) was recorded in 0.1% concentration which was statistically identical with 0.05 and 0.025% concentrations. Significantly highest pupal weight (0.26g) was recorded in control treatment. The second highest pupal weight (0.15g) was recorded in 0.0125% concentration which was statistically different from remaining treatments. Percent pupation was also recorded. The pupation were recorded as 60, 50, 40 and 20% in 0.1, 0.05, 0.025 and 0.0125% concentrations respectively while 100% pupation was recorded in control treatment. Similar reduction of pupal growth, survival and pupal weight were also reported with neem seed oil by different workers.[7,8]

Table 5 : Effect of nimbecidine on the larval development of *Diacrisia obliqua*

Concentration (%)	Larval instar (days)			Total larval Duration (days) (Mean± SE)	Larval weight (g) (Mean±SE)	larval Mortality (%)
	4th instar	5th instar	6th instar			
0	3.2	3.3	4.5	11.00±.0.297b	0.45±0.0372a	0
0.0125	4	4.8	5.1	13.90±0.489a	0.26±0.0323ab	20
0.025	4.4	5.4	5	14.80±0.841a	0.25±0.0223ab	40
0.05	5.1	5.2	6	16.30±0.943a	0.22±0.0374b	50
0.1	5	6	6	17.00±1.075a	0.17±0.0333c	70

Means having same letter (s) in a column did not differ significantly

Table 6: Effect of nimbecidine on the pupal development of *Diacrisia obliqua*

Concentration (%)	Pupal period (days) (Mean± SE)	Pupal weight (g) (Mean±SE)	Pupation (%)
0	8.80±0.25e	0.45±0.0372a	100
0.0125	10.50±0.43d	0.26±0.0323ab	60
0.025	12.40±0.51c	0.25±0.0223ab	50
0.05	13.00±0.41b	0.22±0.0374b	40
0.1	14.25±0.25a	0.17±0.0333c	20

Means having same letter (s) in a column did not differ significantly

Table 7: Effect of nimbecidine on the food consumption of *Diacrisia obliqua*

Concentration (%)	Larval instar (days)			Total (Mean ± SE)
	4th instar	5th instar	6th instar	20497.40±.847.56a
0	4039.40	5197.80	11260.00	16011.80±1962.18b
0.0125	3097.00	4596.30	8318.60	10219.40±1146.85c
0.025	2987.50	3174.40	4057.50	9015.20±1325.27c
0.05	2837.00	2880.70	3297.50	7686.60±937.35c
0.1	2081.50	2605.00	3000.00	20497.40±.847.56a

Means having same letter (s) in a column did not differ significantly

Effect of nimbecidine on feeding response of *Diacrisia obliqua* nimbecidine significantly reduced the food consumption of the test insect when fourth instar larvae were allowed to feed on treated food (Table 7). Food consumption gradually decreased with the increase of concentration of nimbecidine. The total food consumption at 0.0125, 0.025, 0.05 and 0.1% concentrations were 16011.80, 10219.40, 9015.20 and 7686.60 mm^2 respectively. These figures were statistically lower from the control treatment. In control treatment, the total food consumption was 20497.40 mm^2. The lowest food consumption (7686.60 mm^2) was recorded at 0.1 % concentration which was statistically identical with 0.05 and 0.025 % concentrations.

Mosaddeque (1995)[6] reported that neem oil significantly reduced the food consumption of the larvae of D. *obliqua* and the food consumption decreased more with the highest concentration. He concluded that reduction of food consumption might be due to the antifeedant activity of neem oil. Chowdhury (2001)[5] also stated that azadirachtin, the major constituent of neem exhibited significant antifeedant activity against *D.obliqua* larvae.

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Biodiversity of compost mesofauna as an indicator of the composting

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Abstract : One of the key issues in compost research is to assess the quality and maturity of the compost. Biological parameters, especially supported mesofauna, have multiple advantages for monitoring a given system. The mesofauna of compost includes Isopoda, Myriapoda, Acari, Collembola, Oligochaeta, Tardigrada, Hexapoda, and Nematoda. This wide spectrum of organisms forms a posh and rapidly changing community. Up to this, none of the dynamics, in reference to the composting process, of those taxa are thoroughly investigated. However, from the mesofauna, only nematodes possess the specified attributes to be potentially useful ecological indicators in compost. They occur in any compost heap that's investigated and in virtually all stages of the compost process. Compost nematodes are often placed into a minimum of three functional or trophic groups. They occupy key positions in the compost food web and have a rapid respond to changes in the microbial activity that is translated in the proportion of functional (feeding) groups within a nematode community. Furthermore, there's a transparent relationship between structure and function: the feeding behavior is definitely deduced from the structure of the mouth cavity and pharynx. Thus, evaluation and interpretation of the abundance and function of nematode faunal assemblages or community structures offers an in situ assessment of the compost process.

Keywords : Acari, Collembola, Isopoda, maturity, Nematoda, Myriapoda, Oligochaeta, succession

INTRODUCTION

Composting is an aerobic, heat producing, and controlled process by which a wide spectrum of organisms converts a mixed organic substrate into water; carbon dioxide (CO_2); inorganic nutrients; and stabilized sanitized organic matter (humus).[1] In contrast to natural rotting or decomposition, the environmental conditions during composting (e.g., moisture, temperature, substrate composition, and oxygenation) remain controllable.[2,3] Composting is considered important for sustainable agriculture and partly solves the problem of organic waste treatment.[4,5] Multiple benefits of compost addition to the soil's physical and chemical characteristics have been reported extensively in recent literature and include increased soil organic matter content[6] and the subsequent improvement of soil aeration, soil porosity, drainage, and water holding capacity.[7] Furthermore, compost can provide an important source of nutrients for plants by promotion of mineralization of compost nitrogen in the soil.[8] Composts are also known for their disease suppressive activity based on both physicochemical and biological mechanisms.[9,10]

One of the key issues in compost research is to assess compost maturity as the status of the compost determines ultimately the quality of the product and the associated advantages as mentioned above. Nevertheless, compost maturity is often loosely defined as the state when compost is dominated by humic substances[11] or as the state where the temperature reaches a near ambient level. For the past decade, researchers have proposed multiple chemical and physical variables to assess maturity as well as biological parameters.[12] At present and to the best of our knowledge, none of the proposed tests reliably, consistently and

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unequivocally quantify compost maturity.[13] Other important characteristics that need to be considered for evaluation of proposed maturity tests are required expertise, necessary lab equipment, time, and costs. Despite the fact that the compost fauna is not well known, interest in the ecological effects of composting has been growing and recently soil fauna is proposed as a potential tool to evaluate the ecotoxicology of compost. However, the compost is comprised of fauna that can be used in assessing the maturity and quality of the compost. In general, biological parameters have enormous advantages for monitoring a given system, i.e., they are well correlated with ecosystem functioning, they respond sensitively to management practices and climate, and they illustrate the chain of cause and effect. The surplus value of biological parameters in system monitoring holds true, especially for compost being the direct result of a biological process.

Compost supports a diversity of microbes (e.g., fungi, bacteria, actinomycetes, and algae), microfauna (protozoa), and mesofauna (mainly annelids, arthropods, and nematodes).[14] The microbes and microfauna are the chemical decomposers responsible for the organic matter decomposition through aerobic respiration, while the larger organisms (mesofauna) are the physical decomposers important for the (initial) mechanical breakdown of organic materials into smaller particles, thereby increasing surface area for microbial action and/or mesofauna and occupying key positions in the compost food chain.

Although microbes execute the lion's share of the primary decomposition, mesofauna have multiple advantages over soil microbes and microfauna as an indicator for the quality and status of the compost. In general, mesofauna are more integrated in the food web by being one or two steps higher and their response to changes in the environment are more significant on a stable temporal scale because of their longer generation time, this way making it possible to distinguish between "real" environmental changes and transient nutrient flushes. In addition, some mesofauna groups (e.g., nematodes and mites) occur at more than one level in the energy pyramid and are representative of several levels of consumers.

In this paper, we provide an overview of the mesofauna found in compost and discuss for the potential for each taxon to be an indicator of compost quality and process status. In the second step, the use of effective indicators is further elaborated, based on nematodes as the most promising group.

MESOFAUNA IN COMPOST

Together with bacteria, fungi, and other microbes (e.g., actinomycetes), the mesofauna in compost form a complex food web or energy pyramid with primary, secondary, and tertiary levels of consumers. Obviously, the base of the pyramid or the energy source is made up of organic matter. All the levels above are populated, though not exclusively, by mesofauna. The mesofauna of compost includes Isopoda, Myriapoda, Acari, Collembola, Oligochaeta, Tardigrada, Hexapoda and Nematoda. This wide spectrum of organisms forms a complex and rapidly changing community that is not limited to a specific compost but can be found from vermicompost to mushroom compost and from small to large scale controlled or open air composting processes. However, the available information on mesofauna from compost is limited and mainly only present in "gray" literature or from non scientific works. Information on their presence relative to the stage of a compost process (thermophilic phase, cooling phase, and maturation phase) is virtually absent. This limited information on compost mesofauna, ranging from structuring or natural compost inhabitants to more accidental taxa, will be listed in the following text. The known compost taxa can hereby be roughly classified according to the following main focuses: 1) their direct and visible influence on the compost process (i.e., mainly Oligochaeta) and 2) merely accidental records from general biodiversity studies.[15]

Within the phylum Annelida, earthworms occur in diverse habitats with a lot of organic material like manure, litter, and compost.[16] Earthworms are very important physical decomposers in the composting process because they maintain the aerobic condition in the compost by tunneling, and therefore mechanically partitioning the organic material while feeding on this material. As a consequence, they facilitate the transformation of nutrients into available forms and increase microbial activity.[17] When earthworms are artificially added to a composting process, the process is called vermicomposting. Many recent studies focus on this inoculation with epigeic earthworms and their potential to recycle organic waste materials into value-added products is well documented.[18] Inoculation with earthworms accelerated the decomposition process though the composting potential of different species in diverse compost situations remains to be investigated.[19] In a study on forest litter decomposition by Manna *et al.* (2003), the compost potential of *Eisenia fetida* was clearly superior

compared to other species (i.e., *Perionyx excavatus* and *Dicogaster bolaui*). At present, *E. fetida* is the most popular compost earthworm and is known under various common names, including redworms, brandling worms, and tiger worms. Owing to its recognized function in a compost process, *E. fetida* specimens are widely commercially avail- able in temperate regions for use in vermiculture. It is investigated the biology and ecology of the earthworm *Eudrilus eugeniae*, which is indigenous in Africa but bred extensively internationally for the fish bait market. These researchers concluded that this earthworm species might be a good candidate for vermicomposting in tropical climates. Thus, earthworms have a well known positive effect on compost, but they usually only reach high densities after inoculation and/or within specific controlled circumstances. Furthermore, they are not present in all stages of the composting process.

Mites (Acari) are commonly found in compost[18] provided an overview of the species found in the investigated composts (respectively in Canada and in Central Europe). This study reported the presence of either red pepper mites (*Siteroptes mesembrinae*) and/or bacterial feeding mites (*Histiostoma feroniarum*) and associated these mites with the reduced mushroom yield. Believe these species can serve as an indicator species for poor pasteurization or conditioning in mushroom compost processes. Other encounters with mites (i.e., *Pygmephorus* sp., *Histiostoma* sp., and *Parasitus bituberosus*) in mush- room compost were reported.[20-22] Although the contributions of mites in the compost food web are not known, mites cover multiple levels of the food web, ranging from scavenging on organic debris to fungal feeding, bacterial feeding, and predators, and most likely can have a certain structuring effect on the compost process, but this is not yet studied.

Within the arthropods, beetles (Coleoptera) are the most diverse animal group and they can be found in almost all habitats, including compost.[23-25] Beetles might act as decomposers, fungivores, or as predators on other insects present in the compost.[24-27] Compost heaps can be very rich in beetle species: 273 species were found in compost from Lulea in northeastern Sweden[28,29],>260 species in different composts from Uppsala in central Sweden[30,31], and 225 species in seed-composts from Skane in southern Sweden[32]. A number of exotic beetle species (34) have been established in compost heaps in the Nordic countries during the 20th century. The successful establishment of exotic species in compost habitats results from the increased frequency of introductions and warm, thermal stability in large compost heaps during the Nordic winter.

Woodlice (Isopoda) are saprophages that generally live in moist soil, but some species such as *Porcellionides pruinosus* and *P. sexfasciatus* live aggregated in compost and manure heaps.[1] Terrestrial isopods are principally consumers of organic debris in an ecosystem. However, extreme populations of woodlice may cause serious damage to vegetables. But whereas populations of woodlice increased strongly in fresh compost, in mature compost they barely survived. Therefore, stressed the importance of a complete compost process in order to manage woodlice populations at an acceptable level.[33-35]

Centipedes (Chilopoda) and millipedes (Myriapoda) are another group of more accidental compost mesofauna. Centipedes are mainly carnivores, while millipedes (Diplopoda) feed on plant materials and fragments of decaying organic material, though both groups are found in similar habitats. In urban areas, their common microhabitat is often extended by building stacks of materials like compost and garbage piles. The distribution of *Oxidus gracilis* (millepedes) is connected with greenhouses and gardens with compost heaps in higher altitudes. Identified centipedes collected from compost heaps include *Geophilus proximus* and *Haplophilus subterraneus.*

It is reported springtails (Collembola) from compost processed at the Corporación Universitaria Lasallista. Nowadays, springtail reproductions in compost products are often used to develop ecotoxicological tests (e.g., *Folsomia candida*[37] or to investigate the survival of potential biocontrol species after the addition in composts.

There also have been more isolated recordings of other Arthropoda taxa in compost. Finally, tardigrades or water bears, exhibit a high tolerance to extreme environmental conditions such as temperature, making them a likely group to appear in compost. However, published recordings of Tardigrada in compost are scarce. *Hexapodibius reginae* from compost and recorded tardigrades in rice straw compost using DGGE analysis. Our own observations indicate the accidental presence of tardigrades in compost.

All the above mesofauna groups can be found in compost, but none of these groups is always found in all compost types. None of these groups can be found

throughout all successive compost stages. Considering the population dynamics in relation to the composting process, none of the above taxa have been thoroughly investigated. According to the limited current knowledge, only nematodes (Phylum Nematoda) occur in any compost pile that has been investigated and they are active in virtually all stages of the composting process. Recordings of nematodes in isolated compost samples were made by several authors. These primarily bacterivorous nematodes were recorded in general biodiversity studies, but their function within the compost process was not discussed. The structure and succession of the nematode community during the composting process was investigated for the first time.[38]

Major shifts in species composition occur during the composting process. At the beginning of the process (thermophilic phase), immediately after the heat peak, the nematode population is primarily built by bacterial feeding enrichment opportunists (cp-1) (Rhabditidae, Panagrolaimidae, Diplogasteridae), followed by the bacterial feeding general opportunists (cp-2) (Cephalobidae) and the fungal feeding general opportunists (Aphelenchoididae). Thereafter, during the cooling and maturation stages, the bacterial feeding predator opportunistic nematodes (e.g., *Mononchoides composticola*) became dominant. Finally, at the most mature stages, the fungal feeding Anguinidae (mainly *Ditylenchus filimus*) were most prominent. Hence, nematodes are omnipresent during the composting process and the succession of nematodes is associated with major shifts in life strategies and feeding behavior.

POTENTIAL OF MESOFAUNA AS ECOLOGICAL INDICATOR IN COMPOST

Several authors have used microbial communities (bacteria and fungi) to evaluate compost maturity, typical changes in the microbial community but stressed the need for more studies on different composts. However, mesofauna has multiple advantages over microbial communities as indicators for the quality and status of compost. Firstly, it is very difficult and time consuming to identify all bacteria, fungi, and protozoa in a sample and databases of biochemical profiles are either incomplete or inadequate, especially for free living taxa. Secondly, by being one or two steps higher in the food chain, mesofauna serve as integrators of physical, chemical, and biological properties related to their food resources. Third, their generation time (days to years) is longer than metabolically active microbes (hours to days), making them more stable temporally and not simply fluctuating with ephemeral nutrient flushes.[39]

A successful indicator must be able to reflect a past ecological process or predict a future ecological process. Therefore, an indicator should have as many as possible of the following qualities. compost fauna is only fragmentary. Nevertheless, some mesofaunal taxa present in compost, such as Collembola, mites, earthworms and nematodes, have already proven their potential as bioindicators (Table 1).

Table 1. Overview of the mesofauna groups used as bio-indicator and their qualities.

Occurrence	Mites	Collembola	Earthworms	Nematodes
- Ubiquitous				X
- Large population size			X	X
- Present on more than 1 level in food web	X	X		X
Function and Services in Ecosystem				
- Important role in N-mineralization			X	X
- Potential to regulate even suppress magnitude of resources	X		X	X
- Variable sensitivity to stress factors, from extremely sensitive to extremely tolerant				X
Practical Applicability				
- Easily extracted and cultured				X
- Easy observation	X		X	X
- Identification up to species level not needed	X	X		X
- Inexpensive techniques and methodologies				X

Collembola and mites have been widely used to assess the impact of agricultural management practices and disturbances.

Conversely, the potential of earthworms as indicators of environmental changes are more generally accepted. Their possibilities as bioindicators are based on community changes or on a quantifiable bioaccumulation of metals in their tissues. However, earthworms do not occur in every composting stage and process and the accumulation of contaminations is not the main or only focus in assessing the compost status.

Unlike earthworms, Collembola, or mites, nematodes are ubiquitous, even in disturbed or polluted areas. Nematodes live beside the most diverse and abundant soil organisms and are the most important secondary consumers within soil mesofauna. They have been extensively used as indicators of soil biodiversity and functioning and as indicators of environmental disturbances in soil regarding compost nematodes, they can be placed into at least three functional or trophic groups. They occupy key positions in the compost food web because they interact with the species providing their food (e.g., bacteria, fungi, etc.) and other organisms using the same food resources. They are also

involved in predator prey interactions (e.g., protozoa, fungi, and mites as well as nematodes). In these ways, changes in the food web are mirrored in shifts among nematode feeding groups. Most importantly, nematodes show a rapid response, translated into proportions of functional (feeding) groups, to changes in the microbial decomposition pathway and the availability of food sources, due to their relatively short generation time. In addition, the transparent nature of nematodes allows easy observation of the clear relationship between structure and function: their feeding behavior is easily deduced from the structure of the mouth cavity and pharynx. Furthermore, the relative abundance and small size of nematodes makes sampling and extraction easier and less costly than for other soil fauna. Neher *et al.* validated the hypotheses that nematode based soil food web indices are useful indicators of other soil organisms such as mites. Hence, nematode faunal assemblages theoretically have all of the properties needed to enable an optimal *in situ* assessment of the compost process, based on evaluation and interpretation of their abundance and function.

However, in the future, the generality of these first promising results must be assessed by analyzing disparate composting processes with different feedstock materials. Importantly, the nematode characteristics must also be analyzed along with the microbial and abiotic data (temperature, pH, moisture content and C/N ratio) in order to provide a more profound insight into the precise relation/ correlation of nematode community and the composting process. Finally, threshold values of the nematode based ecological indices must be determined in order to obtain useful and reliable indicators indices of the maturity and quality of the compost.

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Effect of storage temperature and relative humidity on the decay of garlic bulb

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Abstract : After harvest in late April or by mid-May of the year, garlic is dried for a few days in the sun and stored in ordinary jute bag or netted nylon bags. No care is seriously taken to store garlic at low temperature and low relative humidity. Due to this, garlic decay during storage. To prevent decay of garlic it should be stored at low temperature and low humidity. In the present work, it was observed that 5^0C temperature completely control damage of garlic bulbs at all RH level. The rate of decay of garlic bulb following the increase of temperature at 50% relative humidity was slower than that stored at 60 and 70% relative humidity. It was maximum at 70% relative humidity level.

Keywords : Relative humidity, temperature

INTRODUCTION

Economically garlic is highly popular agricultural product in every part of the world for the preparation of various recipes such as Soup, Pasta, Meat, Fish, Seafood, Main dish, etc. Garlic is used as medicine. In Ayurveda the use of garlic has been recommended for home remedies such as cold, cough, allergies, toothache, ring worm, diabetes, viral and bacterial infections, etc. The main biochemical constituents found in the fleshy storage leaf of garlic are volatile oil which contains Allyl disulphide, Ally proylsulphide and polysulphide, Allicin and Allocetion I and II. During storage of garlic, it is damaged by the infection of some pathogenic fungi such as *Aspergillus niger, Fusarium moniliforme, Fusarium solani, Fusarium culmorum, Aspergillus sydowi, Fusarium oxysporum, Aspergillus flavus* and *Aspergillus nidulans.* To protect damage of garlic bulb from pathogenic fungi temperature and relative humidity plays an important role. Higher relative humidity and temperature favours growth of pathogenic fungi. It is necessary to store garlic bulbs at low temperature and low relative humidity.

In the present work, the role of relative humidity and temperature in the protection of garlic from damage during storage has been evaluated.

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MATERIAL & METHOD

5^0, 10^0, 15^0, 20^0, 25^0 and 30^0C temperature were maintained in the refrigerator incubator and BOD incubators. Glycerol solution to maintain 50%, 60% and 70% RH was taken in desiccators. The noted RH levels were maintained each for the noted temperature. Four uninjured bulbs of dried garlic as were noted earlier were taken in one petri dish and set in each desiccators. The lid of the desiccators was sealed with Vaseline. The experiment was set in the last week of May, 2013 and the observation for decay was made in the last week of August, 2013 i.e. the storage period was of three month duration.

RESULT

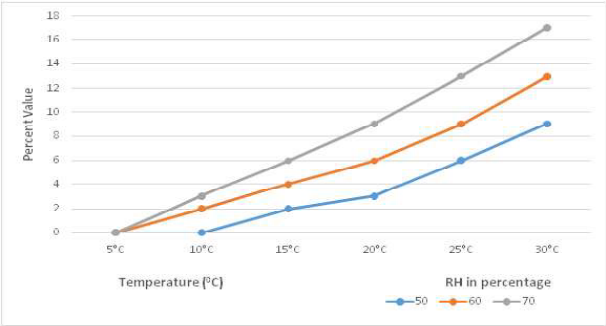
In the present study, it was observed that 5^0C temperature completely control damage of garlic bulbs at all RH level. The rate of decay of garlic bulb following the increase of temperature at 50% relative humidity was slower than that stored at 60 and 70% relative humidity. It was maximum at 70% relative humidity level. The fungi involved were *Aspergillus niger, Aspergillus flavus* and *Fusarium moniliforme.*

The temperature of storage above 10^0C seems not desirable due to considerably high percent of decay of garlic. 30^0C of temperature proved the worst for maximum percent of decay. The result is represented in table 1 and graph 1.

Table 1: Effect of Temperature and RH of storage of garlic on their % decay

Si. No.	RH(%)	Temperature(^{0}C)					
		5	10	15	20	25	30
1.	50	0	0	2	3	6	9
2.	60	0	2	4	6	9	13
3.	70	0	3	6	9	13	17

Graph 1: Effect of Temperature and RH of storage of garlic on their % decay



CONCLUSION

Decay of garlic during storage is due to infection of some pathogenic fungi. In low relative humidity and temperature the pathogenic fungi cannot grow thus, saving the garlic from damage. So, it is desirable to store garlic at low temperature and low relative humidity. During present work, it was observed that at a temperature of 5^0C no any damage occurred in garlic bulb during storage even at high relative humidity. The rate of decay of garlic bulb following the increase of temperature at 50% relative humidity was slower. Then that stored at 60 and 70% relative humidity. It was maximum at 70% relative humidity level. The fungi involved were *Aspergillus niger, Aspergillus flavus* and *Fusarium moniliforme.*

The temperature of storage above 10^0C seems not desirable due to considerably high percent of decay of garlic. 30^0C of temperature proved the worst for maximum percent of decay.

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Spatial variation in the S.W species diversity of oriental garden lizard of mangroves & home garden in Madhepura district of Bihar, India

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Abstract : The natural profile of species diversity at habitat level & biodiversity at biospheric level always displays temporal & spatial variation in time and space, hence is a dynamic function. Therefore it is mandatory for researcher to undertake spatial (geographical) and temporal (time specific) mapping of the species diversity to investigate the degree of variation. The present article being the part of continuous mapping of lizard faunal diversity of Madhepura urban conglomerates done during the years 2014-2016 targets to unravel the spatial variation of garden lizards found in two different habitats - home garden and mangrove. Oriental garden lizards being an important member of lacertilian reptiles are subjected to population wise threat & endangerment due to multidimensional unmindful anthropogenic interference. The article brings on record different species of garden lizards present in specific terrestrial habitats such as home garden & mangroves of Madhepura. Six different species have been identified from these habitats & their cumulative species diversity index through Shannon-Weiner equation (S-W) have been statistically computed on the basis of species abundance & richness recorded numerically during the sampling without sacrificing any live individual (IBML) so that none of the species get harm during the research.

Keywords : Spatial variation, S-W diversity, oriental garden lizard, mangroves

INTRODUCTION

Immense diversity in our biosphere exists not only at the species level but also the biological level that ranges from macromolecules within cells to biomes. As we know that the word biodiversity is the most common word reflecting species diversity and species richness, the rapid change undergoing is due to climatic variations and other human influences.[1] Spatial variation in pattern of biodiversity result in change of their distribution, leaving of the local communities, extinction etc.[2,3] Changes occurring in the community composition of any organism are however not necessarily associated with the changes in species richness if any gain or loss of the species are balanced.[4-7] However it can be adverse in the places where the intensity of land use is seen higher, it could result in decrease of species richness.[8] Significant geological events, such as the formation of mountain ranges, are often treated as a source of long-term biogeographical barriers.[9] In some cases, however, distinct geographical patterns can arise without obvious geographical barriers (e.g ecological divergence, geographical distance, competition) among populations.[10] Ecological interactions of organisms with their environment play an important role in population divergence and speciation.[11] The use of such data in ecological niche modeling has provided insights into speciation and diversification in relation to environmental factors. Therefore it is very important to incorporate ecological data into studies of mechanisms

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driving diversification and demography.[12] Hence disturbance of the habitat is considered as one of the important anthropogenic factor influencing the ecosystem at high speed caused by human activities in natural habitats.[13]

MATERIAL & METHODS

Spatial variation which is considered as one of the best key to study about the ecological status of any organism, here in this paper study on diversity with reference to spatial variation of oriental garden lizard was done so that the result which may be obtained from the analysis would be anyhow useful to know the current status of the species and could bear in taking of special conservative measures for the protection of the fauna of that particular area.

For this, study was conducted in two different sites mangroves and home garden of Madhepura district during 2014-2016 as also mentioned in the previous paper. The study was carried out between 11 am to 2 pm when the temperature was favorable for the reptiles. Since lizards are model organisms for ecological studies and easily handled not too many precautions were required. The photographic image and few videos were also taken so that it could become easy to examine and identify the species without interfering the IBML that says none of the species should be harmed during research activities.

RESULT & DISCUSSION

As observed in table no. 1 six different species were encountered during the field study namely- *Eutropis carinata*, *Ophisops jerdoni*, *Mabuya macularia*, *Eutropis macularia, Calotes versicolor* & *Sitana ponticeriana* from two different sites of the Madhepura district, which includes mangroves & home garden. Among these species sampled *Calotes versicolor* were among the highest species found but only from the site 1 while in site 2 i.e. home garden *Sitana ponticeriana* were among the highest of all the other species. When diversity of both the sites were compared species from different mangroves were statistically highest having S-W diversity index (&) as 0.7594 which shows that population of the lizard species is not rich but their abundance is high as it should be. While the numerical data of the home garden showed lower abundance & higher richness of the species with respect to the former having diversity index 1.1049. As per species diversity interpretation index if, the obtained diversity value of the species is less than 1, all the species of the lizards will diversified with less survival risk whereas the value being more than 1 denotes the less diversified species present in the habitat with greater survival risk. Thus specific measures are compulsory for the protection of these species in urban areas of Madhepura so that ecosystem get balanced.

Table 1-Site specific species diversity of different species of garden lizards sampled from Madhepura district sampling sites-different mangroves

Sl no.	Name of the species	Family	Average no. of sampled individuals per unit area	pi(n/N)	log pi	pi*log pi	H
1	***Eutropis carinata***	Scincidae	5	0.151	-0.821	-0.1239	
2	***Ophisops jerdoni***	Lacertidae	6	0.181	-0.742	-0.1343	
3	***Mabuya macularia***	Scincidae	3	0.090	-1.045	-0.0940	
4	***Eutropis macularia***	Scincidae	6	0.181	-0.742	-0.1343	**0.7594**
5	***Calotes versicolor***	Agamidae	8	0.242	-0.616	-0.1490	
6	***Sitana ponticeriana***	Agamidae	5	0. 151	-0.821	-0.1239	
	Total		**N = 33**				

Table 2- Site specific species diversity of different species of garden lizards sampled from Madhepura district sampling sites-different home garden

Sl no.	Name of the species	Family	Average no. of sampled individuals per unit area	pi(n/N)	log pi	pi*log pi	H
1	***Eutropis carinata***	Scincidae	3	0.12	-0.920	-0.1104	
2	***Ophisops jerdoni***	Lacertidae	2	0.08	-1.096	-0.0876	
3	***Mabuya macularia***	Scincidae	4	0.16	-0.795	-0.1272	**1.1049**
4	***Eutropis macularia***	Scincidae	4	0.16	-0.795	-0.1272	
5	***Calotes versicolor***	Agamidae	5	0.2	-0.698	-0.498	
6	***Sitana ponticeriana***	Agamidae	7	0.28	-0.552	-0.1545	
	Total		**N = 25**				

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Cure of asthma and jaundice disease by the traditional and ethnic people of Mandar Block, Ranchi, Jharkhand, India through medicinal plants

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Abstract : In the recent era, the people of urban areas of India also depend upon the folk knowledge of ethno medicinal plants for the treatment of different diseases. It is therefore necessary to preserve this indigenous knowledge on traditional medicines by proper documentation, identification of plant species used and preparation of herbal also. This paper reports an identification of medicinal plants used for the treatment of Asthma and Jaundice disease by the traditional and ethnic people of Mandar Block.

Keywords : Traditional people, manaki, pahan, horopaths, pahan and ojhas.

INTRODUCTION

Medicinal plants have always been a fundamental resource for human being and also provided health care to the people since the dawn of civilization. In the modern era, the rural as well as urban people have a great inclination towards the use of medicinal plants. They use the medicinal plants for the cure of common diseases of health related issues. The most common remedies among the people as Ulcer, Constipation, Fever, Cough, Cold, Jaundice, Snakebites, Asthma, Piles, etc are cured through medicinal plants. The local people of Mandar Block, Ranchi District of Jharkhand State has explored through their indigenous practices by using medicinal plants for the remedies of Asthma and Jaundice.

The people of Mandar Block have vital knowledge concerned to the ethnomedicinal plant species. The tribal and scheduled population of the Mandar Block comprises of Munda, Oraon, Birhore, Santhals Rajwar, Ganju, Ravidas, Rajak. These people are dependent completely upon the local herbal medicinal plants for the remedies of common diseases. People dwelling Mandar Block are having a intimate and long association with the indigenous flora for their cultural activities and basic necessity.

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METHODS AND MATERIALS

During the study course, field surveys were completed by the help of Pahans, Ojhas, Manakis, Horopaths and traditional people. They were requested to provide the information against plant used for the cure of Asthma and Jaundice. The tribals of Mandar Block were very gawky in nature and they didn't want to disclose the all concerned precious information to other. They got annuity after a long span of time to supply the required information. The tribal person knows only the common name of herbal plants. The all precious information with their common name was recorded as per prescribed

methodology. The all herbarium of concerned plants were shifted and submitted to the P.G. Department of Botany, Maharaja College, Ara (Bhojpur), Bihar.

All the recorded information is being listed as under with their local name, family name, botanical name and others. The reference and the principle of every plant have been listed after consulting The Wealth of India: A Dictionary of Indian Raw Materials and Industrial products (1948-1976).

For Asthma

1. *Papaver somniferum* Linn. (Papaveraceae) "Post"
   D.E.P., VI(I) 17, 150; C.P. 845; Fl. Br. Ind., I, 117.
   Opium is prescribed in Asthma. The principle opium alkaloids are thebaine, papaverine, codeine, narcotine, and morphine. Among them morphine is the most important.
2. *Argemone mexicana* Linn. (Papaveraceae) "Satyanashi"
   Wlth. India- Raw Materials, I, 116; Fl. Br. Ind., 117; Kirt and Basu, Pl. 54.
   The juice of leaf or root is useful in it. The seeds yielded Agemone oil which contains two alkaloids as dihydrosanguinarine and sanguinarine.
3. *Albizia lebbeck* Benth. (Mimosaceae) "Siris"
   With India- Raw Materials, I, 43; Fl. Br. Ind., II, 298.
   The Juice of flower is prescribed. The flower on stem distillation gives colorless sweet smelling oil. The oil yielded benzyl Alcohol, benzoic Acid and p- nitrobenzoate after fractionation.
4. *Piper longam* Linn. (Piperaceae) "Pipli"
   D.E.P., VI(I) 258; C.P., 891; Fl. Br. Ind., V, 78; Kirt and Basu, III, 2128, Pl. 821A.
   The root and its fruit attributed with numerous medicinal uses in Asthma, bronchitis and cough. The presence of piperine, piplartine and alkaloids are occurred in it.
5. *Alpinia galangal* (Linn.) Willd. (Zingiberaceae) "Barakulanjar"
   Wlth. India- Raw Materials, I, 62; Fl. Br. Ind.,VI, 253; Kirt and Basu, Pl. 949.
   Powder of its Rhizomes is prescribed for Asthma. Cineol, Camphor, probably d- pinene and methyl cinnamate occur in the essential oil.
6. *Artocarpus integrifolia* Hook (Moraceae) "Kathal"
   Wlth India- Raw Materials, I, 125; Fl. Br. Ind., V, 541.
   The juice of its root is prescribed. Supogenins occurs in its stems and leaves. The leaves yield cycloartenone, cycloartenol, β- sitosterol and tannins. They show estogenic activity.

For Jaundice

1. *Curcuma longa* Linn. (Zingiberaceae) "Haldi"
   D.E.P. II, 659; C.P. 445; Fl. Br. Ind., VI, 214.
   Rhizome powder is prescribed. The choleretic action of the essential oil of rhizome is attributed to p- tolymethylecarbinol.
2. *Aloe barbadensis* Mill. (Liliaceae) "Ghi- Kuvar"
   Wlth. India - Raw Materials, I, 61; Bailey, 1949, 209; Hodge, Econ. Bot., 1953, 7, 107.
   Juice of whole plant is prescribed. The leaves contain barbaloin, chrysophenol, glycoside and the aglycone and aloe-emodin.
3. *Ricinus communis* Linn. (Euphorbiaceae) "Erandi"
   D.E.P. VI(I), 506; C.P. 915; Fl. Br. Ind., V, 457.
   Juice of its fruit is useful. Ricin and albumin are the principal toxic substance. Ricin is exreamly toxic than all other species of mammals.
4. *Curculigo orchioides* Gaertn. (Amaryllidaceae) "Kali Musli"
   D.E.P., II, 650; Fl. Br. Ind., VI, 279.
   Juice of its tuberous root is useful. The analysis of the drug shows the presence of ether extract, alcohol extract, water, tannin, ash, crude fiber and starch.
5. *Argemone mexicana* Linn. (Papaveraceae) "Satyanashi"
   Wlth. India- Raw Materials, I, 116; Fl. Br. Ind., 117; Kirt and Basu, Pl. 54.
   Latex of the plant is useful in it. The latex contains the alkaloids, berberine and protopine.
6. *Achyranthes aspera* Linn. (Amaranthaceae) "Latjeera"
   Wlth. India- Raw Materials, I, 24; Fl. Br. Ind., IV, 730.
   Juice of its root is prescribed. The seeds yield two types of saponins: Saponin A and Saponin B.
7. *Citrullus colocynthis* Schard. (Cucurbitaceae) "Makal"

D.E.P., II, 328; C.P., 316; Fl. Br. Ind., II, 620; Kirt and Basu, Pl. 460.

Its root juice is prescribed. The active principle of colocynth includes a bitter amorphous alkaloid and a resin, both of which have violent purgative action.

RESULTS AND DISCUSSION

This interim survey confirmed about the vital knowledge of the tribals and other inhabitants of the Mandar Block of Ranchi District. They are veteran to identify the medicinal plants and have a sound knowledge about the uses of these plants in the treatment of Asthma and Jaundice. 13 different plant species form 13 genera were recorded for the uses in the treatment of Asthma and Jaundice.

The general people are not well known about these medicinal plants. Further a brief account on ethnomedicinal uses of documented plant species has been verified by cross checking with the knowledgeable persons, experienced informants, medicine person. For the safety purpose of human being, further these herbal species were also examined by therapeutic and pharmaceutical persons.

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